FUZZY MULTI-ATTRIBUTE ANALYSIS (FMAA) MODEL FOR ENGINEERING-PROCUREMENT-CONSTRUCTION (EPC) CONTRACTOR SELECTION

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Abstract

Engineering-Procurement-Construction (EPC) projects are normally large and complex, and pose significant challenges for owners selecting the most competent contractor. Although various researchers and organisations have proposed contractor selection methods in different project environments, very few provide an insight to the EPC contractor selection despite the fact that EPC has gained popularity in resource and infrastructure projects globally in recent years.

Selecting a competent contractor is a complex decision-making process, which involves diverse criteria, multiple decision makers and various options. The traditional lowest bid selection, solely based on price, is regarded as one of the main causes of project delivery problems and it is less likely that the best contractor is selected. Best-value procurement, which emphasises quality, efficiency/effectiveness, value for money and performance standard, has become an essential concept in the contractor selection. However, how to determine the best value is complex and difficult with the existence of multiple criteria, multiple decision makers, uncertainty and risk associated with incomplete information, imprecise data, and vagueness in decision making. As a result, subjective judgement of multi-criteria, impreciseness in contractor attribute measurement, and the uncertainty often create fuzziness in contractor performance evaluation.

This research was aimed at developing a comprehensive contractor selection model using Multi-Attribute Analysis (MAA) and Fuzzy Set Theory (FST) to achieve the best value procurement for EPC project owners. Three rounds of Delphi survey helped in identifying and prioritising the criteria specific to EPC projects. Among the 16 criteria specifics to EPC projects, past performance, project understanding, technical, key personnel and health and safety are the top five criteria. Multi-Attribute Analysis (MAA) was used to identify optimum choice against multiple objectives. Given that subjectivity, uncertainty and impreciseness exit in multi-attribute assessment, Fuzzy Set Theory (FST) was used to address this fuzzy nature in human decision making when evaluating tenders by using linguistic variables instead numerical values to rate the contractor performance against each criterion.

The final Fuzzy Multi-Attribute Analysis (FMAA) model will help project owners to select the most reliable and capable contractor by eliminating the current weaknesses of over-reliance on subjective methods, the lowest bid, and impreciseness and uncertainty in human decision making, thus leading to the achievement of best-value. The findings contribute significantly to the current body of knowledge of the EPC procurement system and provide owners with practical implications for contractor selection.

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List of Abbreviations

ACA	Australian Constructor Association
ACT	Australia Capitol Territory
CEM	Construction Engineering and Management
EA	Engineers Australia
EOI	Expression of Interest
EPC	Engineering-Procurement--Construction
EPCM	Engineering-Procurement-Construction Management
FEED	Front-End-Engineering-Design
FST	Fuzzy Set Theory
QLD	Queensland
MAA	Multi-Attribute Analysis
MCA	Multi-Criteria Analysis
MCDM	Multi-Criteria Decision Making
NSW	New South Wales
NT	Northern Territory
SA	South Australia
SPSS	Software Package for Social Sciences
TAS	Tasmania
TMR	Transport and Main Roads
VIC	Victoria
WA	Western Australia

Statement of Original Authorship

The work contained in this thesis has not been previously submitted to meet requirements for an award at this or any other higher education institution. To the best of my knowledge and belief, the thesis contains no material previously published or written by another person except where due reference is made.

Signature: QUT Verified Signature

Date: 23/10/2017

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Publications and Presentations

2017 A review journal paper on "The Engineering-Procurement-Construction (EPC) Market in Australia: A Review" - to be submitted

2016 A full conference paper on "Tender Evaluation Criteria for EPC Contractor Selection" has been presented at the 10th International Conference on Project Management (ProMAC2016) at Gold Coast 16-19 November 2016 and published in the Proceedings of the 10th International Conference on Project Management (ISBN 978-4-902378-48-1)

2014 A full conference paper on "Measuring Sustainability Performance within the Australian Energy Industry" has been published by Springer in the proceedings of the 19th International Symposium on Advancement of Construction Management and Real Estate (CRIOCM 2014) in China

Chapter 1: **Introduction**

1.1 RESEARCH BACKGROUND

The construction industry is one of the most significant contributors to the Australian economy in terms of Gross Domestic Product (GDP) and employment (Australian Bureau of Statistics (ABS), 2012b). Project delivery methods such as traditional design-bid-build, integrated design-build, and construction management are used for construction projects with varying degrees of success, depending on project types and skills required (Forbes & Ahmed, 2010). With the rapid growth of construction activities in recent decades, the Engineering-Procurement-Construction (EPC) project delivery method has been used widely for large and complex engineering projects in Australia, mostly in oil and gas, mining and infrastructure major projects (DLA PIPER, 2011).

EPC is a project delivery method where one or more contractors and designers combine their efforts to deliver a full and complete engineering project under a single point of responsibility for the design and construction (Baram, 2005; Forbes & Ahmed, 2010; Galloway, 2009). Particularly for its innovation in design and construction, cost and time certainty, guaranteed performance and a reduced administration burden associated with asset development (EPC Engineer, 2013; Forbes & Ahmed, 2010; Halvorsen, 2009; Meinhart & Kramer, 2004), EPC has been increasingly used to deliver large-scale and complex industrial projects that are driven by engineering designs instead of architectural designs (Forbes & Ahmed, 2010). The EPC project delivery method has weaknesses too. Unexpected additional costs may arise in the form of change orders as the project design and construction evolves. The owner has less design control and intervention opportunities, and allocating almost all project risks to the contractor can lead to unrealistic designs, high costs and reduced quality (Baram, 2005; DLA PIPER, 2011; Forbes & Ahmed, 2010). The nature of engineering design within EPC is multi-disciplinary, creating projects with a high level of risk and complexity.

EPC projects are normally large-scale and complex (typically cost AUD 1 billion or more), which involve multiple stakeholders, take several years to develop and construct, and therefore are classified as major or mega projects. As reported, major

projects in the oil and gas industry regularly overrun cost and schedules as the size and complexity of projects are the most significant factors affecting the variance of cost and schedules (Ruwanpura, Tanveer Nabi, Kaba, & Mulvany, 2006). Procurement of EPC projects is highly dependent on a number of international/domestic sub-contractors, where complex supplies are very difficult to manage (Cagno & Micheli, 2011), have long lead times and execution delays, resulting in cost overruns. The delivery nature of EPC projects is also changing from traditional engineering success to the delivery of more sustainable and economic outcomes (Australian Constructors Association (ACA), 2015). Thus, the operation process, management mode, contractual obligations and risk allocations in EPC are very different from traditional delivery methods (Hui & Qin, 2011).

Given that EPC projects are normally of increased complexity, high budget values, long project timelines, and involve multiple stakeholders, selecting the most competent contractor poses significant challenges for owners. High levels of uncertainty and associated risks inherited in EPC projects also make EPC contractor selection a most crucial and challenging task for project owners. Meanwhile, selecting an appropriate contractor is one of the most important decisions at the early stage of any project to achieve successful project outcomes and specified objectives, which rely on the effectiveness of contractor selection (Alzahrani & Emsley, 2013a; Cheng & Li, 2004; Fong & Choi, 2000; Holt, 1998; San Cristóbal, 2012; Singh & Tiong, 2005; Walraven & de Vries, 2009; Watt, Kayis, & Willey, 2010)

Therefore, EPC contactor selection demands a comprehensive strategy for the contractor performance evaluation. This is not an easy process, as it encompasses many decision parameters (e.g. contractor attributes, client objectives) and a number of outcome options that require owners to confidently entrust the chosen contractor with the responsibility to execute the project satisfactorily (Holt, 2010; Holt, Olomolaiye, & Harris, 1994a; San Cristóbal, 2012). Thus, such a complex decision-making process involves diverse criteria, multiple decision makers and various available options (Alzahrani & Emsley, 2013a; Holt, 2010; Holt et al., 1994a; San Cristóbal, 2012). It is a multi-criteria decision-making process where various factors need to be considered other than price. In addition, modern initiatives such as sustainability, life-cycle costing and standardisation should be also integrated in the procurement process (Ruparathne & Hewage, 2015). But it is identifying the

appropriate criteria for assessment of contractor performance that is the most critical activity in the pre-tender stage. However, the determination of criteria for EPC contractor selection has scarcely been investigated. The existing criteria for contractor selection should be further researched to cater for new demands in the industry.

Meanwhile, as no single dominant contractor performs better than all other contractors in terms of all decision criteria, the owner is faced with a trade-off issue, which requires a structured framework to select the most appropriate tender with a high level of confidence (Mahdi, Riley, Fereig, & Alex, 2002). This contractor performance evaluation process becomes even more complex and challenging with uncertain, imprecise and subjective data (Deng, 1999). Subjective judgement of multi-criteria, impreciseness in contractor attribute measurement, and the inherited uncertainty in EPC projects often create fuzziness in contractor performance evaluation.

Although various researchers and organisations have proposed contractor selection frameworks for different project delivery methods, only a few to date have focused on the EPC delivery method. Therefore, the aim of this PhD study is to develop a new EPC contractor selection model to evaluate tenders objectively using multi-attribute analysis and fuzzy set theory. Multi-Attribute Analysis (MAA) is capable of identifying optimum choice against multiple objectives. It is the Fuzzy Set Theory (FST) that has been developed to solve problems where there are no defined boundaries of set of activities or observations. In fact, FST can assist decision-maker assessments by making the contractor selection process more systemic and realistic (Deng, 1999; Singh & Tiong, 2005).

To achieve the research aim, this research first investigated the tender evaluation criteria appropriate for EPC projects through a comprehensive literature review and three rounds of Delphi questionnaire survey with the experts in the Australian EPC construction industry. Multi-Attribute Analysis then imposes a disciplined structure of criteria and their importance weights, and award algorithms to identify the most preferred option. As subjectivity, uncertainty and impreciseness exit in multi-attribute assessment, Fuzzy Set Theory was used to address this fuzzy nature in human decision making when evaluating tenders. The final EPC contractor selection model will enable owners to achieve the best value of the procurement by selecting

the most reliable and capable contractor using fuzzy multi-attributes analysis that eliminates the current weaknesses of over-reliance on subjective methods, the lowest bid, and overcoming impreciseness and uncertainty in human decision making. The findings add significant insight to the body of knowledge of EPC procurement systems and provide owners with practical implications for contractor selection.

1.2 KNOWLEDGE GAP AND RESEARCH PROBLEM

1.2.1 Research Problem

To support clients in selecting the most competent contractors, researchers have developed different evaluation and selection methods (Fong & Choi, 2000; Holt, 1998; San Cristóbal, 2012), with most exhibiting significant variations, an over-reliance on the lowest bid and on subjective methods (Holt, 1998).

Pertinent issues such as lack of understanding of procurement practices, weaknesses in the traditional selection methods, and inherent problems in identified strategies in existing contractor selection models do not guarantee EPC project owners a quality contractor selection process. The problems mentioned below are the most significant.

- **Problem 1: Lack of understanding of EPC method and the current EPC market**

No clear view on procurement in construction projects leads to poor interaction with venders in the engineering and procurement stages (Ruparathne & Hewage, 2015). A full understanding of the EPC delivery method in relation to the unique nature of projects and prevailing market conditions is needed to fully benefit from its use (DLA PIPER, 2011). Furthermore, a large number of contractors for a limited number of projects and uncertain construction industry environment results in intense competition among competent contractors (Australian Industry Group (AIG), 2015). Thus, sound judgement on tenders using a well-structured contractor selection framework is essential to select the most appropriate contractor.

- **Problem 2: Drawbacks in current selection process and failure to achieve 'value for money'**

Government and private client organisations emphasise procurement framework to achieve 'value for money' (Oyegoke, Dickinson, Khalfan, McDermott, & Rowlinson, 2009; Ruparathne & Hewage, 2015). However, use of the lowest bid

causes project delivery problems such as cost over-runs, delays and poor performance (Abdelrahman, Zayed, & Elyamany, 2008; Darvish, Yasaei, & Saeedi, 2009; Mahdi et al., 2002; San Cristóbal, 2012; Singh & Tiong, 2005; Walraven & de Vries, 2009). Contractors in a volatile construction market desperately quote a low bid price to remain in business, which can be risky especially when the contractor is responsible for both design and construction.

Drawbacks in qualification-based selection processes include high subjectivity on selection and reduced competition, favourable towards non-minorities. This is less socially sustainable as new firms would find it difficult to enter the market merely because they have no experience.

Moreover, the current strategies also have inherent problems such as being time consuming, high cost, academic in nature, etc. Effectiveness of evaluation using such methods highly depends on the skill, experience and knowledge of the decision makers (Mahdi et al., 2002).

- **Problem 3: Lack of investigations on selection criteria specific to EPC to fulfil the need of adopting multi-criteria**

Fong and Choi (2000) comment that an objective approach based on the best combination of multi criteria other than price to evaluate the contractor's all-round performance is significant in the contractor selection process. San Cristóbal (2012) emphasises the need to adopt a multi-criteria approach for contractor selection. However, the factors that influence the selection of an EPC contractor and relative importance of the criteria have barely been investigated (Watt et al., 2010).

- **Problem 4: Failure to capture subjectivity, uncertainty, and impreciseness in multi-criteria assessment**

Subjectivity, uncertainty and impreciseness existing in contractor attribute assessments and the descriptive nature of decisions without well-defined boundaries in relation to contractor performance add fuzziness in the contractor performance evaluation process. This fuzzy nature in contractor selection has posed a significant challenge to EPC project owners.

Although there is plenty of available literature relating to other common project delivery methods (e.g. design-bid-build, design-build, etc.), very little has been done to provide an insight to the EPC contractor selection or to address the fuzzy nature in

the human decision-making process. In particular, there is little, if not any, research on the EPC project delivery method in Australia. A comprehensive contractor selection model should be presented to identify the most appropriate EPC contractors to deliver the best value for money for EPC projects in the Australian construction market.

1.2.2 Research Questions

To fill the gaps in the literature, this research focused on developing a comprehensive contractor selection model for EPC projects. The following research questions were addressed accordingly:

- What is the current EPC market circumstance in Australia?
- What are the appropriate evaluation criteria for EPC contractor selection and how important are they in the decision-making process?
- How can decision makers evaluate EPC tenders objectively with the identified criteria, and minimise subjectivity, uncertainty and impreciseness in EPC contractor selection process?
- How well does the EPC contractor selection model fit for industry use?

The above research questions provide the span of research context.

1.3 AIM AND OBJECTIVES

The aim of this research is to develop a fuzzy multi-attributed contractor selection model, characterised by comprehensive evaluation strategies that align with owner objectives and contractor attributes, while minimising the weaknesses of current practices in Australian EPC market. The research objectives include:

Objective 1: Understanding the EPC project delivery method and EPC market in Australia

Objective 2: Developing an EPC contractor selection framework

Objective 3: Identifying and prioritising the criteria for EPC contractor selection

Objective 4: Developing a new EPC contractor selection model using Multi-Attribute Analysis and Fuzzy Set Theory

Objective 5: Validating the EPC contractor selection model for industry use

1.4 SIGNIFICANCE

This research study proposes a Fuzzy Multi-Attribute Analysis Model for EPC contractor selection in the Australian construction industry. This model captures the fuzziness in multi-attribute evaluation by modelling uncertainty, impreciseness and subjectivity in contractor selection practice.

This model includes the most appropriate criteria through qualitative assessments with industry practitioners and addresses the fuzzy nature in multi-criteria assessment with uncertain, imprecise and subjective data, enabling project owners to select the most reliable, economical and capable contractor with best-value-for-money for EPC projects.

As hardly any criteria specific to EPC contractor selection have been investigated, this research fills this knowledge gap with a list of important criteria and their importance weighting for EPC tender evaluation through a rigorous Delphi study with experts in the EPC industry.

More importantly, the current study contributed to the body of knowledge with application of theoretical concepts such as fuzzy set theory and practical evidence within the EPC industry for successful project implementation.

1.5 THESIS OUTLINE

The thesis starts with an introduction of the thesis (Chapter 1), describing the background, research problems and questions, research aim and objectives, and the significance of the research.

Chapter 2 discusses the findings from a comprehensive literature review. It covers project delivery methods, EPC method, contractor selection process, existing contractor selection models, multi-criteria approach, and fuzzy approach adopted in contractor selection models. The literature review findings are used to conceptualise the topic, identify research problems, and define research questions.

The research design and the methodology adopted in this research are discussed in Chapter 3. The research techniques used in this study include literature review, Delphi survey, fuzzy set theory, and interviews. Research approach, research method and frameworks and methods of data analysis are also discussed.

Chapter 4 presents the overview of the EPC market in Australia using secondary data analysis and literature review findings. It is set to achieve objective 1 of the research.

Chapter 5 presents the EPC contractor selection framework.

Chapter 6 details the identification of contractor selection criteria and their weights through three rounds of Delphi survey.

Chapter 7 presents a comprehensive EPC contractor selection model based on multi-attribute analysis and fuzzy set theory.

Chapter 8 introduces model validation through interviews with industry experts and establishes the final EPC contractor selection model for practical use.

Chapter 9 outlines research findings and conclusions, recommendations for future research, and contribution of this research to the existing body of knowledge.

Chapter 2: **Literature Review**

A literature review is the collection of background information of a research study, which aims to consolidate all previous studies related to the research topic and to form understanding of the current practice (Chow, 2005). A comprehensive literature review helps researchers to identify research problems or gaps through critical thinking, while reading existing works (Yeung, 2007). This chapter provides a comprehensive literature review to identify research gaps in EPC contractor selection decision making and to gain clear understanding of the current contractor selection models.

2.1 OVERVIEW OF PROJECT PROCUREMENT PROCESS AND PROJECT DELIVERY METHODS

2.1.1 Construction Project Procurement Process

The project procurement is defined as the process of buying and obtaining the necessary property, design, contracts, labour, materials, and equipment to build a project (Construction Management Association of America (CMAA), 2012; Molenaar, Sobin, & Antillón, 2010; Wang et al., 2013). Procurement links the highly fragmented supply side of the construction industry (engineers, architects, contractors, suppliers, etc.) with the less fragmented demand side (project owners and representatives) (Ruparathne & Hewage, 2015). Furthermore, procurement is an integral part of a construction project (e.g. sourcing, purchasing, etc.), and represents the purchasing steps that an owner must take to gain the services and commodities required under the chosen project delivery method. Ruparathne and Hewage (2015) documented the four phases of the construction procurement process and the important activities occurring in each phase (see Table 2.1)

Table 2.1 Procurement phases

Pre-contractual phase	**Contracting phase**	**Contract administration phase**	**Post-contractual phase**
• Define requirements • Project delivery method selection • Plan procurement process • Obtain necessary approvals • Bid solicitation	• Tender Invitation • Pre-bid meeting • Tender evaluation • Contract execution	• Issue contract amendments (variation orders/extension claims) • Monitor progress • Follow upon delivery • Administer progress payments	• Issue final claim/ amendments • Ensure completeness of documentation • Complete Financial audit • Return performance bonds

Transport and Main Roads (TMR) delivers many public EPC projects in Australia and their procurement planning phase comprises setting prequalification level, selecting contract type and contractor selection as the main tasks after selecting a project delivery method (TMR, 2014). Miller, Furneaux, Davis, Love, and O'Donnell (2009) identified the selection of procurement methodology (overall approach), procurement strategy (means of achieving project objective) and delivery method (sometimes known as procurement route/system) as being the three main activities of construction procurement.

Essentially, three major pre-contracting activities of design and construction project procurement are selection of project delivery method, procurement strategy, and contracting method. As these terms often confuse professionals, close investigation is required for clarity. Project delivery methods and contracting methods are discussed in this section while procurement strategies are described under contractor selection process (section 2.3).

2.1.2 Project Delivery Methods

Project delivery is a comprehensive process by which designers, constructors and various consultants provide services for design and construction to deliver a complete project to the owner (Molenaar et al., 2010). The project delivery method consists of components of design and construction, as well as responsibilities for cost, schedule, quality, and management, which are combined under an agreement that results in a completed facility (Beard, Loulakis, & Wundram, 2001).

Selection of the most appropriate project delivery method has been a major project success factor and any failures can cause client dissatisfaction. The principals (clients and consultants) and constructors often have different views on how best to procure major infrastructure projects. However, a lack of understanding of different project delivery methods causes problems (Miller et al., 2009), suggesting a clear view of delivery methods is also important.

A number of project delivery methods have been developed for the process of designing and constructing facilities and commonly used project delivery methods include (CMAA, 2012; Forbes & Ahmed, 2010; Molenaar et al., 2010): Design-Bid-Build (DBB), Design-Build (DB) or Design and Construct (D&C), Construction Manager at Risk (CM at Risk), Engineering-Procurement-Construction (EPC), Design-CM (Construction Management), Fast-Track Construction, Partnering, Turnkey, and Build-Operate-Transfer (BOT). These different project delivery methods are distinguished by the structural variations between the owner, the designer, and the constructor, financing methods, and operational variations within the parties (Ghavamifar, Touran, Molenaar, & Gransberg, 2011; Oyegoke et al., 2009).

There are four main project delivery categories that have been widely accepted in the construction industry. These are traditional procurement, integrated procurement, management procurement and collaborative (or relational) procurement (Table 2.2). EPC is categorised as an integrated procurement method.

Table 2.2 Major project delivery categories

Categories	Project delivery methods
Traditional (Separated) Procurement	DBB/Construct only
Integrated Procurement	DB plus variants (Design, novate and construct (DN&C), Design, development & construct (DD&C), DC&M) EPC
Management procurement	Managing contractor CM Direct Managed
Collaborative (or relational) procurement	Alliance Private Financed Project (PFP) (includes BOOT, BOT and DBFM) Public-private partnerships (PPP) includes DBO, DBFO, DBOM and DBFM Fast-Track Early Contractor Involvement (ECI)

Understanding the unique features of the EPC delivery method compared with other delivery methods, in particularly DB, is important. Even though the term of Design & Construct (D&C) is more often used for DB projects in Australia, the term of DB is used in this thesis hereafter to keep consistency with the majority of literature. Common features of the main project delivery methods are given in Table 2.3.

Table 2.3 Different project delivery methods

Delivery method	Major Features

DBB	• Traditional method • Owner conceptualises the project and plans • Responsibility for design and construction spread between three parties (Owner, Designer, Contractor) • Design and Construction are two distinctive contracts • Completed project is of an acceptable quality however cost overruns and schedule delays may occur
DB	• Owner conceptualises the project and plans, owner design team prepares scope document and/or schematic designs, which forms basis for 'Request for Proposals' • DB contractor develops design, cost and schedule proposals • DB contractor responsible for both design and construction • Design-build contract closes out, similar to closing out of a DBB contract
EPC	• Very similar to DB • Primarily used for industrial projects driven by engineering design • Owner's design team or FEED contractor develop FEED, which forms the basis for inviting proposals • EPC contractor responsibilities are extended for commissioning and maintenance beyond design and construction • EPC contractor hands fully operational facility over to client at close out
Turnkey	• One business entity performs design, construction, and construction financing • Payment is typically made on completion thus the contractor turns over the 'key' to the owner, enabling owner to start operation/occupancy
CM at Risk	• Construction manager assumes risk of pricing and directly contracts with trade contractors • Not quality initiative but is achieved through a selection process

CM	• Owner hires CM organisation to provide professional management services • Owner has separate contracts for design professionals and trade contractors • Owner has heavy responsibilities for work coordination
Fast-Track Construction	• Used when accelerated schedules are demanded by the owner • Contractor commences work immediately after contract award and simultaneously designer completes construction documents
BOT	• One business entity performs the design, construction, long-term financing, temporary operation and transfer to owner at end of operation period
Partnering/alliancing	• Two or more entities undertake work co-operatively on the basis of sharing project risk and reward for the purpose of achieving agreed outcomes

It is noted that DB and EPC are distinguished as two different project delivery methods even though they share critical similarities. Typical DB structure in Fig. 2.1 and EPC structure in Fig. 2.2 clearly show the structural variations of both delivery methods, which are determined by the roles of the main parties involved in projects.

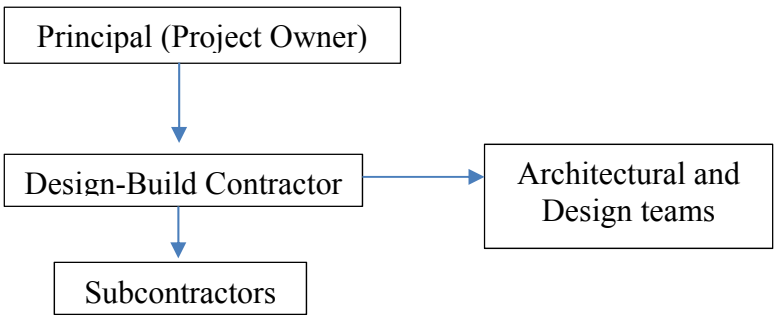
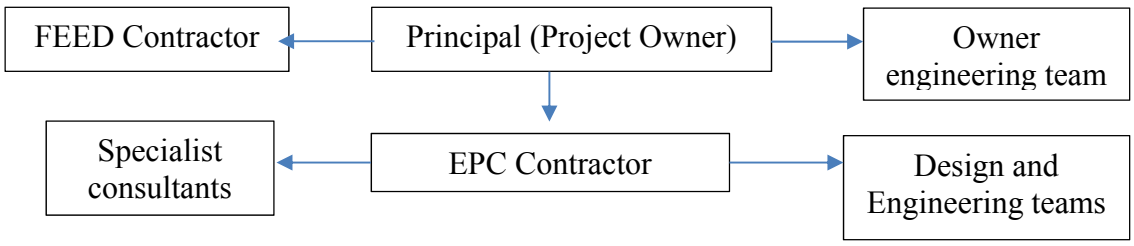


Figure 2.1 Typical design-build structure



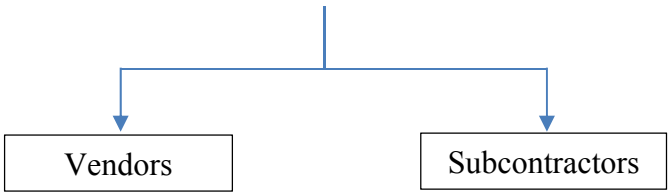


Figure 2.2 Typical EPC Structure

2.1.3 Contracting Methods

Project contracts, different from project delivery method or procurement strategy, represent the way in which the delivery systems are packaged and how contractors are paid by the owner, as well as who accepts the risks of performance based on the standards and conditions stipulated in the contract (Galloway, 2009). The main categories of construction contracts to be integrated with the project delivery methods (Forbes & Ahmed, 2010; Gordon, 1994; Huse, 2002) include

- Fixed Lump-Sum Price
- Fixed Unit Prices (Bill of Quantities)
- Cost Plus a Fee (Cost Reimbursable), and
- Guaranteed Maximum Price

Lump sum (Fixed Price)

The lump sum provides the contractor a fixed sum of money for completion of work stipulated in the contract regardless of contractor's as-built work (Forbes & Ahmed, 2010; Gordon, 1994; Huse, 2002). The lump sum commonly includes all labour, materials, project overhead, company overhead and profit (Gordon, 1994). The difference between the fixed sum and actual cost of the works to the contractor constitutes the contractor's loss or profit (Huse, 2002). However, the fixed sum can be adjusted in certain circumstances such as the changes in work as required by the owner, unforeseeable adverse site conditions, incorrect data supplied by the owner, suspension or delays and defective specifications etc. (Forbes & Ahmed, 2010; Huse, 2002). Generally, the contractor is paid in instalments based on a schedule of payments or at specified stages of completion (Huse, 2002). The lump sum contract is generally easier and less expensive to administer than other contracts (Huse,

2002). However, a fixed cost contract exposes high risk to a contractor (Ruparathne & Hewage, 2015). Despite some researchers having regarded 'fixed price' as a traditional procurement method, 'fixed price' can be distinguished as a contracting method.

Unit Price (Bill of Quantities)

Unit price is common and the traditional method of contracting for construction works. The price is established per unit of quantity with reference to a bill of quantities or a schedule (Huse, 2002). The contractor agrees to be paid a set cost per unit of each item, and the actual amount paid is based on the actual measured units constructed on the project, times the unit price agreed to (Gordon, 1994). The unit price for each item commonly includes all labour, materials, project overheads, company overheads, and profit. Sometimes overhead items are paid separately (Gordon, 1994). Since units are not fixed, this allows renegotiation for a unit price for substantial variation in unit quantities from the estimated quantities (Forbes & Ahmed, 2010).

Cost-plus (Cost Reimbursable)

Under cost plus contracts, the owner pays the contractor for costs incurred plus a predetermined margin of profit (Huse, 2002). The contractor reimburses the cost of the work including labour, material, and project overhead, plus a fee (Gordon, 1994). The cost is defined as the contractor's direct job site costs and is essential to avoid dispute over the cost items that can be reimbursed. The fee covers the company overhead and profit in terms of a fixed sum, a percentage of the cost or a formula incorporating both. This form of payment may not be economical as the greater the cost, the greater the profit irrespective of progress. To reduce the risk to the owner, one possibility is to have a condition that any cost above the target cost will not incur a fee or materials and services actually used by the contractor will not be paid for. This method can generally be used as a last resort where it is impossible to calculate the construction costs and for situations where the design requirements are constantly changing or unknown (Ruparathne & Hewage, 2015). However, this approach presumes the total financial risk for the client (Ruparathne & Hewage, 2015).

Guaranteed Maximum Price (GMP)

The GMP contract is normally based on scope of work outlined in preliminary design documents and the contractor provides cost estimates throughout the design process. The contractor reimburses the cost of work, which includes labour, material, and project overhead plus a fee (Gordon, 1994). The fee includes company overhead and profit up to a prearranged maximum price. Once that price is reached, the contractor must finish the work at no additional cost to the owner, unless there are owner-directed scope changes (Forbes & Ahmed, 2010). If the work is finished under the maximum price, there is often a sharing of the cost difference between the owner and the contractor as an incentive to the contractor to reduce costs. The client accepts a part of the financial risk within this contract (Ruparathne & Hewage, 2015).

Although there are several forms of contracts, some are better suited for particular project delivery methods (Forbes & Ahmed, 2010). Gordon (1994) indicated that the selection of a contracting strategy is basically governed by project delivery method and organisation type (see Fig. 2.3). The lump sum contracting method is often used for EPC contracts and enables the use of fixed payments by stages of completion (Huse, 2002).

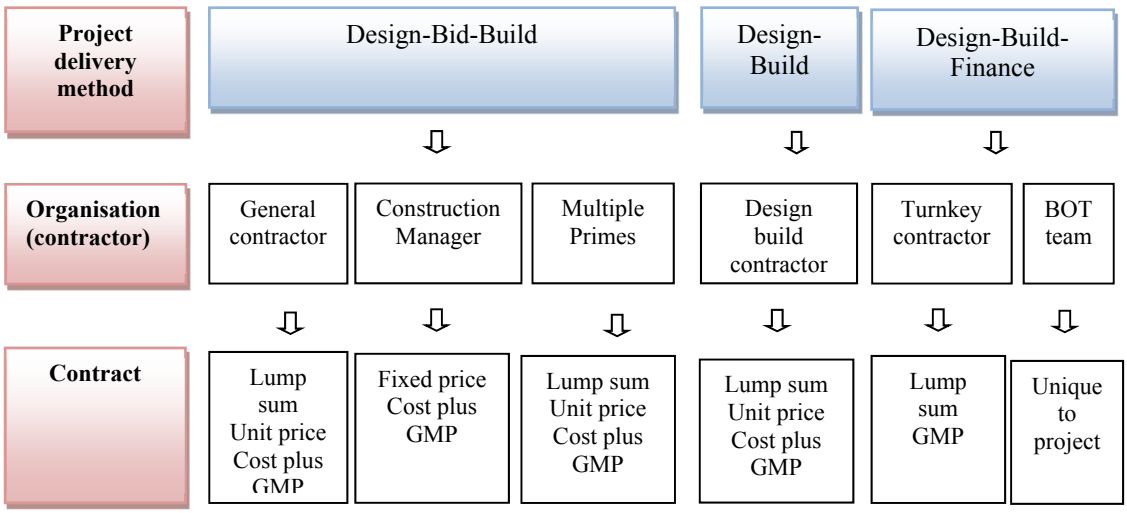


Figure 2.3 Appropriate contracts for respective project delivery

Source: Gordon (1994)

2.2 EPC PROJECT DELIVERY METHOD

Engineering-Procurement-Construction (EPC) is a delivery method where one or more contractors and designers combine their efforts to deliver a fully operational facility under a single responsibility (Galloway, 2009). It has many advantages, such as innovation in design and construction, cost and time certainty, guaranteed performance and reduced administration burden associated with asset developments (EPC Engineer, 2013; Forbes & Ahmed, 2010; Halvorsen, 2009; Meinhart & Kramer, 2004).

EPC is increasingly used to deliver large scale and complex industrial projects that are driven by engineering designs instead of architectural designs (Forbes & Ahmed, 2010). The current construction industry extensively uses the EPC delivery method for major mining, oil, gas and infrastructure projects, particularly in the private sector (DLA PIPER, 2011) and it is popular in the energy and natural resources sector in the global market (KPMG International, 2015).

Understanding the EPC structure clearly benefits EPC project stakeholders for planning and implementing of an EPC project. Two main phases in EPC are planning and implementation. Work packaging and EPC contractor selection (pre-EPC activities) occur in the planning stage. Pre-EPC works typically take 2-3 years to complete. Another important activity that occurs in the planning stage is that the client or client's agent develops Front-End-Engineering-Design (FEED) to a level sufficient for inviting tender proposals (Mayer Brown, 2008). Then, the EPC contractor is often selected in competitive bidding environment and is typically engaged by a lump sum contract.

After the contract award, the EPC contractor starts the implementation phase, which normally spans for at least three years, and the EPC contractor becomes the single point of responsibility for completing the detailed engineering, procurement and construction and delivery of functioning facility to the client/owner within the agreed time and budget (Baram, 2005).

EPC implementation is apparently a major challenge for owners because:

- Integration of multi-discipline of engineering creates a high level of risk and complexity (Engineers Australia (EA), 2013c)
- Projects become increasingly larger, longer and more complex

- Size and complexity are significant factors that affects cost and schedules (Ruwanpura et al., 2006)
- Nature of project delivery changing to more sustainable and economical outcomes (Australian Constructors Association (ACA), 2015)
- Procurement within EPC is highly dependent on international and domestic sub-contractors-thus complex supplies are very difficult to manage (Cagno & Micheli, 2011)
- EPC contracts are complex agreements
- Long lead times in supply chain can cause cost overruns
- Project teams are supported by large, often geographically dispersed sub teams
- Critical asset data and project information must be created and transferred in multiple stages - thus, sophisticated systems and methods are required
- Involves multiple stakeholders - decision making process more complicated (Li, Ng, & Skitmore, 2016), and
- Poor interaction with vendors in engineering and procurement stages causes project delays (Ruqaishi & Bashir, 2015).

Therefore, capability and capacity of the EPC contractor are important to ensure successful execution of projects (Baram, 2005; Lunde, 2001; Xia, Chan, & Yeung, 2009). Reportedly, two-thirds of large projects fail at twice the rate of smaller projects thus placing greater risk on EPC project owners as well as investors. The Australian Constructors Association (ACA) reported that the Australian mega-project (>$1 billion) performance is poor, as 20% of projects overrun budgets and 28% of projects experience schedule overrun (ACA, 2015) alerting the EPC project owners to their challenges.

Wrong contractor selection is identified as a key factor that results in major project problems (Rothman, 2000). Selecting a competent EPC contractor is of ultimate importance for project clients/owners given that EPC contractors faced labour supply restrictions, loss of intellectual knowledge and limited funds availability for infrastructure projects (DLA PIPER, 2011; Galloway, 2009) as a result of weak global economies that have not yet fully recovered.

2.3 CONTRACTOR SELECTION PROCESS

Contractor selection is one of the critical tasks for project owners or their representatives dealing with the early stages of a project. Given the importance, this topic has been broadly discussed in the literature. It is described under the following themes in subsequent sections:

- Procurement strategy (low bid/best-value/qualifications based and tendering options)
- Stages of contractor selection, and
- Tender evaluation.

2.3.1 Procurement Strategy

Procurement strategy consists of policies and procedures which govern the selection of a satisfactory contractor to support a preferred project delivery method (Ruparathne & Hewage, 2015). The procurement strategy selection is typically governed by project drivers, owner drivers and market drivers. Therefore, when designing the appropriate procurement strategy, the factors such as legislative context, compliance obligations, organisational risk appetite, market availability, project complexity, in addition to scale, scope, and risk of the project, should be considered.

Various procurement strategies are used in design and construction contracts. Conventional procurement strategies are mostly based around the concept of time, cost and quality. Such criteria are inevitable and are the basic attributes for selecting a contractor. Other factors have been considered with the complexity associated with client needs and priorities, the contradictory nature of the performance and lack of agreement to measure these criteria when assessing contractor performance. Thus, a comprehensive procurement strategy that clearly defines contractor selection process is important in achieving the best value outcomes. Ruparathne and Hewage (2015) identified three main procurement strategies that govern contractor selection as price-based procurement (Low-Bid), qualification-based procurement (Best Qualifications), and value-based procurement (Best-Value) where Molenaar et al. (2010) and El Wardani, Messner, and Horman (2006) identified 'sole-source procurement' as the fourth procurement strategy; however, this has been categorised along with 'Best Qualifications' procurement strategy because selection is based on

subjective and qualitative procurement (El Wardani et al., 2006). Thus, three main procurement policies that govern contractor selection are:

- Low-Bid (LB) - selection is solely based on price
- Best-Value (BV) - selection is based on a weighted combination of price and qualifications, and
- Best Qualifications (BQ) or Sole-source - selection is solely based on qualifications.

Low-Bid (LB) selection

Traditional Low-Bid selection is solely based on price and usually regarded as the key to winning a contract (San Cristóbal, 2012; Walraven & de Vries, 2009). Holt explained that the majority of contractor selection methods exhibit constraints and rely overly on acceptance of lowest bid (Holt et al., 1994a). For some public sector clients, the lowest bid selection is a legislative requirement (Abdelrahman et al., 2008). Otherwise, public sector clients are facing more difficulties as they are accountable for their decisions if they have selected a bidder other than the lowest (Hatush & Skitmore, 1998). In general, it is considered to be a low-bid selection if the cost criteria represents more than 90% of the evaluation criteria (El Wardani et al., 2006).

However, selection based on low price is regarded as one of the major causes of project delivery problems such as project completion delays, poor quality and financial losses (Darvish et al., 2009; Ruparathna & Hewage; San Cristóbal, 2012; Walraven & de Vries, 2009; Yu, Wang, & Wang, 2013).When a contractor has a shortage of work, it is more likely to bid for a lower price to secure the business in the short term and raise additional income through claims or cutting costs to compensate (Walraven & de Vries, 2009). The low bid system encourages contractors to implement cost-cutting measures instead of quality enhancing measures (Abdelrahman et al., 2008). Selection based on lowest bid price can be risky, especially when the contractor is directly responsible for both the design and construction of more complex projects (Walraven & de Vries, 2009). As a result, the

lowest bid selection practice has been criticised over the years because it involves high risk exposure of the client (Darvish et al., 2009).

With these reasons, it is less likely that contracts will be awarded to the best-performing contractors, who will deliver the optimum quality projects under low-bid selection. The recent trend is away from the lowest price policy (San Cristóbal, 2012). Undoubtedly, the bid price is still an important factor in contractor selection, however there are other important factors that should merit consideration (Darvish et al., 2009; Walraven & de Vries, 2009). Government agencies also believe that the lowest bid, even under competitive bidding, may not result in the best value for money or the best performance during construction (Abdelrahman et al., 2008).

Best Qualification based selection/ Sole Source selection

In best qualification-based selection, the owner selects the contractor through a request for qualification (RFQ) for the project and negotiates directly with the contractor who has the most qualifications at a reasonable price (El Wardani et al., 2006). Selection of the contractor is primarily based on qualitative criteria, which include past performance, reputation, technical competence, and financial stability. The owner ranks the contractors and begins negotiations with the number 1 ranked contractor to reach a 'fair and reasonable' price for the services required. Gransberg and Senadheera (1999) indicated that the administrative burden can be considerably eased when owners select the contractor based on qualification. A procurement strategy is to be considered as the best qualification based selection if the non-price criteria represent more than 50% or more of the evaluation (El Wardani et al., 2006).

However, sole-source selection can be a direct selection of the contractor based on established relationships through previous projects (El Wardani et al., 2006). Private sector clients especially, who have established long-term relationships with contractors, use sole-source or direct selection. Public sector clients may also employ sole-source selection when there are no other potential bidders or in an emergency, which allows the waiver of strict procurement rules (Beard et al., 2001).

Drawbacks in qualification-based selection processes include high subjectivity on selection, reduced competition, and favourability towards non-minorities. This is less

socially sustainable, because contractors who are new to the market would find it difficult to be sustained in the market merely because they have no experience or lack of relationships with clients.

Best-Value selection

Best value is defined as a procurement strategy where price and other key factors are considered in the contractor selection process to enhance the long-term performance and value of construction (Abdelrahman et al., 2008; Yu & Wang, 2012). Ashworth (1998) describes 'Value for money' as a combination of subjective and objective judgements. In construction procurement, 'Value for money' is defined as an achievement of the optimum combination of entire lifecycle cost and quality to meet the customer's requirements (Ruparathne & Hewage, 2015). The best value is therefore regarded as a balance between the price and qualifications to select the 'best value for money' bids through a structured, multi-criteria approach that allows evaluation of different criteria simultaneously to obtain optimum outcome (Abdelrahman et al., 2008; El Wardani et al., 2006; Walraven & de Vries, 2009). In the multi criteria method, the contractor who can maximise the client's investments wins the tender (Walraven & de Vries, 2009). There is greater recognition and an increasing acceptance of factors other than price as key determinants of the contractor selection process in obtaining 'Best-Value'.

Best value procurement consists of four primary concepts of parameters, evaluation criteria, rating systems, and award algorithms (Molenaar et al., 2010). Cost, time, qualifications, quality, technical, environmental aspects, maintenance and operability, and managerial safety are potential parameters of best-value procurement (Yu et al., 2013). Abdelrahman et al. (2008) and Yu and Wang (2012) highlighted that the factors other than price can be typically placed under key groups of technical and managerial merit, financial health, and past performance. Abdelrahman et al. (2008) indicated that by inclusion of key factors that can address the specific needs of a project, increases the possibility of selecting the best contractor. Gransberg and Senadheera (1999) revealed that the best value selection is the most flexible procurement strategy that allows evaluation of project specific factors such as project size, facility complexity, design and technical requirements. As such, best value

procurement emphasises quality, efficiency/effectiveness, value for money, and performance standard (Yu et al., 2013).

Best value procurement process involves assigning a score for each contractor with consideration to price and quality (Migliaccio, Bogus, & Chen, 2010). Usually such evaluations are performed separately and the best value algorithms combine each score to find the best value to the owner (Migliaccio et al., 2010). Molenaar et al. (2010) indicated that the project owners rate contractor ability to meet project objectives that are defined by the best-value parameters.

As described extensively, the best-value procurement is more advantageous to the client than traditional price-dominated low-bid procurement (Yu & Wang, 2012). Abdelrahman et al. (2008) indicated that the low-bid selection fails to offer the lowest overall cost, however best-value offers a reduction in cost growth from 5.7 to 2.5% and in claims and litigation by 86%, which are important to project owners. El Wardani et al. (2006)'s findings given in Table 2.4 emphasised that the best value selection achieves overall best performance for projects with high complexity with the least cost growth and schedule growth.

Table 2.4 Comparison of procurement strategies of high complexity projects

Performance indicators	**Procurement Strategy**			
	Sole source	Qualification based	Best value	Low bid
Cost Growth (%)	8.4	0.5	2.5	7.3
Schedule (%)	0.0	0.0	1.0	14.8

Source: El Wardani et al. (2006)

Moreover, best-value selection contributes many values such as innovative designs, additional facilities, better aesthetic value, cost and time savings, better quality control, lower life-cycle costs and reduced risks (Palaneeswaran, Kumaraswamy, & Zhang, 2012). Therefore, best value is the most common procurement strategy for design-build project according to the Construction Management Association of America (CMAA) (2012). This is evidenced in the report of Molenaar et al. (2010), that the largest proportion of public DB projects were acquired through best-value procurement.

Additionally, the Australian government project procurement framework clearly stated that the government clients or agencies should aim to achieve 'value for

money' whenever they procure building and construction industry services, as the traditional method of awarding construction contracts to the lowest price may not select the most suitable contractor for the project (Department of Treasury and Deregualtion (DTD), 2012). The Victorian Auditor-General's report highlighted that it requires obtaining the optimum combination of quality, quantity, risk, timeliness, on a whole-of-contract and whole-of-asset-life basis, to find the best value for money. Australian government clients include 'value for money' as a one of the key objectives because it is advantageous to consider both price and non-price criteria, as assessment of prequalification may not be relevant to the specific project circumstances, complexity, and /or risk factors.

As the lowest price is not the promising procurement strategy to attain the overall lowest project cost upon project completion, researchers and practitioners commonly use multi-criteria selection (Darvish et al., 2009). However, it is essential to identify an appropriate way of achieving the best value using the best tendering method.

2.3.2 Tendering method

Tendering (or sometimes called 'bidding') is a process that will be undertaken in every procurement route that attracts offers from suitable tenderers (Oyegoke et al., 2009). The tendering process needs to be tailored to the client requirements, supplier market, and commercial realities. Additionally, the initial transaction cost inherent in a tendering method can impact the selection of the appropriate tendering method (Oyegoke et al., 2009).

Tendering involves strategic considerations, and is governed by project drivers, owner drivers and market drivers, therefore the procurement strategy, its level of risk, time and cost implications needs to be analysed prior to selecting an appropriate tendering method for any procurement activity. NSW government procurement guidelines (2011) highlight that it requires consideration of the following factors when selecting an appropriate tendering method:

- Advantages and disadvantages of the different tendering options
- Availability of pre-qualified or pre-registered tenderers
- Capabilities of the market
- Risks identified and their implications, and

- Approach adopted for managing risk and ensuring best value for money, process probity, fair dealing and effective competition.

Nieto-Morote and Ruz-Vila (2012) identified open tendering, selective tendering/restricted tendering, prequalification or negotiation as common tendering methods that are used to select a contractor. Gordon (1994) revealed that the major procurement strategies over the years are competitive bid, cap, negotiation, qualification and price proposal, time and price proposal, qualification, time and price proposal, and design and price proposal which appear to be a mixture of procurement strategies as well as tendering methods. Enshassi, Mohamed, and Modough (2013) referred to the following two approaches as procurement strategies even though they are more likely to be tendering methods. These include single source (includes direct hiring, negotiation, and restrictive bid), and open competition (lowest bidder or non-lowset bidder). Apparently, the researchers have implicated tendering methods very differently. The main tendering methods are described in Table 2.5 below.

Table 2.5 Tendering methods

Tendering method	**Description**
Open Tendering	Every contractor can bid and is very similar to selection of lowest bid in a market competition
Selective tendering/restrictive tendering	Only contractors who fill project requirements can bid and used when special expertise and high technology are necessary
Prequalification	Only prequalified contractors based on predetermined criteria are invited to bid
Negotiation	Single contractor is selected by negotiation in an emergency when other processes are not available

Ashworth (1998) indicated that there are two ways of selecting a contractor, either by competition or negotiation, into which the above tendering methods can be grouped. Competition includes 'selective competition' where a few selected contractor organisations are invited to bid, or 'open competition' where bidding is opened to any contractor organisation that wishes to submit a tender. Contractor selection by negotiation involves a single contractor organisation. The negotiated selection

process is included in the best qualification based selection method (El Wardani et al., 2006). The negotiated approach is also beneficial to clients for a number of reasons such as long-term business relationships, early start on site, continuation of contract, contractor specialisation, financial arrangements, etc.

Historically, construction procurement was based on sealed bidding where the lowest responsible bidder was selected, simplifying the awarding process and avoiding bid protests in courts (Enshassi et al., 2013). The lowest bidder selection through competitive bidding is a routine practice within the construction industry worldwide (Enshassi et al., 2013). This practice promotes healthy competition and ensures the lowest contract price for the project is achieved, but does not guarantee the lowest cost on completion as well as the best contractor.

Moreover, the Australian government or their agencies use three tendering methods such as open, limited, and selective. In the limited-offer process (also known as a limited procurement method; limited tender or direct sourcing), the client invites a contractor(s)/supplier(s) of its choice to submit an offer in response to an approach to market. The selective offer process (also known as prequalified tender, selective procurement method, select tender) invites contractors or suppliers that have pre-established criteria to submit an offer in response to an approach to market. The definition of each tendering methods from a government perspective is given in Table 2.6.

Table 2.6 Australian government tendering methods

Open Tender	• One-stage-open approach to the market • Evaluation of all tender received • Must be advertised in media as e-Tender or newspapers • Used when there is a broad competitive market and is not efficient or cost effective to establish pre-qualified or pre-registered tender lists
Prequalified Tender	• Two-stage process • Multi-use list • List consist of all potential suppliers who meet predetermined criteria
Limited Tender	• Invited tendering and direct negotiation • Known service providers (those on pre-qualified list and list of

	other agencies) are invited to tender. • Evaluation based on quotes directly from one or more suppliers • Referred as 'sole source', and 'select' or 'restricted source' procurements • More appropriate for emergency situations or specialist work or when a limited number of service providers are known.

Source: Department of Finance- (http://www.finance.gov.au/procurement/procurement-policy-and-guidance/buying/procurement-practice/process-considerations/practice.html#ref4)

There is not much difference in tendering methods used across states and territories in Australia. However, the NSW government interprets 'selected tendering' as 'multi-stage tendering'. Multi-stage tendering includes prequalified/selected methods and the process involves expression of interest (EOI) first, and a selected number of tenderers from a prequalified list (or all prequalified tenderers), pre-registered list, or short list are then invited for tendering.

A summary of tendering methods identified from the scholarly publications is documented in the Table 2.7 below. Common tendering processes used in achieving the best value outcome for design and construction vary from sole-source selection to open tendering within two distinctive approaches; either competition or negotiation.

Table 2.7 Tendering method referred to in journal articles

Authors	**Tendering methods**		
	Competition		**Negotiation**
El-Reedy (2011, pp. 187-218)	Open	Limited (Registered)	Direct /Negotiation
Twort and Rees (2004, pp. 63-71)	Open	Selected	Negotiation
Nieto-Morote and Ruz-Vila (2012)	Open	Selective restricted/ Pre-qualification	Negotiation
Ruparathne and Hewage (2015)	Open	Nominated-list of bidders/ Qualified or selected from the respondents/ Two stage	Negotiation (single bidder)/ Restricted competitive negotiations
Enshassi et al. (2013)	Open		Direct hiring/restricted

			bid/negotiation

One stage vs two stage selection process

Apparently, there are various stages during which contractor potential is checked and verified in project planning and executing phases, to ensure the best value for money is achieved. Best value procurement can be performed with a one-stage or two-stage selection process (Migliaccio et al., 2010). Construction Management Association of America (CMAA) (2012) describes the one-stage and two-stage processes as:

- One stage process: includes single round of submittal that determines the selection, and
- Two stage process: includes a qualification submittal (pre-qualification) as a first step and price proposal as a second step.

In a one-stage contractor selection process, the eligible bidders require submitting both a technical and a price proposal in a single round of submission that determines the contractor selection (Palaneeswaran & Kumaraswamy, 2000). One stage selection process determines the best value as a combination of price and quality considerations in a single round and appropriate for simple projects where the proposal evaluation is not expensive (Migliaccio et al., 2010).

Two stages that greatly impact contractor selection occur in pre-tendering and post-tendering stages (Plebankiewicz, 2012). Hatush and Skitmore (1997b) explained that contractor's capabilities should be assessed in two stages. These two steps are technically referred to as prequalification/ short listing and bid evaluation respectively (El Wardani et al., 2006; Hatush & Skitmore, 1997b; Migliaccio et al., 2010; Palaneeswaran & Kumaraswamy, 2000). Pre-qualification is a decision-making process involving a wide range of decision criteria and multiple decision-makers that occurs in pre-tender stage, while bid evaluation occurs post-tender stage where it considers both bid amount and the contractor capabilities (Hatush & Skitmore, 1997a).

Prequalification is the screening process of potential contractors according to predetermined criteria that assess the contractor's competence or ability to be invited into the tendering process (Puri & Tiwari, 2014). It aims at the elimination of

incompetent contractors form the bidding process (Enshassi et al., 2013), and provides the client a list of qualified contractors from amongst those declaring willingness to participate in the tendering (Puri & Tiwari, 2014). Hatush and Skitmore (1997a) highlighted the importance of having a prequalification step in both negotiated and competitively bid contracts. A prequalification process ensures that clients obtain a number of competitive and reasonable bids that are easy to evaluate, because they are submitted by equally suitable and experienced contractors. Most prequalification methods use criteria and a weighted scoring system to prequalify bids (Enshassi et al., 2013). Hatush and Skitmore (1997a) indicated that criteria selection depends on the project delivery method or contract type. Prequalification and bid evaluation processes involve development of necessary and sufficient distinct criteria to evaluate contractor performance (Hatush & Skitmore, 1997a; Puri & Tiwari, 2014).

The decision to select a contractor using a two-stage selection process is subject to various factors. As projects become more complex and proposal evaluation becomes more expensive, it is suggested that owners short list interested bidders first on qualification before either inviting them for bidding or evaluating their proposals (Migliaccio et al., 2010). Moreover, project complexity has been confirmed to significantly impact on project duration, cost, and quality (Xia & Chan, 2012). The two-stage contractor selection process is recommended for complex projects and it includes a qualification submittal as a first step and price proposal as a second step (Palaneeswaran & Kumaraswamy, 2000).

With the above background, contractor selection processes that can be tailored to individual owner requirements are (Migliaccio et al., 2010)

- Low-bid
- One stage best value
- Two stage best value, and
- Negotiated selection.

2.3.3 Tender Evaluation

Tender evaluation assesses the submitted tenders and assists in selection of the successful tenderer. Tender evaluation is a decision-making process that considers

necessary and sufficient decision criteria for assessment of contractor capabilities. Puri and Tiwari (2014) stated that there is no consensus on a common set of selection criteria and the criteria varies according to the characteristics of the project, especially the client's objectives.

Evaluation criteria (also known as selection criteria) are the measures used by the decision makers for selecting the most appropriate response to an approach to market (commonly called invitation to offer, request for proposal, or request for tender). When a client or client representative approaches the supply market calling for offers, it is necessary to clarify the evaluation method and predetermine appropriate contractor selection criteria prior to inviting tenders (Hatush & Skitmore, 1997b).

Generally there are three selection criteria categories (Municipal Association of Victoria, 2013): (1) confirmation requirements (also called submittal requirements) where a tender response is checked that the information requested is in the tender documentation (e.g. schedules, statements of conformance, etc.) (2) mandatory requirements where submission of insurance policies, compliance with occupational health and safety standards, provision of financial information are confirmed, and (3) scored selection criteria (which are scored and weighted) against which contractor performance is measured (scored) to obtain optimum value for money. One and two criteria categories are generic to any project, however scored criteria are more project specific.

Hatush and Skitmore (1997a) identified the criteria (scored) for bid evaluation of prequalified contractors as bid price, quality assurance, existing workload, experience (on projects of a similar nature), experience of working with the owner, financial stability, local knowledge, and responsible attitude towards the work. Contractor selection criteria should be capable of identifying optimum choice and be suitable for the multi criteria/multi alternative nature of contractor selection.

The Australian government emphasises the importance of adopting weighted criteria to determine the tender that offers best value and a system of rating to facilitate consistency of scoring against criteria. These criteria contribute in different degrees to the project success factors such as cost, time, and quality (Walraven & de Vries, 2009) in different project environments. Singh and Tiong (2005) indicated that the criteria may have not been changed largely over the years, though their priorities can change in different project environments.

Researchers have extensively investigated and continued to identify appropriate contractor selection criteria with the rapid changes in project procurement laws, increased complexity of projects and client needs. EPC contractor selection criteria have not been specifically investigated and the existing literature on EPC contractor selection criteria is very limited. An industry research study based on interview survey conducted in North America, Europe and Asia reveals 18 criteria specific to EPC delivery method and the importance of these criteria (Transmar Consult Tnc., 2006). Criteria for EPC contractor selection need to be further investigated.

2.4 CONTRACTOR SELECTION MODELS

Contractor selection is a challenging task for most project owners. The construction industry uses various contractor selection models in different project environments using various contractor selection criteria.

To name a few of the existing models, there are Bespoke approaches (BA), Multi-attribute analysis (MAA), Multi-attribute utility theory (MAUT), Cluster analysis (CA), Multiple regression (MR), Fuzzy set theory (FST), Multivariate discriminant analysis (MDA), Analytical hierarchy process (AHP), Elimination and choice expressing reality III (ELECTRE III), Technique for order preference by similarity to ideal solution (TOPSIS), and VIKOR (means multi-criteria optimisation and compromise solution) (Cheng & Li, 2004; Darvish et al., 2009; Fong & Choi, 2000; Holt, 1998; San Cristóbal, 2012):

Among those, the models utilising multi-criteria theories for combining bid price and non-price criteria are considered more objective. For example, Holt et al. (1994a) combined the price criterion with non-price criteria in a proportion of 60:40 to be made on a single criterion to evaluate tenders. The confirming tender with the highest best value is recognised as the tender that best meets the value for money. When the difference between the first and second ranked scores is less than 3%, the lowest price tender of the two is taken as the preferred tender in TMR tender evaluations.

However, multi-criteria methods may encounter some difficulties when comparing different criteria measured in different scales. Evaluation of these criteria is ambiguous, subjective and not an easy task to determine one common scale to all

criteria (Plebankiewicz, 2012). Therefore, a comprehensive literature review was focused on two main approaches in existing contractor selection models, namely

- Multi-criteria approach, and
- Fuzzy approach.

Section 2.5 gives a broad overview of the full range of existing contractor selection techniques that use multi-criteria theories, while section 2.6 discusses the fuzzy approach for contractor selection, which is relatively new to the construction industry.

2.5 MULTI-CRITERIA APPROACH FOR CONTRACTOR SELECTION

2.5.1 Multi-Criteria Analysis (MCA) or Multi-Attribute Analysis (MAA)

Contractor selection is a complex multi-criteria decision-making (MCDM) problem in which decision makers evaluate the contractor's attributes to deliver the project against a large number of the decision criteria (Plebankiewicz, 2012; Singh & Tiong, 2005). Among the well-known multi-criteria methods, MCDM aims at using a set of criteria for a decision problem (Cheng & Li, 2004). Increased project complexity and higher requirements have recently demanded the use of multi-criteria decision-making (MCDM) methods for contractor selection (San Cristóbal, 2012). Various forms of MCDM technique can be used to perform a comprehensive evaluation and are broadly classified in terms of Multi-Objective Analysis (MOA) and Multi-Attribute Analysis (MAA) (Cheng & Li, 2004; Holt et al., 1994a). Holt et al. (1994a) stated that MOA is a problem-solving technique where objectives are not predetermined, therefore considerably high quality data accruing and a perfect solution are practically impossible. MAA based on predetermined objectives has superior use.

MAA is sometimes referred to as Multi-Criteria-Analysis (MCA) because the words 'criterion' and 'attribute' are often used synonymously in the literature. Attribute is referred to as a measurable criterion. The definitions of criteria and attributes given by Holt et al. (1994a) are used in this research.

- Criteria: measures of effectiveness - the fundamental elements of any MCDM, they may surface as either (client) objectives or (contractor) attributes

- Attributes: performance parameters providing the means of evaluating a decision option in respect of an objective

All MCA approaches make the options and their contribution to the different criteria and require the exercise of judgement (Dodgson, Spackman, Pearman, & Phillips, 2009). However, they differ in how they combine the data. MCA techniques usually provide an explicit relative weighting system for different criteria. The main role of the techniques is to deal with the difficulties that human decision-makers have in handling large amounts of complex information in a consistent way.

MCA techniques can be used to identify a single most preferred option, to rank options, to short-list a limited number of options for subsequent detailed appraisal, or simply to distinguish acceptable from unacceptable possibilities. MCA is a way of looking at complex problems that are characterised by any mixture of monetary and non-monetary objectives, of breaking the problem into more manageable pieces. In this way, MCA provides different avenues for disaggregating a complex problem, of measuring the extent to which options achieve objectives, of weighting the objectives, and reassembling the pieces. This systemic procedure greatly improves the evaluation process and consequently the potential success of the project (Enshassi et al., 2013).

A Multi-Criteria Decision Making (MCDM) process typically follows the sequence below:

1. Identifying objectives
    - Objectives should be specific, measurable, agreed, realistic and time-dependent.
2. Identifying options for achieving the objectives
    - Options (e.g. potential contractors)
3. Identifying the criteria to be used to compare the options
    - Criteria should reflect performance in meeting the objectives.
    - Each criterion must be measurable, be possible to assess how well a particular option is expected to perform in relation to the criterion.
4. Analysis of the options

- Preference scoring and weighting stages
- Preferences can be measured and averaged.

5. Making choices
    - Final stage of the decision-making process is the actual choice of option

Identifying the appropriate criteria and sub-criteria (as in step 3) by which the contractors' performance is measured and assessing performance levels (with scoring) are important activities within MCA. MCA aims at either finding the single most appropriate option or short-listing a set of options for subsequent, more detailed investigation. However, human judgements may not always work well, especially when dealing with complex problems. Humans are biased in their assessments of alternatives (contractors) that are more linked to familiar, recent, memorable, or successful experience. MCA techniques are designed to help overcome these limitations by imposing a disciplined structure of criteria and their importance weights.

Developing a performance matrix, where a row describes an option and each column describes the performance of the options against each criterion, is an important step. MCA techniques commonly apply numerical analysis to a performance matrix in two stages (1) scoring, and (2) weighting. Firstly, each option is assigned a numerical score on strength of preference scale for each criterion. The use of an interval scale measurement permits a full MCA application. Secondly, numerical weights are assigned for each criterion with the relative valuations of each criterion. The relative importance or weight of a criterion indicates the priority assigned to the criterion by the decision-maker techniques, since low scores on one criterion may be compensated by high scores on another.

2.5.2 Multi-Criteria Models

Researchers and practitioners have developed comprehensive multi-criteria decision-making models to employ in design and construction projects. These multi criteria tools in literature are presented in Table 2.8.

Table 2.8 Multi-criteria contractor selection models

Evaluation Tool	Features/or principal characteristics	Author
Multi-Criteria Decision Making (MCDM)	Estimates values of contractors	Cheng and Li (2004)
Multi Attribute Analysis (MAA)	Academic and industrial use but can be subjective Estimates values of contractors Assigns weights to selection criteria Uses simple scoring for rating criteria	(Cheng and Li (2004); Holt (1998))
Multi Attribute Utility Theory (MAUT)	Academic use, scope for derivation of utility functions Estimates values of contractors Develops a relationship between utility and cost incurred as a sequence of a decision Assigns utility values by evaluating multiple criteria and combines to obtain overall utility value	(Cheng and Li (2004); Hatush and Skitmore (1998); Holt (1998); San Cristóbal (2012))
Multivariate Discriminant Analysis (MDA)	Academic use, broader scope for research and industrial application Estimates values of contractors Project-specific criteria are derived to discriminate contractor performance into good and poor groups	(Cheng and Li (2004); Holt (1998); Wong, Nicholas, and Holt (2003))
Multiple Regression (MR)	Evidence of academic use, scope for further research and industrial application Estimates values of contractors Statistical technique; predicts the effect of several independent variables upon a dependent variable	(Cheng and Li (2004); Holt (1998))
VIKOR method	Based on a measure of closeness to the positive ideal solution Suitable for situations in which the decision maker wants maximum profit and the risk of less important decision	San Cristóbal (2012)

Evaluation Tool	Features/or principal characteristics	Author
Technique for order preference by similarity to ideal solution (TOPSIS)	Based on the principle that the optimal point should have the shortest distance from the positive ideal solution and the farthest from the negative -ideal solution Suitable for cautious decision makers who like to have a decision that makes profit and avoids risk	San Cristóbal (2012)
AHP	Prioritises selection criteria based on decision maker judgement - subjective in nature Assigns weights to selection criteria Inherent uncertainty and imprecision Able to structure a complex, multi-person, multi-criteria problem hierarchically Derives dominance priorities from paired comparisons Systematically and logically provides a structured solution	(Cheng and Li (2004); Deng (1999); Enshassi et al. (2013); Mahdi et al. (2002); Mousavi, Tavakkoli-Moghaddam, Heydar, and Ebrahimnejad (2013); San Cristóbal (2012))
Weighted average method (WAM)	Assigns weights to each criterion Obtains the summation of weight (relative weights)	Abdelrahman et al. (2008)
Dimensional Weighting Method	Contractors are ranked against selection criteria Contractor's total score is calculated by multiplying ranks by criteria weights	(Enshassi et al., 2013)

Multi-criteria weighting models use the value of relative weights to assess criteria (Abdelrahman et al., 2008). Weighted score is calculated from the individual scores that the tender assessment panel allocate to the tenders thus weighted score for each criterion is calculated by multiplying by criteria weighting. Combined score is used as the basis of ranking contractors.

2.6 FUZZY APPROACH FOR CONTRACTOR SELECTION

A decision problem becomes complex and difficult with the existence of multiple criteria, multiple decision makers, uncertainty and risk associated with incomplete information, imprecise data and vagueness in decision making (Singh & Tiong,

2005). Typically, contractor performance is evaluated using crisp values. Verbal evaluation could be used instead of numeric values when it is relatively difficult for decision makers to provide precise numerical values for the criteria or attributes.

Fuzzy Set Theory (FST) allows decision makers to use linguistic terms rather than crisp values in assessments for contractor performance on criteria (Holt, 1998; Singh & Tiong, 2005). Fuzzy sets attempt to capture the idea that our natural language in discussing issues is not precise. As such, assessment of different attributes in contractor selection can be obtained by subjective judgement such as good or fair. It suggests a need to implement a method that incorporates subjective judgement to capture both the qualitative and the quantitative aspects of these imprecise terms. Fuzzy Set Theory that can capture this subjectivity is an appropriate method for addressing this need (Alhumaidi, 2015). Since verbal evaluations are explained by approximate values, it is useful to implement the fuzzy set theory, especially triangular and trapezoidal membership functions, to reduce the ambiguity of such evaluations (Mazaheri-Zadeh & Naji-Azimi, 2015).

2.6.1 Fuzzy Set Theory (FST)

A fuzzy set is defined as "a class of objects with a continuum of grades of membership" by Zadeh (1965). Main concepts of FST include membership function, linguistic variables, natural language computation, linguistic approximation, fuzzy integrals and fuzzy weighted sum as described by Singh and Tiong (2005). A fuzzy set is a set whose elements have varying degrees of membership and is defined mathematically by assigning a value representing its grade of membership in the fuzzy set (Singh & Tiong, 2005; Xia, Chan, & Yeung, 2011). This grade represents the degree to which that individual is similar or compatible with the concept represented in the fuzzy set. It emphasises the degrees of belonging and transfers the subjective consciousness of human beings to objective quantities and provides clear results. These membership grades are very often represented by real number values ranging between 0 and 1. Fuzzy arithmetic then captures these qualified assessments using the idea of membership function, through which an option would belong to the set of options with a given degree of membership, lying between 0 and 1.

Membership function of an element represents a degree to which the element belongs to a set (Singh & Tiong, 2005). Two types of membership functions that are

commonly used are trapezoidal membership function and triangular membership function, and they are represented graphically in figure 2.4 and 2.5 respectively.

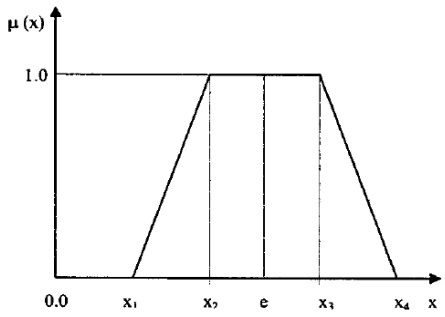


Figure 2.4 Trapezoidal membership function

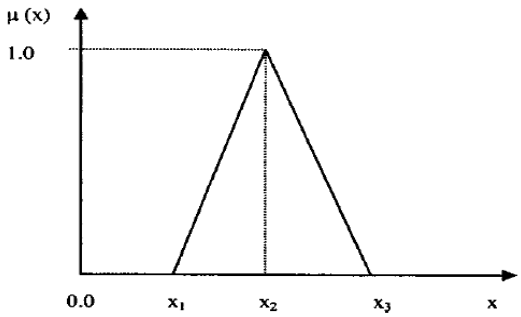


Figure 2.5 Triangular membership function

Furthermore, FST is useful for analysing the human evaluation process and specifying the preference structures, as human beings find it difficult to make a precise decision when facing complex decision-making situations. Subjective judgements by humans are usually fuzzy and imprecise in nature. FST has been used to address problems dealing with incomplete and imprecise data as it uses linguistic terms to model vagueness and subjectivity. The fuzzy set therefore introduces vagueness with the aim of reducing this complexity. Fuzzy sets can capture the idea of natural language in discussing issues that are not precise as it uses linguistic variables to model vagueness intrinsic to the human cognitive process.

A linguistic variable differs from a numerical variable so that its values are not numbers but words, such as low, ‘fair’, and ‘good’, which are subjective in nature. As subjective judgement is the preferred solution for non-quantifiable variables, linguistic variables are used to capture the subjectivity inherent in non-quantifiable variable.

Rating of different attributes, their weights, and the relative weights of different decision makers are non-quantifiable, incomplete, and subjective, making precise judgement impossible. Ranking alternatives in a multiple-attribute decision is subjective in nature and involves linguistic terms. Linguistic variables whose values are defined using linguistic terms such as low, fair, good can be defined quantitatively, or fuzzified, using fuzzy numbers. Fuzzification is the process of converting crisp and deterministic values into fuzzy and uncertain ones (Ross, 2010). Fuzzy number represents the meaning of each generic verbal term. The values of linguistic variables were then transformed into fuzzy numbers. Each fuzzy set overlaps its neighbouring sets to a certain extent. Since the verbal evaluations are explained by approximate values, it is useful to implement either triangular or trapezoidal membership functions to reduce ambiguity of evaluation. Then, fuzzy arithmetic captures these qualitative assessments using the membership function belonging to the set lying between 0 and 1.

FST has been widely used to address subjectivity, uncertainty and impreciseness in multi-criteria decision making (MCDM) problems, not only in contractor selection but also in prequalification, project delivery method selection, technology selection, and risk assessment (Chang & Chen, 1994; Guo, Wu, & Wang, 2009; Nasab & Ghamsarian, 2015; Plebankiewicz, 2009).

2.6.2 Linguistic terms and fuzzy membership functions for decision makers

Various linguistic terms and their corresponding fuzzy membership functions have been used for decisions makers' experience rating, attribute weight assessment and attribute rating in contractor selection, and these are given in Tables 2.9- 2.12 below.

Table 2.9 Linguistic scales and fuzzy rating using alpha (α) cuts

Linguistic terms of decision makers' experience level	Linguistic terms for attribute weight assessment	Linguistic terms for attribute rating	Fuzzy rating (using alpha (α) cuts)
Unexperienced	Unimportant	Poor	(0, 0.333)
Fairly experienced	Fairly important	Fairly poor	(0.167, 0.5)
Fair	Fair	Fair	(0.333, 0.667)
Fairly experienced	Fairly important	Fairly good	(0.5, 0.833)
Experienced	Important	Good	(0.667, 1)

Source:(Alhumaidi, 2015)

Linguistic terms and their corresponding fuzzy membership functions given in Table 2.9 are represented graphically using triangular fuzzy numbers in Fig. 2.6.

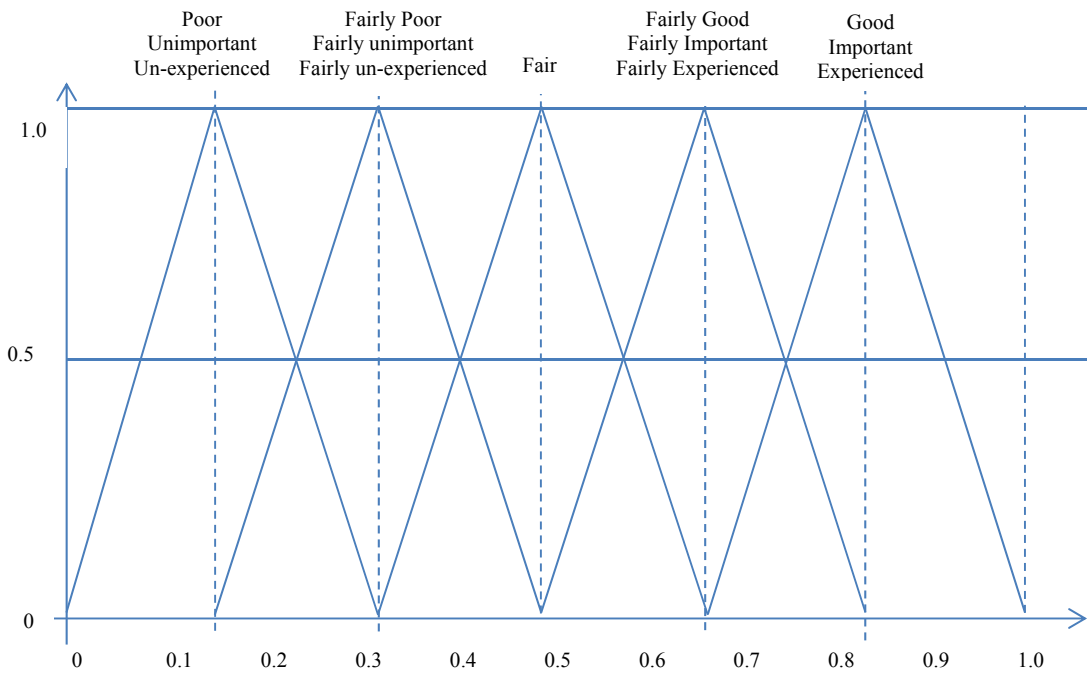


Figure 2.6 Graphical presentation of fuzzy numbers of triangular membership function

Fuzzy numbers for various linguistic variables for trapezoidal membership function are shown in Table 2.10:

Table 2.10 Linguistic variables and fuzzy ratings using trapezoidal fuzzy numbers

Linguistic variables	Trapezoidal Fuzzy number
VG/VI (Very good/important)	(0.8, 0.9, 1.0, 1.0)
G/I (good/important)	(0.6, 0.7, 0.8, 0.9)
AA (above average)	(0.5, 0.6, 0.7, 0.8)
A (average)	(0.4, 0.5, 0.5, 0.6)
BA (below average)	(0.2, 0.3, 0.4, 0.5)
P/LI (poor/low important)	(0.1, 0.2, 0.3, 0.4)
VP/VLI (very poor/very low important)	(0.0, 0.0,0.1,0.2)

Source: (Singh & Tiong, 2005)

Graphical presentation of fuzzy numbers of trapezoidal membership function is given in Fig. 2.7.

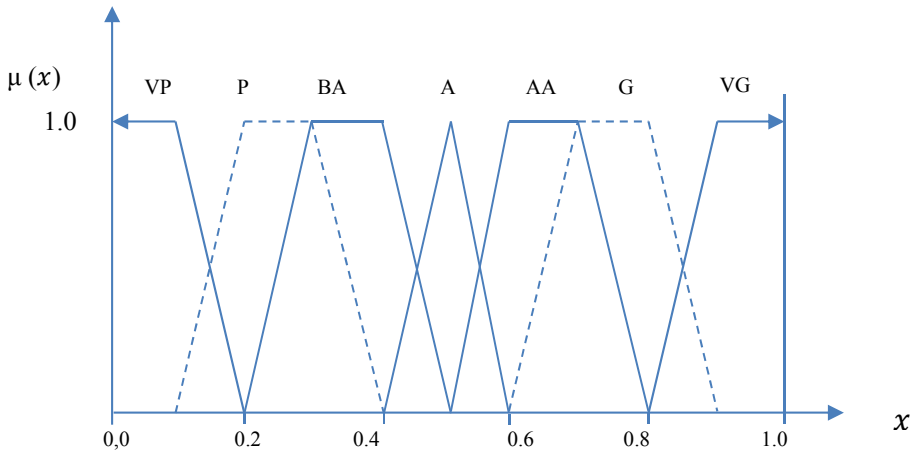


Figure 2.7 Graphical presentation of fuzzy numbers of trapezoidal membership function

Source: (Singh & Tiong, 2006)

Table 2.11 Triangular fuzzy numbers used in Fuzzy VIKOR method for contractor selection

Preference	**Number**	**Triangular Fuzzy number**
Equal	1	(1, 1, 1)
Approximately equal	2	(1.2, 1, 3.2)
Slightly more important	3	(1, 3.2, 2)
More important	4	(3.2, 2, 5.2)
Much more important	5	(2, 5.2, 3)
Quite more important	6	(5.2, 3, 7.2)

Source: (Vahdani, Mousavi, Hashemi, Mousakhani, & Tavakkoli-Moghaddam, 2013)

Table 2.12 Fuzzy numbers used in Fuzzy AHP

Preference	**Number**	**Triangular Fuzzy number**
Equal	1	(1, 1, 1)
Approximately equal	2	(1.2, 1, 3.2)
Slightly more important	3	(1, 3.2, 2)
More important	4	(3.2, 2, 5.2)
Much more important	5	(2, 5.2, 3)
Quite more important	6	(5.2, 3, 7.2)

Source: (Mazaheri-Zadeh & Naji-Azimi, 2015)

2.6.3 Fuzzy MCA Models

The multi criteria decision-making process becomes complex and challenging with the presence of uncertain, imprecise and subjective data. As mentioned earlier, this uncertainty, imprecision and subjectiveness can be addressed by fuzzy set theory as this concerns the subjectiveness and imprecision of human behaviour (Deng, 1999). To evaluate the degree of importance of each criterion in turn, in reaching the given objective, the linguistic variables such as 'very important', 'important', 'average', etc. can be used. Meanwhile the linguistic variables of very good, good, average, etc. can be used to evaluate the degree of the contractor satisfying a given criterion (Plebankiewicz, 2012).

Fuzzy MCA models develop procedures for aggregating fuzzy performance levels using weights that are sometimes also represented as fuzzy quantities. Alhumaidi (2015) mentioned several fuzzy models that other researchers have developed over the years and these include fuzzy number recognition method, weighted-centre method, Fuzzy Technique for Order Performance by Similarity to Ideal Solution (TOPSIS), simple defuzzification method, and fuzzy number recognition method (the most complex method of all).

A few of the fuzzy models that are used in multi-criteria decision-making problems are given in Table 2.13.

Table 2.13 Existing fuzzy models for contractor selection

Model	Description	Researcher

Fuzzy AHP + Fuzzy PROMETHEE	Fuzzy AHP - Fuzzy AHP to determine the weight of each criterion (triangular fuzzy numbers/pairwise comparisons/largeness degree and weight of criteria/ final weight vector by normalisation) Fuzzy PROMETHEE - Fuzzy PROMETHEE to transform the verbal into numeric information (where PROMETHEE is used for prioritising a large number of alternatives)	Mazaheri-Zadeh and Naji-Azimi (2015)
Fuzzy TOPSIS	- Multi criteria contractor selection based on single decision maker	Nieto-Morote and Ruz-Vila (2012)
	- Final ranking is based on fuzzy Euclidean distance	Vahdani et al. (2013)
	- Linguistic variables for the importance weight of criteria and linguistic rating for supplier - Closeness coefficient determines the ranking order of suppliers	C.-T. Chen, Lin, and Huang (2006)
Fuzzy Decision Model	- Use fuzzy Delphi method to achieve group consensus on criteria importance - Average of trapezoidal fuzzy number across all the decision makers are considered - Relative importance of each criterion is determined by 'Shapley value' - Simple additive method to find the total score - Highlighted challenges include complexity and difficulty of establishing preference scale structure when there are multiple stakeholders	Singh and Tiong (2005)
Multi-Attribute Fuzzy Weighted Average Ranking Method	- Use triangular fuzzy numbers to describe subjective judgement on decision makers' experience, attribute weight assessment, attribute rating - Weighted average method for grouping of decision makers	Alhumaidi (2015)

Fuzzy Regression Model	- Fuzzy Random variable based multi-attribute decision making - Fuzzy random regression to aid in determining the weight importance of selection attributes - Confidence-Interval-Based Fuzzy Random Regression model	Nureize and Watada (2011)
Fuzzy AHP	- Uses total integral value as method for ranking fuzzy numbers as extent analysis method cannot estimate the true weights from a fuzzy comparison matrix	Alias, Noor, Saman, Abdullah, and Selamat (2011)
Fuzzy VIKOR	- Triangular fuzzy numbers are used to represent rating values of complicating criteria as well as criteria weights and collective index being proposed to rank the alternative	Vahdani et al. (2013)

2.7 OTHER CONTRACTOR SELECTION MODELS

There are various other contractor selection models that are used for contractor selection as described in Table 2.14. However, this research focus is limited to both multi-criteria and fuzzy models.

Table 2.14 Other contractor selection models

Evaluation Tool	Features/or principal characteristics	Author
Cluster Analysis (CA)	Limited academic use, scope for further research and application to prequalification Aims at reducing the original set (of contractors) into a series of smaller, manageable sub-sets to observe contractor quality by which the best subset(s) is identified for subsequent tender invitation if prequalified	(Cheng and Li (2004); Holt, 1998)
Bespoke Approaches (BA)	Prolific industrial use but very subjective Contained in two decision stages namely prequalification criteria and project-specific criteria	(Cheng and Li (2004); Holt (1998))

Elimination and choice expressing reality III (ELECTRE III)	Based on the principle that the decision maker is not perfectly rational, therefore express for each pair of actions (no preference)	San Cristóbal (2012)
Decision support systems	Computerised decision systems Helps decision makers to handle a huge amount of information	San Cristóbal (2012)
Evidential Reasoning (ER)	Uses the concept of degree of belief in situations with uncertain/or incomplete information to make a rational decision based on real preferences of the decision maker Integrates both quantitative and qualitative hierarchal methods to solve the contractor selection problem.	San Cristóbal (2012) (Enshassi et al., 2013)
Goal programming	Works well when there are multiple and conflicting objectives	San Cristóbal (2012)
Program Evaluation and Review Technique (PERT)	Evaluates contractor data against client goals (time, cost and quality)	(Hatush & Skitmore, 1997a)

2.8 SUMMARY

This chapter has reviewed the literature on the construction project procurement process, delivery methods, EPC delivery method and contractor selection process. One of the most important decisions that occurs at the early stage of an EPC project is selecting the most suitable contractor to which the client can entrust the responsibility of all aspects of the project without any hesitation. Given the importance, a comprehensive literature review on contractor selection has been conducted. Sub-topics include procurement strategy (low bid/best-value/qualifications based and tendering options), stages of contractor selection, and tender evaluation where existing contractor selection models were broadly discussed. Then, the literature review was focused on contractor selection models which either use a 'multi-criteria approach' and/or 'fuzzy approach'. Application of multi-attribute analysis and fuzzy set theory were also investigated in depth.

Chapter 3: **Research Design**

Research is a 'voyage of discovery' and it consists of systematic investigation in order to establish facts and reach new conclusions (Fellows & Liu, 2015). What is discovered depends on research questions, techniques used for searching, quality of information collected, analyses carried out and more importantly, reflection by the researcher on the results of the analyses in the context of the theory and literature, and methods employed (Fellows & Liu, 2015). The majority of the research is a combination of theories and applications, contributes to the body of knowledge and follows a scientific method.

This chapter describes the research design to achieve the aims and objectives as stated in Chapter 1. Section 3.1 discusses the methodology used in the study; section 3.2 outlines the research approach for this investigation, the stages by which the methodology was implemented; section 3.3 discusses limitations; finally, section 3.4 provides chapter a summary.

3.1 METHODOLOGY

Basic research methods include experimental approach, surveys, secondary data analysis, case research, focus group research, action research, etc. Research can be exploratory, explanatory or descriptive. Every research project has an exploration phase to identify the research questions and conduct a comprehensive literature review. The purpose of the literature review is reviewing existing knowledge in the area, identifying gaps in knowledge, and identifying theories that help addressing the research questions (Anol, 2012). After reviewing literature, research gaps were identified, research questions were framed to address the research gaps, and objectives were set. The next step is to identify the research methodology addressing the research questions and meeting the set objectives. Fig 3.1 outlines the overall research process with the important steps that need to be completed to deliver the EPC contractor selection model.

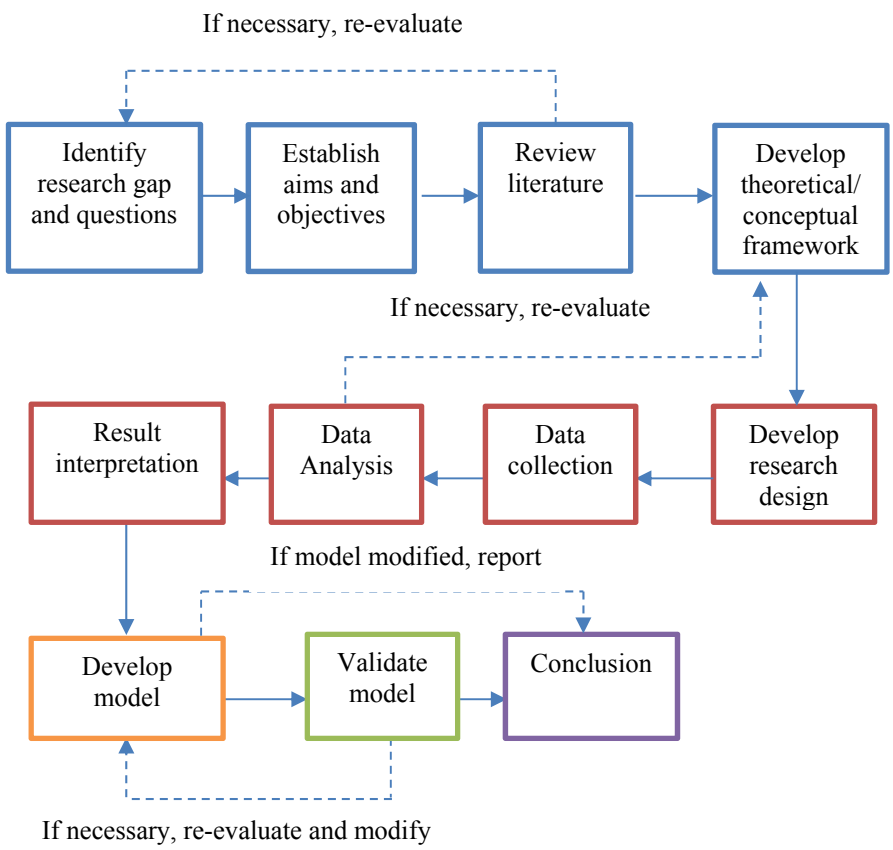


Figure 3.1 Research Process

Three main types of research questions in a research study can be (1) difference (to compare values of independent variable with scores on the dependent variable) (2) associational (to associate or relate the independent and dependent variable) and (3) descriptive (to describe or summarise data) (Morgan & Griego, 1998). The questions to be answered in this research are associational and descriptive in nature. This research uses a combination of both associational and descriptive approaches as each method has unique strengths that are significant in answering research questions.

As data is an essential part of research, it requires identifying what data are required, data sources and mechanisms for data collection during the planning stage and it highly depends on the research questions. Researchers collect quantitative and qualitative data using combinations of techniques such as questionnaires, interviews, observations, documents or secondary data collection as much and diverse as possible to provide the best possible insight to the subject of interest. Combinations of qualitative and quantitative methods are very powerful in gaining insights and results as they reduce or eliminate disadvantages of each individual approach while

gaining the advantage of each (Fellows & Liu, 2015). Qualitative analysis can be based on quantitative estimates and makes quantitative evaluation, both in-depth and specific (Hammond & Wellington, 2012).

In this research, several research methods including literature review, secondary data analysis and survey methods (Delphi questionnaire survey and interviews) were adopted for data collection in response to the research questions. In addition, theoretical applications such as multi-attribute analysis (MAA) and fuzzy set theory (FST) have been adopted to develop a new EPC contractor selection framework and tender evaluation model. Identified research methods and theoretical applications with their alignment to research objectives are given in Table 3.1.

Table 3.1 Research objectives and methods

	Research Objective	Research Method
1	Understanding the EPC project delivery method and the EPC market in Australia	Literature Review Secondary Data Analysis
2	Developing an EPC contractor selection framework	Literature Review
3	Identifying and prioritising the criteria for EPC contractor selection	Literature Review Delphi questionnaire Survey
4	Developing a new EPC contractor selection model using Multi-Attribute Analysis and Fuzzy Set Theory	Multi Attribute Analysis (MAA) Fuzzy Set Theory (FST)
5	Validating the EPC contractor selection model for industry use	Semi-structured Face-to-Face Interview

Literature Review

A literature review is a collection of background information of a research study, which aims to consolidate all previous studies related to the research topic and to form understanding of the current practice (Chow, 2005). A comprehensive literature review helps researchers to identify research problems and knowledge gaps as it presents critiques of existing works (Yeung, 2007).

The purpose of the literature review in this research is reviewing existing knowledge of EPC contractor selection and identifying gaps in knowledge, and identifying theories that help in addressing the research questions. After comprehensive

literature review, background information associated with the research topic was collected, the research gaps were identified, the research questions were formulated, the research methods to obtain data were selected, and ways of analysing data were identified. For this purpose, various data sources such as academic journals, conference proceedings, industrial and government publications, and books, were extensively mined.

Secondary data analysis

Secondary data analysis is defined as an analysis of data that has previously been collected and tabulated by other sources. Such data was obtained from government agencies (e.g. Australian Bureau of Statistics), data collected by other researchers, or publicly available third-party data. This secondary data analysis is used in addition to literature review to answer the research question 1 (What is the current EPC market circumstance in Australia?). A limitations of this method is that there is a possibility that the data was not collected in a systematic manner and may be intended for another purpose rather than that for which this research uses it.

Content analysis was used to analyse the secondary data, and to gain understanding and create empirical knowledge. Under content analysis, printed and electronic documents were reviewed and evaluated. Documents used in this method can be in any form; advertisements, background papers, letters, books, survey reports, organisational and institutional reports, newspaper articles, and press releases. The procedure used for retrieving the documents include: (1) key word searches in the search engines (2) scanning abstracts/prefaces of documents to assess whether the documents fit the study objectives. Then, the documents were critically reviewed, and the relevant information was compiled.

Delphi method

The Delphi method is considered as a research tool for investigating research questions, issue identification or prioritisation (Okoli & Pawlowski, 2004). Delphi is preferred over the subjective research methodologies such as traditional surveys or focus groups because of the exceptionally high quality of the participants, ability to minimise judgement-based bias, and ease of implementation (online Delphi surveys).

It was first developed by Rand Corporation in the 1940s and has been increasingly used for technology assessment, measuring and aiding forecasting, and decision making in a variety of disciplines (Grisham, 2009; Linstone & Turoff, 1975; Liu, Xie, Yuan, & Fang, 2012; Rowe & Wright, 1999). It can be used in judgement and forecasting situations where pure model-based statistical methods are not practical (Rowe & Wright, 1999). Moreover, Hallowell and Gambatese (2010) indicated that the Delphi method has strong potential for widespread application in Construction Engineering Management (CEM) research for obtaining highly reliable data from qualified experts.

To answer the research question 2 (What are the appropriate evaluation criteria for EPC contractor selection and how important are they in decision making process?), an online (web and email) Delphi questionnaire survey method was selected after considering the inherent advantages in this method over the other research methods.

As Delphi is a systematic analysis approach involving opinions and value judgements (Hammond & Wellington, 2012), its structured group communication process effectively allows a group of individuals, as a whole, to deal with a complex problem (Linstone & Turoff, 1975). It is capable of avoiding some problems associated with group dynamics or direct interactions (Crisp, Pelletier, Duffield, Adams, & Nagy, 1997). Key features of the Delphi method include (Hammond & Wellington, 2012; Liu et al., 2012; Rowe & Wright, 1999)

1. Anonymity
2. Iteration
3. Controlled feedback
4. Statistical aggregation of group response (consensus)

The Delphi procedure involves a panel of anonymous experts to whom intensive questionnaires and controlled feedback were given to obtain the most reliable consensus on the subject being investigated (Crisp et al., 1997; Rowe & Wright, 1999). Anonymity benefits in overcoming the psychological vulnerabilities inherent in meetings of experts. Controlled feedback in each round is typically given by means of statistical average (mean/median/upper or lower quartiles). These Delphi studies are varied by the requirement for expert qualification, and methods of data

collection, data analysis and feedback, and number of rounds of surveys, and measures of consensus (Hallowell & Gambatese, 2010).

Structure of the Delphi method is intended to allow access to the positive attributes of interacting groups (knowledge from a variety of sources, creative synthesis) while pre-empting their negative aspects, removing bias by engaging a diverse group of experts (Grisham, 2009; Rowe & Wright, 1999). Even though input from experts can be gathered from a traditional survey, use of the Delphi method, which has a stronger methodology, is advantageous for rigorous query (Okoli & Pawlowski, 2004). Hallowell and Gambatese (2010) suggested a Delphi structure (Fig. 3.2) that is applicable to many types of CEM research. It is, with minor changes, adopted in this research.

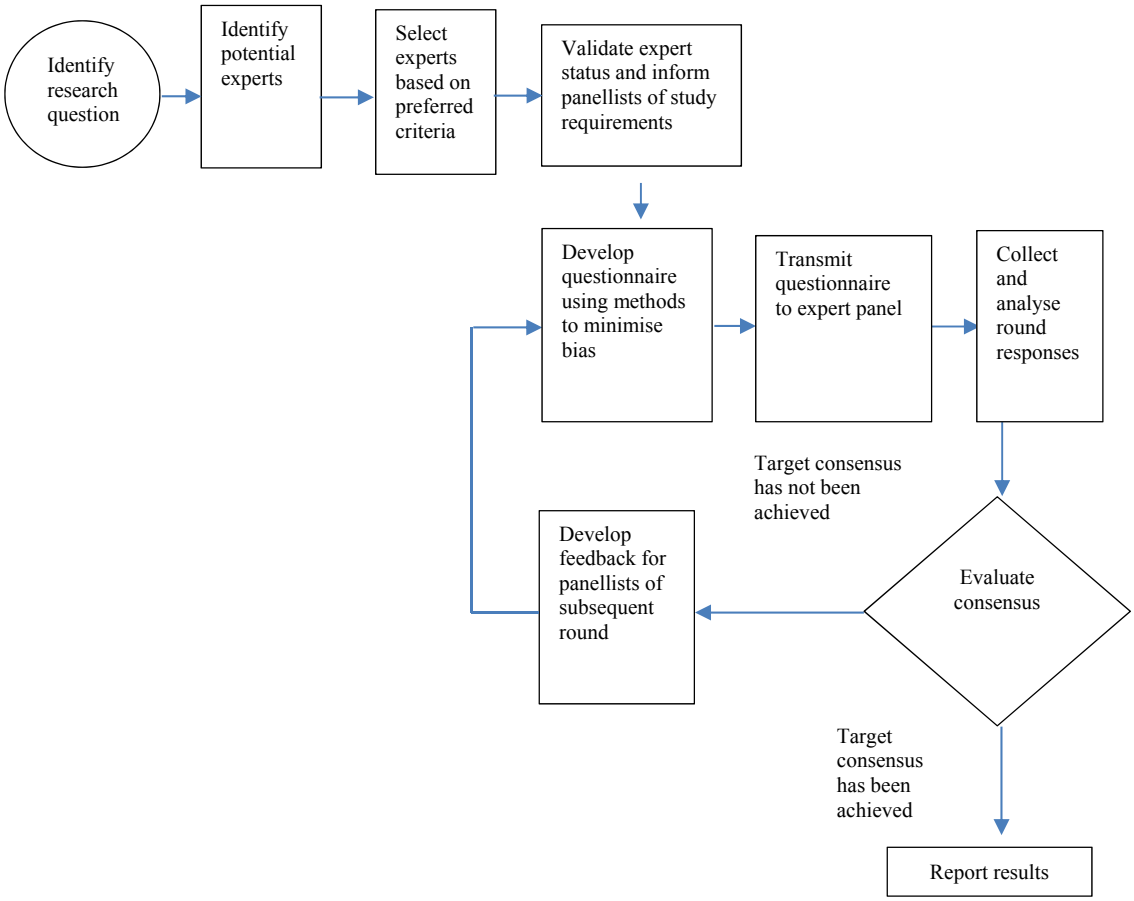


Figure 3.2 Delphi structure

Source: Hallowell and Gambatese (2010)

Delphi typically consists of two or more rounds, however the ability to administer the questionnaire survey through mails, and online (email or web-based), eases the implementation process and does not require physical meeting of participants as such

members can response from several locations. The number of experts vary from 3-80 for Delphi studies, which takes several months to complete (Hallowell & Gambatese, 2010). Employing the Delphi method to a larger sample is advantageous as it increases the reliability of the survey responses.

Interview

Interview is a very personalised form of data collection. Interviews can be face-to-face interviews, telephone interviews, and online methods like skype interviews and video conferencing. The most typical form is face-to-face interview, which is capable of collecting the most qualitative data and yields the best results. The interviewer has the opportunity to clarify issues raised by the respondent or to follow-up on questions instantly. Skype interview and video conferencing also have the same advantages, however poor reception or technical faults can obstruct communication. However, all interviews are time consuming and resource-intensive (Taylor-Powell & Hermann, 2000). A variation of the personal interview is a group interview, which is also called as focus group.

During the validation process, the interview research method was used to collect validation data. The type of interviews includes face-to-face interviews with the construction professionals who are highly experienced in EPC contractor selection process, to validate the model. The interviews were conducted in a semi-structured manner that allowed interviewees to extend their opinions, and the interviews were conducted either at the interviewees' office or at QUT.

Multi-Attribute Analysis (MAA)

Contractor selection is a multi-criteria decision-making process, which involves rating of alternatives against measurable criteria determined by the decision makers to achieve value for money. After identifying criteria for EPC contractor selection using the Delphi questionnaire survey, it needs to determine the scale for rating these criteria, determine the best value and rank the alternatives based on the best value.

Holt et al. (1994a) suggest that MAA is suitable for the multi criteria/multi alternative nature of the contractor selection problem. As discussed extensively in the literature review in Chapter 2, MAA is appropriate to meet objective 3 of this

research - to evaluate contractor performance using multi-criteria (objectively). However, MAA is also inherent with advantages and disadvantages. The advantages worth mentioning are (Holt et al., 1994a)

- Facilitates decision making of multiple conflicting criteria
- Considers multiple attributes in respect of multiple client objectives where preferences are incorporated quantitatively by assigning weights
- Reflects real-life decision problems encompassing client judgements
- Assesses options systematically to produce aggregated results where highest score indicates the optimum choice, and
- Procedure is reliable, and results are reproducible.

Disadvantages exist in MAA include (Holt et al., 1994a):

- Each selection problem has its own multiple objectives that need to be identified
- Multiple objectives are often complicit
- Results of attribute evaluation often yield incommensurable units. For example, units can be binary (yes/no) answers, numeric (number of years, project etc.), or descriptive (limited/ adequate/ excellent).

There are two forms of MAA, i.e. linear model and additive model. In a linear model, the attributes are quantified on a commensurable scale and an aggregated score for each contractor is determined. Aggregate score is the total of variable (attribute) score multiplied by importance or utility weight. Linearity assumes a constant rate of trade-off between conflicting attributes. In an additive model, an aggregated score for each contractor is determined by the total of variable (attribute) score, which is a function of attributes. Importance weights can be incorporated in an additive model.

An MAA additive model can address (1) selection criteria, (2) importance weights, and (3) attribute evaluation in matrix form (Holt et al., 1994a). Optimal choice is defined as the maximum possible value for each of its objective functions within the matrix. Objective functions are determined by assigning them with importance weights to simplify the decision process. The MAA process furnishes the ‘combined

scores' and these scores are achieved by considering the importance of weights, which are normally determined by the clients or their agents. Therefore, MAA is deemed suitable for the EPC contractor selection task.

MAA emphasises the judgement of the decision makers in establishing objectives and criteria, estimating relative importance weights and, in judging the contribution of each option to each performance criterion. MAA is inherent with many advantages such as consideration of multiple attributes in respect of multiple client objectives; however, subjectivity prevailing in MAA can be a matter of concern. In addition, multi-attribute evaluation is relatively difficult for decision makers to provide precise numerical values for criteria (or attributes). It is essential that these weaknesses are appropriately addressed in the proposed model.

Fuzzy Set Theory (FST)

Multi-attribute assessment is challenging for decision makers with the presence of uncertain, imprecise and subjective data. Rating alternatives in a multiple-attribute decision making is also subjective in nature and involves linguistic terms. Researchers suggest that statistical analysis cannot adequately handle the subjectiveness and imprecision of the human decision-making process. This fuzzy nature in the decision-making process is quite challenging. FST is primarily useful in modelling uncertainty (Zimmermann, 2001 pp.6-7). Zimmermann (2001) highlighted that FST is superior to numerous other methods and theories available to model uncertainty because they were not defined sufficiently or in specific circumstances. Fuzzy set theory can be used to model specific types of uncertainty under specific circumstances. The significance of fuzzy set theory and its potential to address the fuzziness in the decision-making process was extensively described in Chapter 2.

FST concerns the subjectiveness and imprecision of human behaviour (Deng, 1999). As such, it can capture the idea of natural language in discussing issues that are not precise (Deng, 1999). A linguistic variable differs from a numerical variable as its value lies not in numbers, but words or sentences in a natural or artificial language. Linguistic variables such as 'poor management', 'good performance' and 'moderate risk' describe the vague concepts. Linguistic variables such as 'very good', 'good', 'above average', 'average', 'below average', 'poor', 'very poor' can be used to evaluate the degree of the contractor satisfying a given criteria (Plebankiewicz, 2012)

and these can be defined quantitatively using fuzzy numbers. Two types of membership functions that are commonly used are trapezoidal membership function and triangular membership function. A triangular fuzzy number ($\tilde{A}$) is defined by three numbers a<b<c where the base of the triangle is the interval [a, c] and its vertex (membership value 1) is at x=b, then, $\tilde{A}$= (a, b, c).

Aggregation of fuzzy sets in a decision-making context can be done using the aggregating techniques used in utility theory or multi-criteria decision theory. These procedures allow trade-off between conflicting goals, when compensation is allowed. Weighted and unweighted arithmetic or geometric means are examples of nonparametric examples of averaging operators.

Pilot study

It is important to conduct a pilot study before actual data collection is started. Often qualitative data forms the basis of a pilot study. Results of the pilot study are then used to produce a relatively more quantified approach (e.g. from an open-ended interview to a particularly structured questionnaire).

3.2 RESEARCH DESIGN FOR THIS INVESTIGATION

As it has already been mentioned, several research methods were selected to meet the objectives of this research. As this research aims to develop a new EPC contractor selection model, overall research was completed within four phases as in Fig. 3.3.

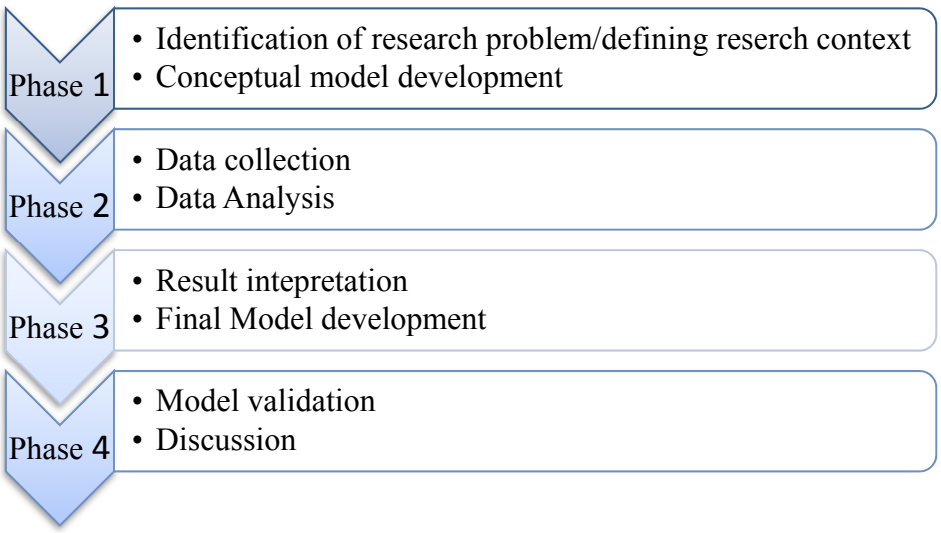


Figure 3.3 Main research phases

Fig. 3.4 describes overall research process of this investigation.

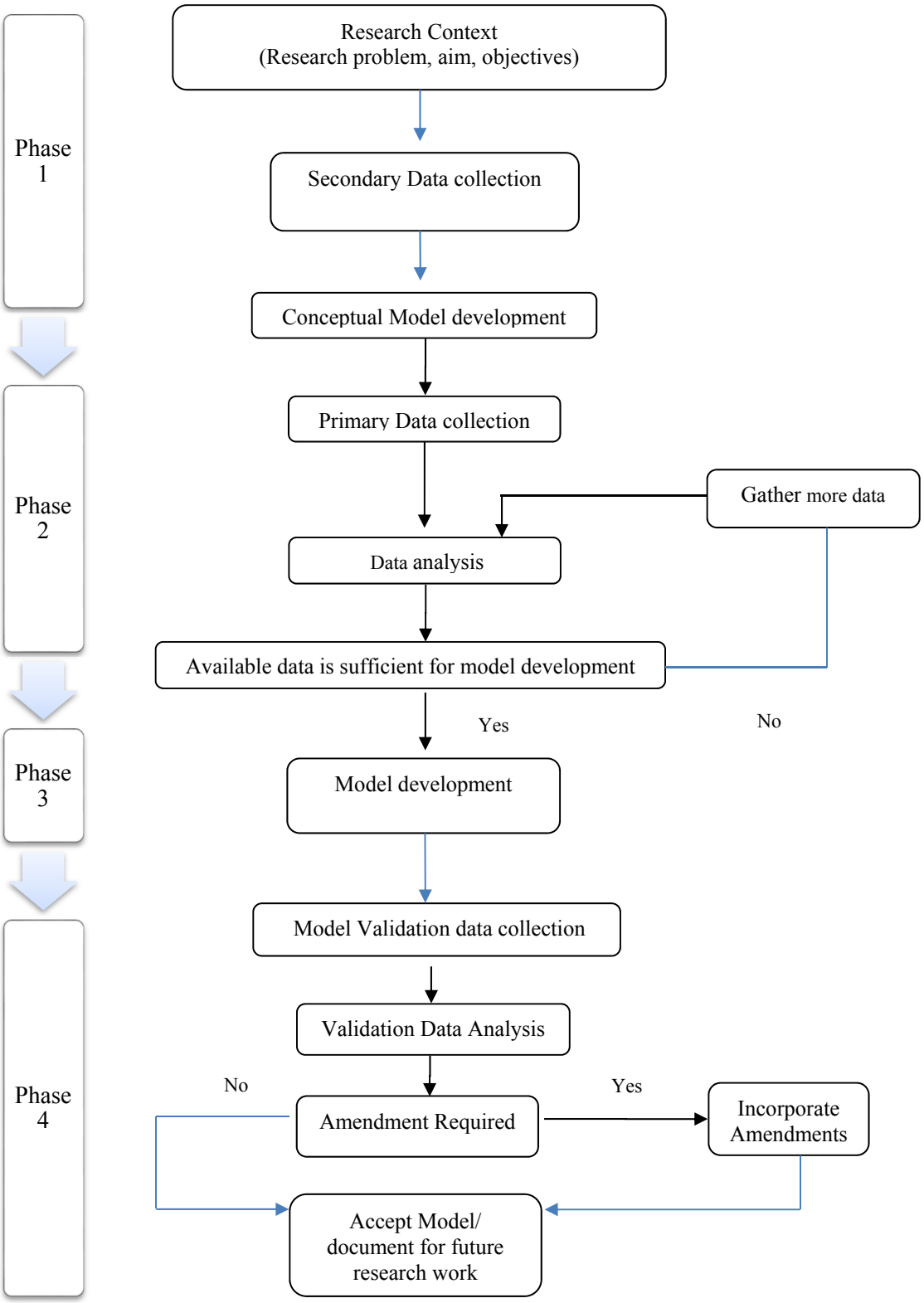


Figure 3.4 Research flow diagram

Fig.3.5 describes the research method flow chart which relates the research methods to the research objective and research outcomes.

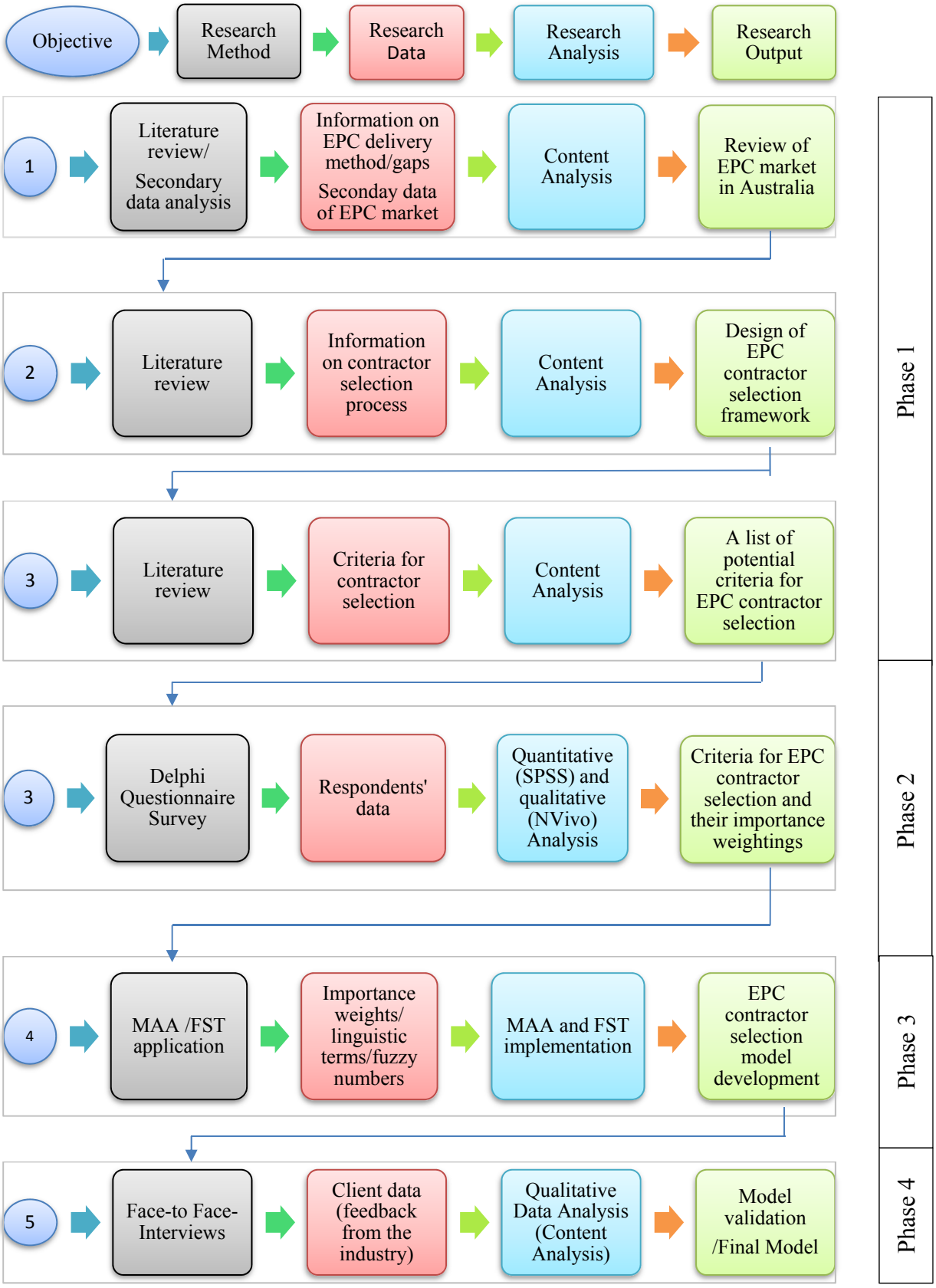


Figure 3.5 Research method flow

Phase1: Identification of research problem, knowledge gaps in existing contractor selection methods and developing preliminary contractor selection framework

Literature review: A comprehensive literature review on project delivery methods, EPC delivery method and existing contractor selection process was conducted to define the research problem and understand the knowledge gaps connected with the problem. In a review of the EPC market in Australia, secondary data were mined and analysed. A literature review aids in identifying contractor selection criteria and identifying strengths and weaknesses in existing contractor selection models, as well as theories/tools that have been used in those models. Preliminary EPC contractor selection framework was designed using literature review findings. Moreover, potential criteria for EPC contractor selection were identified from the literature review.

Pilot study: A pilot study consisting of four different professionals from different organisations has been conducted to ensure clarity appropriateness of criteria identified from literature for EPC contractor selection and completeness of the questionnaire Necessary amendments were included prior to implementing the Delphi questionnaire survey in the next phase.

Phase 2: Data collection and Data Analysis

Research data is compiled from primary and secondary sources. A mixed method (both quantitative or qualitative) approach was selected for data collection. Literature review findings and the Delphi questionnaire survey research methods predominantly facilitated data collection in phase 2.

The Delphi questionnaire survey method was intended to collect subjective data precisely to identify and prioritise the criteria which are more important in EPC contractor selection as the Delphi method provides rich data in quality and accuracy beyond the literature review. One of the key features of the Delphi method is giving 'feedback' in each round. The feedback process is the mechanism for informing panel members of the opinions of their anonymous counterparts. The most common feedback provided in subsequent rounds includes simple statistical summaries such as median, mean, or quartile ranges. To implement the Delphi survey for this

research, the Delphi structure (Fig. 3.2) suggested by Hallowell and Gambatese (2010) was adopted.

Participants were identified using multistage sampling (extension of cluster sampling) and referral sampling (snowball sampling) methods from publicly available data bases, social networking platforms like LinkedIn and researcher's personal contacts. Delphi study panels vary from as low as three panel members to as high as 80 (Hallowell & Gambatese, 2010). However, significant finding might be more significant if the samples were larger. The most important facet of a panel member is their level of expertise in a Delphi study. Therefore, a relatively large unbiased sample of 64 participants were selected from those who are knowledgeable, have relevant experience and can commit to multiple rounds of the Delphi Survey.

The Delphi study comprises three rounds of a survey to identify and prioritise criteria for EPC contractor selection. The main objective of having multiple rounds of a Delphi survey is to reach consensus by reducing variance in responses and secondly to improve precision through controlled feedback and iteration. It is well accepted that convergence to a collective opinion and precision are improved as a result of each round. Previous Delphi studies indicate that the number of rounds ranged from 2 to 7 (Rowe & Wright, 1999) and more Delphi studies found acceptable convergence after two or three iterations (Ameyaw, Hu, Shan, Chan, & Le, 2016). As described in the research of Hallowell and Gambatese (2010), the Delphi results are more accurate after round 2 and become less accurate as a result of additional rounds. The use of at least three rounds allows the researcher to obtain reasons for outlying responses as a part of the second round and report these reasons as feedback in round three. With due consideration to the above facts and time constraints, the Delphi study for this investigation was limited to three (03) rounds and each round objective, and measure of feedback are given in Table 3.2.

Table 3.2 Delphi rounds' objectives

# Round	Objective	Feedback
Round 1	To identify criteria for EPC contractor tender evaluation	Criteria identified from the literature review
Round 2	To determine importance for the criteria identified from Round 1	Response rate (% frequency) of round 1

Round 3	To re-rate the selection criteria considering the overall results of Round 2	Round 2 group mean

'Level of measurement' (or 'rating scale') is the most commonly used measurement tool in a research study (Anol, 2012). Four different levels of measurement from lowest to highest include nominal, ordinal, interval and ratio. Nominal scale can be used for measuring categorical data. Benefit of use of interval scale is that advanced techniques can be employed when analysing data. Both nominal and interval scales were used in the questionnaires of three rounds of the Delphi survey. Most common, the Likert scale which typically has 5 or 7 choice points (Anol, 2012; Cummins & Gullone, 2000), has been selected to measure interval data. In addition, a "Do not know" category was included if issues of item relevance applied to a potential respondent.

Round 1 Questionnaire

In the first round of the Delphi survey, the respondents were requested to identify the criteria that needed to be included in EPC contractor selection. The Delphi round 1 questionnaire was comprised of two parts where the first part of the survey was intended to collect the background information of respondents, and part two was designed with both close-ended and open-ended questions to identify the criteria for EPC contractor selection.

Closed-ended questions included in the first part of the questionnaire were to identify the respondent's profile - work experience, industry, organisation type, ownership - and are of categorical data measured on nominal scale. SPSS statistical programme can be used to analyse the data. The second parts of the survey, also included a closed-end question in which the selection criteria identified from the literature were given as feedback for round 1 respondents and they were asked to determine the criteria that are important to include in EPC tender evaluation. Additionally, in this round, the participants were given the opportunity to comment on each criterion and also to suggest new criteria that were not mentioned in the questionnaire by the researcher in an open-ended question. Open-ended questions provide respondents the opportunity to express their comments in their own words. Open-ended questions provide richer data, however more time and effort are required to analyse the data and sometimes it is difficult to interpret.

Round 1 questionnaire is given in Appendix A.

Round 2 Questionnaire

Round 2 questionnaire survey was developed based on the outcome of the round 1 survey. In round 2, the respondents were requested to determine the level of importance of the EPC contractor selection criteria identified by round 1 using the 7 point Likert scale (1= Not at all important, 2=low important, 3=slightly important, 4=neutral, 5=moderately important, 6=very important, 7=extremely important or essential). In the second round, the response rate as a percentage (frequency) of each criterion was given as the feedback. The Round 2 Questionnaire is given in Appendix D.

Round 3 Questionnaire

Respondents were asked to re-rate the criteria in round 3 in the light of the outcome from round 2 using the same scale used in round 2. The respondents were given the round 2 feedback as the group mean for each criterion in the last round. The round 3 Questionnaire is given in Appendix E.

Analysis of Delphi questionnaire survey data

To analyse data from Delphi questionnaire surveys of this research, the following techniques were adopted:

1. **Eye-balling:** To eliminate outliers and replace the missing values by '999' or NA as appropriate thus preparing an ordered data sheet that can be exported to SPSS software programme to carry out statistical testing
2. **Summarising data:** To summarise the data in terms of frequency and present nominal scale data graphically using bar graphs/charts and interval data using histograms
3. **Describing data:** To describe the structure of data using descriptive statistics, which include central tendency by means of mean, and standard deviation
4. **Measuring internal consistency:** To determine scale reliability in terms of Cronbach's alpha coefficient

5. **Measuring the consensus:** To measure the group consensus of experts in each round of the Delphi survey using a Kendall W non-parametric statistical test in terms of Kendall coefficient of concordance (W)

It is essential to prepare data of each round survey responses for analysis with a consideration to data analysing tools to be used and the format (e.g. spread sheets, text mode, csv format) of data as required by respective analysing tools. To prepare a quality data set, missing data were replaced by discrete value of '999' for numeric variables and discrete value of 'NR' string variables. In addition, the data that reads 'Don't know', 'Not Applicable' or 'Other responses' were handled using 'ditto'.

There are different statistical techniques to analyse the quantitative data however use of these techniques highly depends on various factors such as type of data and scales that were used to measure, number of responses, etc. Parametric statistics have a great deal of power, nonparametric statistics have relatively low power. Distribution of the data (structure) also contributes to deciding whether to use a parametric or non-parametric statistical technique. Non-parametric tests are appropriate for this investigation (round 2 & 3 data) because it is less likely to get a normally distributed data set when the criteria importance, which is measured on 1-7 Likert scale, can be varied significantly.

On the other hand, Qualitative data is collected from responses of open-ended questions in a questionnaire survey. Qualitative analysis technique- NVivo qualitative data analysing software can be used to analyse the text data (string data) from open-ended questions and comment sections of questionnaire survey 1. Unlike quantitative analysis, qualitative analysis highly depends on the researcher's analytical skills and the knowledge of the subject.

Data from all three rounds of Delphi survey were processed as required by respective analyses. Table 3.3 presents the data analytical methods for analysing the data of each round of the surveys.

Table 3.3 Analytical methods of survey data

Instrument	**Analytical tool**	**Purpose**
Delphi Questionnaire 1	Excel spreadsheets	To organise the responses and to calculate response rate on each criterion (close-ended question) as a feedback to next round

	SPSS Quantitative Data Analysis software	To describe and summarise participants profile data of close-ended questions
	NVivo Qualitative Data Analysis software	To organise and analyse the responses of open-ended questions (e.g. comments and suggestion) and identify new criteria
Delphi Questionnaire 2	SPSS Quantitative Data Analysis software	To analyse the responses measured on interval scale (ratings) and prepare feedback to the participants on the central tendencies (means) of criteria for next round To describe the data structure (distribution) To determine internal consistency and to measure group consensus
Delphi Questionnaire 3	SPSS Quantitative Data Analysis software	To analyse the responses measured on interval scale (ratings) and find the central tendencies (means) of criteria To describe the data structure (distribution) To determine internal consistency and to measure group consensus

Scale Reliability and Validity

Scale reliability can be determined by assessments of either internal consistency reliability, inter-rater reliability, test-retest reliability or split-half reliability (Anol, 2012; Donohue & Cooper). Internal consistency reliability is a measure of consistency between different items of the same construct. It is important to ensure that a sufficient number of items are there to capture the concept adequately (Salkind, 2010). Both questionnaires (2&3) contain multi-criteria (multiple item) and respondents were asked to rate the criteria importance using 1-7 Likert scale; measuring reliability in terms of internal consistency is therefore the most appropriate reliability measure for this construct. Internal consistency reliability is tested using Cronbach's alpha (called as coefficient of reliability). Getting an alpha value to 0.7 or higher is important.

Additionally, validity of the scale and construct validity need to be assessed. Validity (known as construct validity) refers to the extent to which a measure adequately represents the underlying construct that it is supposed to measure. Pilot study data

can be used to ensure the validity of the construct in addition to the group agreement with the content of round 1 of the Delphi survey, in which respondents were asked to identify the criteria relevant to EPC contractor selection. The round 2 questionnaire was developed based on the round 1 results. As such, construct validity and scale validity can be confirmed by the group agreement of the respondents.

Measuring consensus

One of the more difficult aspects of the Delphi process is 'measuring consensus'. It is common to use 'variance' as a measure of consensus. In the Delphi study completed by (Hallowell & Gambatese, 2010), consensus was measured in terms of absolute deviation (i.e. +/-5% deviation about the median). Absolute deviation has been used instead of the standard deviation because it measures variability in response about the median rather than the mean and median and is less likely to be influenced by biased results. Additionally, Grisham (2009) suggested that if 80% percent consensus is achieved, the iteration can be stopped. Statistical techniques that are used to measure consensus in Delphi studies include Kendall's coefficient of concordance (W) and Chi-square (χ^2) (Ameyaw et al., 2016). Kendall's W Test is a nonparametric test that measures agreement among raters, and is important for this investigation as the core aim of conducting a Delphi survey is to obtain the group rating for EPC contractor selection criteria with respect to their importance.

Therefore, Kendall's W (or Kendall's coefficient of concordance) non-parametric test, which does not assume normal distribution, is appropriate for measuring group consensus of this investigation. Kendall's W Test is used to assess the trend of agreement among the respondents. Kendall's W value ranges from 0 to 1. The value '1' refers to the complete agreement among the raters, and value '0' refers to the complete disagreement. Kendall's coefficient of concordance for ranks (W) calculates agreements between three or more rankers as they rank a number of subjects (n criteria) according to particular characteristics (importance). The idea is that n subjects (criteria) are ranked (0 to n-1) by each of the rankers, and the statistics evaluate how much the rankers agree with each other.

Phase 3: Model Development

To address the multi criteria/multi alternative nature of the contractor selection problem and develop a new model, Multi-Attribute Analysis (MAA) technique was used. Application of MAA normally involves eight (08) steps. MAA also brings a degree of structure, analysis and openness to the decision making. Those eight steps given below were incorporated in the proposed EPC tender evaluation model that evaluates multiple criteria objectively.

1. Establishing the decision context (aims, objectives, decision makers)
2. Identifying the options (e.g. listing of potential contractors)
3. Identifying the criteria to reflect the objectives
4. Rating performance of each option against the criteria
5. Assigning weights for each criterion to reflect their relative importance to the decision
6. Combining the weights and scores of the options to derive an overall value
7. Ranking the options
8. Final decision making

In MAA, the attribute evaluation results can often be presented in incommensurable units, numeric, or descriptive. To address this uncertainty and impreciseness, Fuzzy Set Theory is introduced. Therefore, MAA and FST are used to develop the new model for EPC contractor selection.

Phase 4: Model Validation

Semi-structured face-to-face interviews

The model is then validated by construction industry professionals' feedback from face-to-face-semi-structured interviews for appropriateness to use in the EPC industry in Australia. Potential interview participants were identified using a referral sampling method from public and private sector organisations. The Delphi survey participants who had the most experience working on EPC projects were asked to nominate potential interview participants who have extensive experience in selecting EPC contractors. Three interview participants who have experience in selecting EPC

contractors for EPC projects in Australia were selected. A single round of interviews with each participant was conducted, considering the time constraints and this was felt to be adequate to elicit insight from the participants.

Analysis of audio data of semi-structured face-to-face interviews

Qualitative data analysis is challenging with the sheer volume of audio data from the interviews. However, audio data can be transcribed to text data using transcript services which are very expensive but time saving in comparison to manual transcribing. As already mentioned, the NVivo qualitative data analysis tool can be used to analyse the qualitative data in the form of text data. NVivo can do queries on word frequency, coding, text search, and create models; in addition, it can be used for describing data graphically with reference codes.

Therefore, raw data (audio data) from interviews is required to be transcribed and transcripts exported as a word document in the form of text data for content analysis. Then, the text data of interview transcripts are classified using coding techniques according the semi-structured interview questions. Manual analysis of coded data is used to validate the model for industry use.

3.3 LIMITATIONS

The major limitations of this research include a sampling limitation. The sample for the interview was limited to the participants who were in and around Brisbane in Queensland for the convenience of conducting face-to-face interviews at either the researcher's premises (QUT in Brisbane) or at interviewees' places permitting less travel time and conducting the interviews economically. This was not a limit for the questionnaire surveys as they were conducted online, enabling Australia wide participation.

Sources for identifying potential participants from the EPC industry were also limited with the lack of publicly available data sources specific to EPC. However, the researcher explored all possible avenues, obtained support through personal contacts and used networking skills to identify potential participants.

Another limitation that could not be avoided was the timeline of the questionnaire survey administration and interviews scheduled to suit every participant. This becomes a matter of concern because some participants were impossible to contact

because either they were travelling on work or on personal breaks, especially during major holiday breaks in Australia. Therefore, personalised invitations or reminders were sent to the participants from whom automatic replies were returned and the surveys were kept open for further time than anticipated thus allowing the majority to participate, given that committing to three rounds of Delphi survey is not an easy task for professionals with their work commitments. As a precaution, the researcher aimed for a larger sample even though a lesser sample (<20) is adequate for a Delphi study.

Limitation within the sources that were used to distribute the survey was a matter of concern. For example, when the survey link is posted on LinkedIn, there is no way of sending reminders.

3.4 SUMMARY

To establish research context and reach conclusions, several research methods including literature review, secondary data analysis, Delphi method, and face-to-face interviews, were adopted for data collection in response to the research questions. In addition, theoretical applications such as multi-attribute analysis (MAA) and fuzzy set theory (FST) have been adopted to develop a new EPC contractor selection framework and tender evaluation model. To analyse the quantitative data and qualitative data, use of analytical tools such as SPSS and NVivo in this research and relevant statistical measures were discussed.

Chapter 4: **Overview of EPC Market**

4.1 UNDERSTANDING THE EPC DELIVERY METHOD - ITS BENEFITS AND CHALLENGES

4.1.1 Definition of EPC Project Delivery Method

To meet the different requirement of clients/owners, a number of project delivery systems have been developed for designing and constructing facilities. The most common of these are Design-Bid-Build (DBB), Design-Build (DB), Engineering-Procurement-Construction (EPC), Design-CM (Construction Management) Contracts, Fast–track construction, Partnering and Relational Contracting/Lean project delivery (Forbes & Ahmed, 2010; Molenaar et al., 2010). These delivery methods have been used for many years with varying degrees of success, depending on the type of project involved and the skills required (Forbes & Ahmed, 2010).

A recent survey by KPMG International (2015) reveals that EPC is the most popular project delivery method in the energy and natural resources sector in the global market. However, obtaining a clear definition of EPC is challenging, as there are a number of different definitions currently in use. Added to this confusion, is the use of different terms from different perspectives. As shown in Table 4.1, EPC is sometimes called 'Turnkey' or DB, where the contractor delivers a full and complete facility. While some studies refer to EPC as a variation of DB, others term them collectively as EPC/Turnkey/DB (Baram, 2005), EPC/DB (Galloway, 2009) or EPC/Turnkey (Godwin, 2012, p.18; Grimmitt & Vera, 2007; Huse, 2002, p.5-9).

Table 4.1. EPC Definitions

Author /Organisation (year)	EPC Definitions
Huse (2002)	EPC is a Turnkey contract that places all design, procurement and construction responsibilities on one contractor. The Turnkey arrangement is also known as the 'package deal', DB, DC or EPC.
Ellsworth (2003)	Under an EPC contract, the contractor is obliged to deliver the client/owner a fully functional project ready for commercial operation on time under a single contract, with

	the only function necessary for commencement being the 'turning of the key'.
Baram (2005)	EPC is a project delivery method where one or more contractors and designers work collaboratively to deliver a full and a complete facility fit for its intended use under a single point responsibility for the design and construction with a fixed and agreed price. EPC is sometimes referred as Turnkey or DB.
Grimmitt and Vera (2007)	EPC is a procurement route in which the contractor does everything in return for the contract price, undertakes all aspects of the project, provides a single point of communication and responsibility for the client/owner, assumes a greater proportion of risk, and hands over the keys on completion to the client/owner to operate.
Forbes and Ahmed (2010)	EPC is configured in a very similar way to DB and is used in industrial project delivery where one organisation performs or manages the design and construction functions of the project that emphasises engineering design, as opposed to architectural design, and typically has commissioning and maintenance phases.
DLA PIPER (2011)	Under an EPC contract, a contractor is obliged to deliver a complete facility to a developer for a guaranteed price by a guaranteed date to the specified level and who needs to only 'turn a key' to start operating the facility. EPC is sometimes referred as a Turnkey contract.
Godwin (2012)	EPC is a Turnkey contract where the contractor provides a complete full package that is ready for operation when completed and has full responsibility for all features of the work to satisfy the client/owner's defined requirements.
EPC Engineer (2013)	EPC stands for Engineering, Procurement and Construction and is a prominent form of contracting agreement in the construction industry where the engineering and construction contractor carries out the detailed engineering design of the project, procures all the equipment and materials necessary, carries out the construction work and delivers a functioning facility or asset to the client/owner.

As the terms of DB, Turnkey and EPC are often confused, it is worth comparing these concepts closely (see Table 4.2). Unlike DB projects, contractor's responsibility in Turnkey and EPC projects is extended beyond construction. Turnkey contractors deliver a fully functioning project and the owner purchases it on

completion (Ellsworth, 2003) as the turnkey contractor finances the project. The completed project is ready for commercial operation and the only function remaining for commencement is to 'turn the key of the door' and take possession. However, EPC projects are typically financed by the project owners or the lenders.

Table 4.2. EPC/DB/Turnkey-Definitions

Type	Definition	Project types
Turnkey	The Turnkey method usually involves a single contractor being responsible for the total project life cycle from design through post-construction functions such as commissioning and handover. The client signs the contract and expects that 'turning the key' is the only function to open a fully functional facility (Ahola, Laitinen, Kujala, & Wikström, 2008). This option shifts some risk to the developer. It reduces the economic return to the facility owner or limits the types of technologies or equipment - very similar to client procuring a completed project.	Most common in building and industrial projects
DB	Design build/Construct system is an integrated approach that delivers design and construction services under one contract with a single point of responsibility (Palaneeswaran & Kumaraswamy, 2000). Owner selects a DB contractor to develop a construction project, which is governed by architectural designs.	Most common in building and infrastructure projects
EPC	EPC system where one or more contractors and designers work collaboratively to deliver a full and a complete facility fit for its intended use under single responsibility for the design and construction under an agreed fixed time and fixed price (Baram, 2005). Owner selects an EPC contractor to develop an engineering construction project.	Most common in major industrial projects and infrastructure projects

Some researchers identify BOT (Build-Operate-Transfer), BOOT (Build-Operate-Own-Transfer), BOO (Build-Own-Operate), and BLT (Build-Lease-Transfer) as

variations of EPC (Baram, 2005; DLA PIPER, 2011; Godwin, 2012, p.18; Huse, 2002, p.7). However, these are financial models in which the contractor, in addition to designing and constructing the project, finances the project and leases or operates and then transfers the project to the owner at the end of the agreement. This type of arrangement - also sometimes considered as a 'Turnkey contract' - is more common with owners who do not wish, or are unable, to finance the project.

A number of EPC subcategories also exist in the construction industry, namely EPCI (Engineering-Procurement-Construction-Installation), EPCC (Engineering-Procurement-Construction-Commissioning), EPCIC (Engineering-Procurement-Construction-Installation-Commissioning) and EPCM (Engineering-Procurement-Construction Management) (EPC Engineer, 2013; Galloway, 2009; Huse, 2002, p.7; Meinhart & Kramer, 2004; Schramm, Meißner, & Weidinger, 2010). These EPC subcategories contractually emphasise the different project stages of EPC contractor involvement.

For clarification, this study defines EPC as *a project delivery method, in which one or more contractors and designers combine their efforts to deliver a full and complete industrial project.* This is typically associated with providing an operating facility under a single point of responsibility for the design and construction, and normally for a guaranteed price, within a guaranteed time, to a specified quality and with managed risks.

4.1.2 Understanding the EPC process, benefits and drawbacks

The two main phases of EPC are the development phase and implementation phase. Important planning activities within the development phase, namely investigation, scope definition, work packaging and contract award are crucial to project success. This pre-EPC work normally takes 2-3 years (Rothman, 2000). The main activities in the implementation phase are detailed engineering, procurement and construction. The EPC process normally starts with the client/owner defining the project scope and specifications, and communicating these to either in-house design personnel, external engineering consultants or the EPC contractor to develop Front-End-Engineering-Design (FEED) to a level sufficient for inviting tender proposals (Mayer Brown, 2008). FEED includes feasibility studies, process design, project cost and schedule development (West, 2011). After FEED development, the client/owner can consider different options: to continue with the same contractor or select another contractor or

FEED contractor using a preferred method of bidding. The EPC contractor is commonly selected by competitive bidding and engaged by a lump sum/fixed price contract (M. T. Chen, 1993).

Once the EPC contract is awarded, the implementation phase commences and spans normally at least three years. The contractor then becomes the single point of responsibility for completing the detailed engineering design, procuring all services, equipment and materials, and the construction and delivery of the functioning facility to the client/owner within the agreed time and budget (Baram, 2005). Construction usually encompasses civil, mechanical, electrical, piping and instrumentation (Hammad Ud Din, 2004). EPC contracts typically involve numerous in-house experts and subcontractors (Halvorsen, 2009) and place the responsibility for cost and schedule risks directly onto the contractor (Baram, 2005; Forbes & Ahmed, 2010; Grimmitt & Vera, 2007; Masi, Micheli, & Cagno, 2013). To ensure that the contractor's work is in accordance with the agreed scope of work and the standards and conditions stipulated in the contract, the project client/owner may oversee the contractor by means of an in-house project management team or under a separate contract (EPC Engineer, 2013).

The EPC project delivery method has emerged as the preferred choice for many industrial projects and provides many technical and commercial benefits to client/owners. EPC contracts aim to avoid delays and cost overruns by providing optimised collaboration within integrated disciplines (Halvorsen, 2009). Innovation in design and construction is also encouraged, as the contractor and the designers work together throughout the entire process. The contractor's direct responsibility for all aspects of the project facilitates early ordering of long-lead items, with potential reduction in overall schedules (Meinhart & Kramer, 2004). Furthermore, the single point of responsibility reduces the client/owner's administrative burden associated with the asset development, which enables the client/owner to focus on its core business (EPC Engineer, 2013; Meinhart & Kramer, 2004). In addition, EPC provides better communication between the client/owner, designer and the contractor, which is vitally important in enhancing project productivity. Another advantage to client/owners and contractors alike is the improved risk management process that aligns with business objectives, as EPC offers a comprehensive package that fully integrates the requirements of the lenders, who assess the technical and

commercial project risks involved in project financing (Baram, 2005; DLA PIPER, 2011; Halvorsen, 2009).

While there are numerous advantages for client/owners using an EPC contract, there are also disadvantages associated with the EPC method (Baram, 2005; DLA PIPER, 2011; Forbes & Ahmed, 2010), including:

- a higher contract price due to the allocation of almost all the design and construction risks to the contractor
- less opportunities for design control/ intervention by the client/owner
- the possibility of unrealistic designs, emphasising cost over quality
- difficulty in making post-contract variations except by client/owner change orders
- being unsuited for work, such as refurbishment, where the scope is ambiguous
- not eliminating the need for a client/owner representative

In addition, engineering design within EPC is multi-disciplinary, creating a high level of risk and complexity. For example, there are a number of engineering disciplines involved in oil and gas projects, including Process/Chemical Engineering, Mechanical Engineering, Piping and Instrument Engineering, Electronic Engineering, Electrical /Power Engineering, Information Technology/Telecommunication, Civil and Infrastructure Engineering, Structural Engineering (onshore), Structural Engineering (offshore) and Subsea Engineering. The high level of integration of various disciplines increases the level of risk (Engineers Australia (EA), 2013a).

Furthermore, EPC project delivery is highly dependent on a number of international/domestic sub-contractors, where complex supplies are very difficult to manage (Cagno & Micheli, 2011). Thus, the operation process, management mode, contractual obligations and risk allocations in EPC are very different from traditional delivery methods (Hui & Qin, 2011). Therefore, EPC project implementation has become a major challenge for project owners.

4.2 OVERVIEW OF THE EPC MARKET

4.2.1 The global EPC market

The construction industry is vital for the development of any country and is a measure of its economic growth. For example, the construction industry is one of the most significant industrial contributors to European economy in terms of Gross Domestic Product (GDP) and employment (Alzahrani & Emsley, 2013b). In the UK, the construction industry contributed around 8% of total GDP in 2010. The construction industry, including manufacturing and mining, contributed 17% of total GDP in the U.S. in 2013. Fig. 4.1 provides a snapshot of the countries with the most EPC projects in 2014 according to EPC Engineer data, indicating the U.S. to be the clear leader. EPC is also one of the fastest growing industries in the Middle East, contributing 10.3% of GDP in 2011.

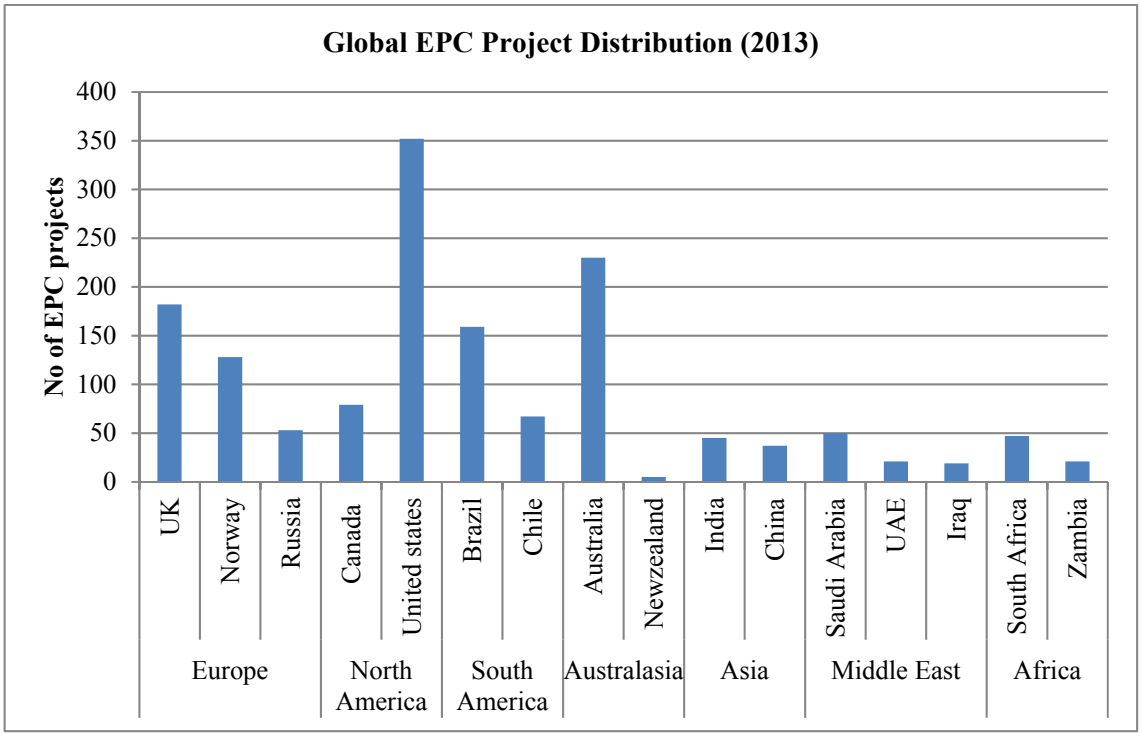


Figure. 4.1. Global EPC project distribution

Source: (EPC Engineer, 2013)

Given that EPC is normally regarded as a variant of the DB system, understanding the global DB market is also vitally important. In the U.S., DB accounted for 15% of the total construction market in 2012 (Construction Management Association of America (CMAA), 2012). Furthermore, nearly 40% of non-residential construction

projects (e.g. military, medical, industrial, commercial, community, educational and government projects) have been delivered through DB in the U.S. since 2005 (Design Build Institute of America (DBIA), 2010). In China, around 10% of all construction projects were delivered through DB (Xia & Chan, 2008).

Despite the EPC market not being specifically addressed, these statistics cover EPC as part of the DB market for large and complex industrial projects, most of which are in the oil and gas, power, processing and mining industries. With the increase of project activities in the energy sector, AUD 234 billion worth of EPC contracts have been awarded in last five years (2008-2013) for oil, gas and petrochemical projects in the Middle East and North Africa Region (Kevin Baxter, 2013). According to (Xia & Chan, 2008), 50% of DB projects in the petrochemical, metallurgical and electronic industries in China are delivered by the EPC method, which account for 15-20% of the market share of these industries. EPC is also forecast to account for around 40% of Indian infrastructure investment for 2012-2017 (Earnest & Young (E&Y), 2011). The Indian EPC market is expected to grow steadily due to unprecedented investment in public and industrial infrastructure, especially in the oil and gas, metals and mining industries (CHEMTECH Foundation, 2011).

The Middle East is one of the most important regions for the global EPC market with USD 65 billion of EPC work in 2011 in the hands of the top 30 EPC contractors in the region (Oil & Gas Middle East, 2012). Furthermore, EPC projects in the UAE and Saudi Arabia have led the Gulf Corporation Council (GCC) region with USD 143 billion worth of EPC contracts since 2008. However, the EPC market in the Middle East has become more competitive in recent years due to the global financial crisis and soaring oil prices - greatly reducing the number of mega projects. However, a steady growth is expected beyond 2013 (Oil & Gas Middle East, 2012).

4.2.2 The Australian construction industry and the EPC market

The construction industry plays a major role in the Australian economy in terms of GDP and employment. The Australian Bureau of Statistics (ABS) reports that the construction industry contributed 7.8% to GDP in Australia in 2014-15 and directly employed over one million people. In addition to being the largest employer of local labour, the construction industry feeds many other industry sectors, such as manufacturing and professional services, during the design, construction and fit-out stage of major projects. Every dollar that is invested in construction returns three to

four dollars to the rest of the economy (Australian Industry Group (AIG), 2015; BIS Shrapnel, 2013b).

The construction industry's contribution to GDP fluctuates significantly. From a low level during World War II, major developments such as the Snowy Mountains Hydro-electric Scheme, post-war buildings and reconstruction work during the immediate post-World War II period, the construction share increased and peaked at 9.5% of GDP from mid-1960s to mid-1970s. This was the period when EPC emerged in the Australian construction industry for oil refineries, pipelines, power plants, chemical facilities, mining, oil and gas, and infrastructure projects (FLUOR, 2013). Since then, it declined to 6% in the early 1990s and remained at that level until 2001. Recent figures show the current construction industry share of GDP is around 7% (6.8% in 2007, 7% in 2008, 6.8% in 2009, 7.7% in 2011-12, 8.3% in 2012-13, 8.4% in 2013-14 and 7.8% in 2014-15) over recent years (ABS, 2006-2015; ABS, 2008; ABS, 2010; ABS, 2012a). The construction industry had particularly strong growth in 2011-2012, when it recorded the second largest increase in total income (AUD 21 billion) (ABS, 2012b)

Table 4.3. Construction work done (trend estimate) in AUD billions

Sector	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
Residential Building	49.90	47.72	46.34	47.21	48.58	46.49	51.37	49.58	47.41	48.81	53.71	61.01
Commercial Building	18.56	20.53	22.28	24.67	27.31	23.51	22.53	22.40	23.80	24.34	32.27	35.11
Engineering Construction	29.53	38.33	48.00	56.47	70.84	76.13	77.96	101.64	129.89	131.43	118.47	100.78
Total	98.0	106.58	116.62	128.34	146.72	146.13	151.86	173.52	131.43	201.10	207.25	196.69

Source: (ABS, 2016, cat.no.8755.0)

The value of construction work completed over the years 2004 to 2015 is summarised in Table 4.3, which indicates a steady growth until 2013 in both commercial building and engineering construction. However, the financial crisis in the recent years has affected the domestic construction market too. As EPC is commonly used in engineering construction, a closer examination is provided in the following section.

4.2.3 Engineering construction

Engineering construction covers infrastructure, mining and heavy industry construction, where infrastructure construction includes electricity generation and supply, sewerage, drainage and water storage and supply, roads, railways, telecommunication infrastructure, pipelines and other civil projects. Heavy industrial construction includes chemical, petro-chemical plants, oil refineries, gas-processing facilities and other industrial plants.

In 2010-11, the largest contribution to the value of engineering construction work was from the facilities for oil, gas, coal and other minerals (19% increase from 2009-10) (ABS, 2012). It witnessed a high level of annual growth of 13.9% in 2012, as heavy industry construction maintained a strong growth, with its main driver being from the oil and gas industry where growth was 28.7%. However, circumstances changed dramatically in 2013 to a 2.1% growth, which was far below the expected 7.6% growth, due to the slower growth in road construction, water and electricity and mining projects (Australian Constructors Association (ACA); Australian Industry Group (AIG) (2013), 2014)).

A mining boom, solid work in pipeline and oil and gas processing, mining related projects such as harbour upgrades and infrastructure projects contributed significantly to the growth in both the engineering and commercial construction sector during 2011-2013. The growth in 2014-2015 was largely impacted by major Liquefied Natural Gas (LNG) developments moving to their completion phase and prospects for new investments weakening over the years (AIG, 2012). As evidenced in Fig. 4.2, the mining and heavy industries within which oil and gas projects are delivered provide the largest contribution to engineering construction during the biggest mining boom in Australia's history (Tulip, 2014) - beginning around 2003 with iron and coal prices rising and high demand from Asian countries like China. The construction industry is one of the sectors that most benefited from the boom, when the LNG sector delivered several large oil and gas projects. Again, the main driver of the LNG market in Australia is the demand from Asian countries.

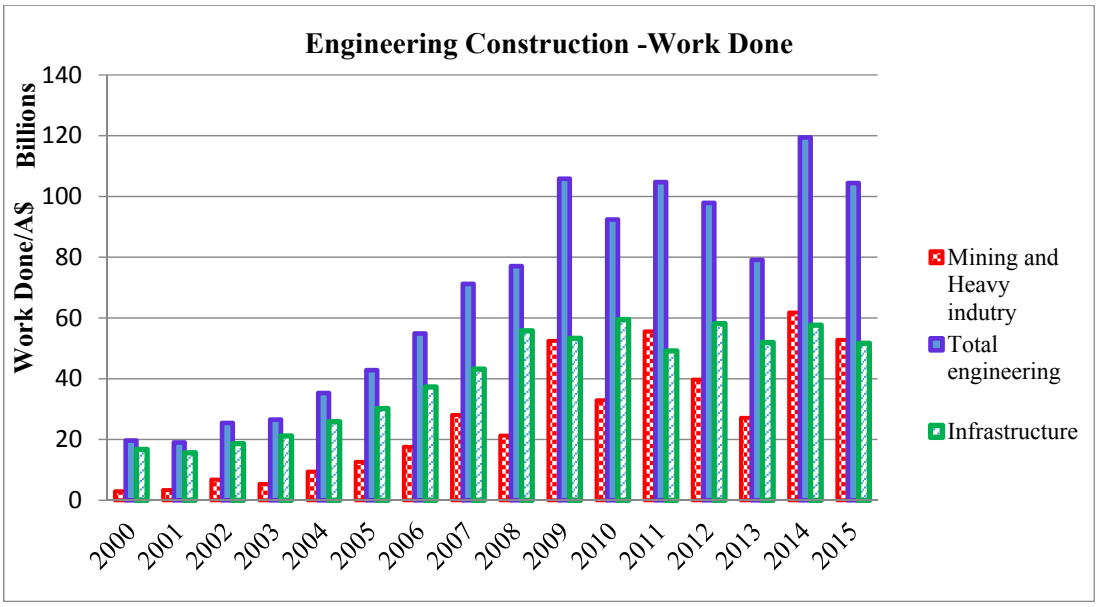


Figure. 4.2. Engineering construction work done (AUD millions)

Source: (ABS, 2016b, cat.no. 8762.00)

According to Construction Outlook reports of 2016, engineering construction was down-turned by 15.0% in 2015 due to persistent weakness in the mining and heavy industrial sector. The negative outlook continued to decline by a further 2.9% in the first half of 2016, however, it is expected to rise by 3.6 % in 2016-17. The fall in construction work for the oil and gas processing sector is reportedly 40.9%, as a number of large scale projects were completed in 2016. However, it is expected to recover in 2017-18 with a 3.7% rise mainly from non-resources infrastructure activities (Fig.4.3). Resources-related engineering construction is expected to fall in 2017 as low prices in oil and gas curtail further investments.

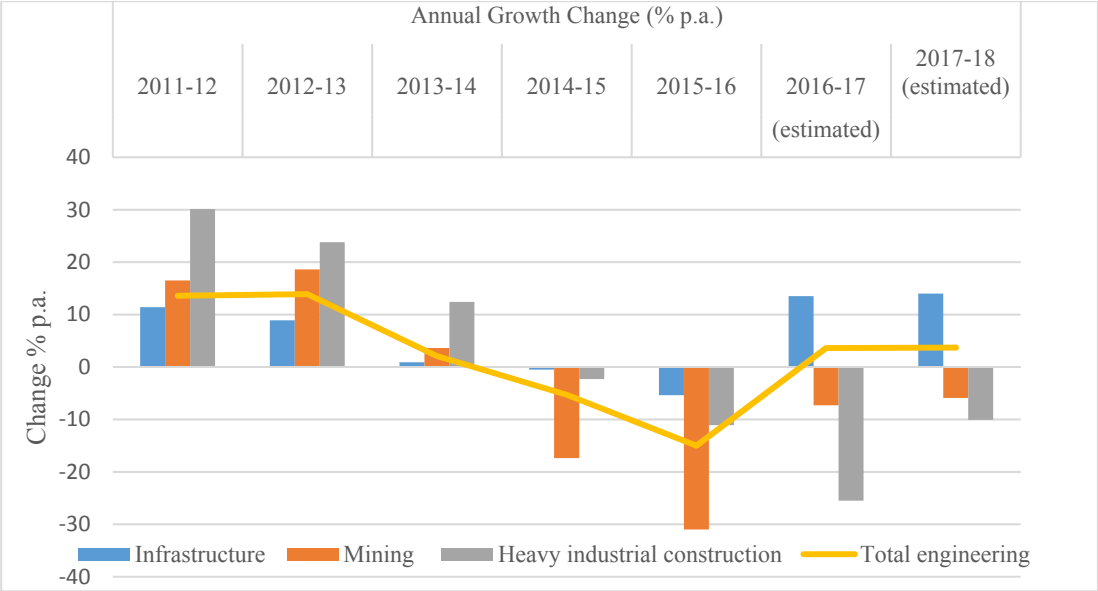


Figure. 4.3. Engineering Construction Outlook (2011-2018)

Source: (Australian Constructors Association (ACA); Australian Industry Group (AIG), 2013, 2014, 2016)

The resource industry still plays an important role in engineering construction at the State and Territory level, and the Northern Territory had a record growth rate in 2012, with major projects including the giant Ichthys gas project, expansion activity in mine sites, and Montara oilfield works (EA, 2013b). The Ichthys gas project is expected to be completed in 2017 (West, 2011). In Queensland, engineering construction activity that increased by AUD 10 billion (activity increased by 43.6%) during 2011-2012 was driven by mining and related infrastructure freight transport and facilities (AIG, 2012). Engineering construction has become the largest component of the construction industry and has been the fastest growing sector over the years since 2001.

Not surprisingly, the resource and energy sector has been the main driver of engineering construction. As most mining and heavy industrial construction projects are delivered through the EPC project delivery method, the EPC market is expected to recover in the next few years. As reported, more than AUD 300 billion has been invested in resources and infrastructure projects over the next decade (ACA, 2015).

4.2.4 EPC in the public sector

EPC is adopted in public sector project delivery mainly for power and large infrastructure projects. The Australian Government currently uses the 'Commonwealth Procurement Rules (CPR)-2012' to procure capital projects that

include building and infrastructure developments (CPR, 2012). All States and Territories have developed procurement frameworks aligned with the core principles of CPR to achieve value for money. For example, Queensland Government departments use a 'Capital Works Management Framework (CWMF)' for the selection of the most appropriate procurement strategy and contracts for government building projects, while the NSW Government uses the 'Procurement Practice Guide' to procure construction projects and contracts to deliver construction projects (NSW Government-Procure Point, 2008). Table 6 summarises the delivery methods of public construction projects procurement in the Australian mainland states.

As shown in Table 4.4, all governments use a range of delivery methods to procure capital projects, including D&C, PPP, managing contractor and alliancing. Additionally, Western Australia, Victoria and New South Wales use various delivery models that are similar to EPC (e.g. Design, Build and Operate (DBO), Design, Build, Finance and Operate (DBFO), Design, Build, Finance and Maintain (DBFM), and Design, Build, Operate and Maintain (DBOM), BOOT and BOT).

As a variant of DB (or D&C), EPC is a particular preference for the construction of power and large infrastructure projects, although it is not specifically prominent in public project procurement. The Infrastructure Australia Magazine reveals several significant government projects being delivered through EPC or EPC variants. For example: the Musselore wind farm project (AUD 394 million) in Tasmania uses EPC; the Legacy Way road project (AUD 1.5 billion) in Queensland is being delivered by EPC variant – Design, Construct, and Maintain and Operate (DCMO); the Beaconsfield substation refurbishment work (AUD 140 million) in New South Wales has been delivered through Design-Procure-Construct (very similar to EPC); and the Christies Beach upgrade project in South Australia uses EPCM.

Some commentators (Baram, 2005; DLA PIPER, 2011; Godwin, 2012, p.18; Huse, 2002, p.7) have also interpreted EPC as a variant of BOT or BOOT, and Western Australia, New South Wales and Victoria use different models of PPP/PFP that have similar concepts to BOT or BOOT. These are common methods used in the construction industry when the project client/owner cannot, or does not wish to, finance large projects itself (Godwin, 2012; Huse, 2002, p.9). Large-scaled complex public sector projects are typically delivered through D&C or relationship contracts (e.g. alliances and PPP) where the EPC project components are present. PPP involves

not only the design and construction of the infrastructure but also the operations and finance (Carbonara, Costantino, & Pellegrino, 2015) that differentiate the PPP from EPC. PPP project are executed on an agreement between the public and private sector organisations and typically are financed by the private sector for public sector projects.

Table 4.4. Public sector project delivery methods

Procurement Systems	Queensland	New South Wales	Victoria	Western Australia	South Australia	NT	ACT	TAS
Traditional Procurement	Traditional–Lump Sum	Construct Only (CO) Guaranteed Maximum Price (GMP)	CO	CO	DBB	CO	CO	DBB
Integrated Procurement	Design and Construct (D&C)	D&C Design, novate and construct (DN&C) Design, development & construct (DD&C)	D&C (variants include DN&C, DD&C, DC&M)	D&C	D&C	D&C	D&C	D&C
Management Procurement	Managing contractor	Managing contractor	Managing contractor Construction Management (CM)	Managing contractor CM Direct Managed	Managing contractor CM		Managing Contractor	
Collaborative Procurement (Relational)	Alliance	Alliance Private Financed Project (PFP) (includes BOOT, BOT and DBFM) Public-private partnerships (PPP)	Alliance contracting PPP (includes DBO, DBFO, and DBFM)	Alliance contracting PPP (including DBO, DBFO, DBFM, DBOM)	Private Finance PPP Fast-Track	PPP	PPP	PPP
	Bundling	Design, construct and maintain (DC&M) (including DCO, DDC&O)	DC&M	DC&M	--			
	Early Contractor Involvement (ECI)	ECI		ECI	ECI	ECI	ECI	ECI

4.2.5 *EPC project procurement in the private sector*

EPC has become the most common project delivery method in the private sector for large-scale and complex projects that include power, oil and gas, transport, water and telecommunications (DLA PIPER, 2011). For example, most industrial projects (e.g. power, processing plant) and almost all large private sector wind farms use EPC (DLA PIPER, 2011). In addition, EPC has been widely adopted for oil, gas, mineral, energy, power, and infrastructure projects (EPC Engineer, 2013).

Until 2001, the public sector was the predominant funder of engineering construction activity in Australia. Since that time, privatisation and contracting-out strategies have led to the private sector, carrying out engineering-related work previously undertaken by the public sector so that the value of engineering construction work in the private sector has exceeded that of the public sector (see Fig. 4.4). This has increased to the point where the engineering construction work completed during 2011-2012 was 28% and 72% for the public sector and private sector respectively with total work by the private sector valued at AUD 102.8 billion, the largest proportion of which is mining and heavy industry – accounting for 46% of the engineering work completed (BIS Shrapnel, 2013a).

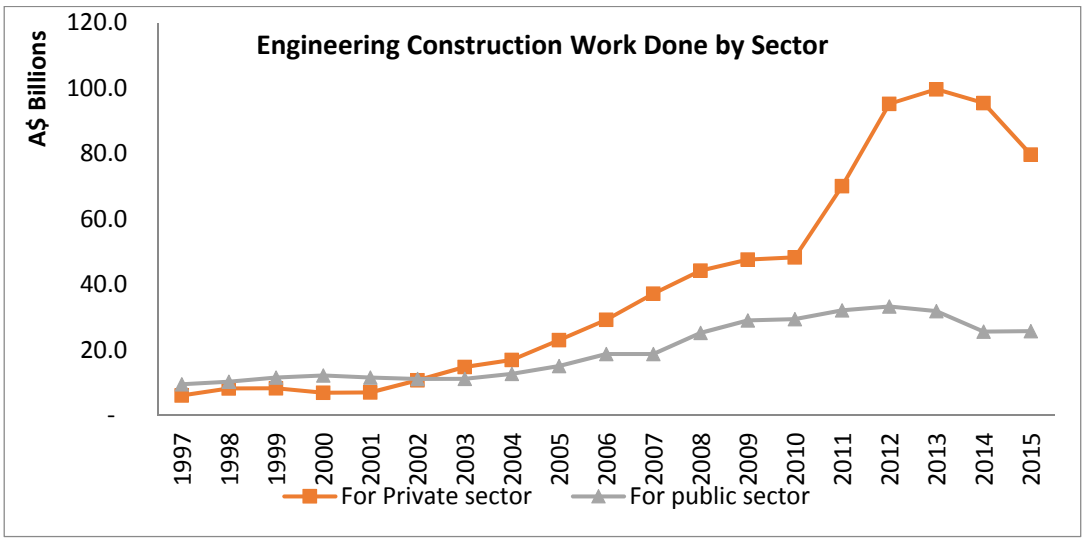


Figure. 4.4**.** Engineering construction work done by sector

Source: (ABS,2016a, cat.no. 8762.0)

To obtain information relating to the use of EPC, it is necessary to consult project data from the ENR100 list of international DB contractors conducting business in the Australian construction market (Engineering News-Record (ENR), 2013). FLUOR,

Bechtel, Jacobs and KBR are the largest on this list and Table 4.5 summarises the available project information from their company project portfolio and web-based sources. Although PCL Construction (Rank 12) and URS Corporation (Rank 16) also conduct business in Australia, they do not provide enough EPC project information for the analysis.

According to Table 4.5, most EPC projects are from the oil and gas industry and involve large capital costs. Furthermore, EPC contractors are sometimes joint ventures (JV) that consist of different EPC contractors offering the integrated services required. Western Australia and Queensland dominate the EPC market, with many ongoing oil and gas projects being the key drivers of Australia's State economies.

Table 4.5. Major EPC projects in Australia

Project	Industry	Contractor	Project Delivery by	Capital Cost (AUD billions)	State	Year
Cadia East Mine	Mining	FLUOR	EPC	1	SA	2009
Prominent Hill Copper mine	Mining	FLUOR	EPC	1	SA	2009
Gladstone LNG (GLNG) Project	Oil & Gas	FLUOR	EPC	16	QLD	2015
Curtis LNG Project (QCLNG)	Oil & Gas	Bechtel	EPC	20.4 (USD)	QLD	2014
Curtis LNG Project (APCLNG)	Oil & Gas	Bechtel	EPC	24.7	QLD	2015
Wheatstone LNG project	Oil & Gas	EVT JV (EV LNG Australia/Thiess)	EPC	29 (USD)	WA	2016
Darwin LNG	Oil & Gas	Bechtel	EPC	--	QLD	2006
Caval Ridge project	Mining	Bechtel	EPCM	--	QLD	2013
Boddington Gold mine	Mining	JACOB	EPCM	--	WA	2010
Hope Down North Mine	Mining	KBR	EPCM	--	WA	--
Major Copper – Uranium mine	Mining	KBR	EPCM	--	SA	--
Ichthys LNG	Oil & Gas	JKC JV (JGC Corp/KBR/ Chiyoda Corp.)	EPC	34 (USD)	WA	2016
Gorgon LNG project	Oil & Gas	KJVG (KBR/JGC/Clough/ Hatch)	EPCM	2.7	WA	2015

4.3 SUMMARY

This chapter included a comprehensive overview of the EPC market in Australia, as the EPC delivery method has gained much popularity for engineering projects in infrastructure and resource sectors, which have driven the Australian EPC market in recent years. EPC delivery method and benefits and challenges of EPC implementation were broadly discussed. Moreover, the EPC market particularly in Australia was extensively investigated.

Chapter 5: EPC Contractor Selection Framework

Selecting the most appropriate contractor is of paramount importance in achieving successful project delivery. Given that EPC projects are normally of increased complexity, high budget values, multiple stakeholders and longer project timelines, EPC project owners face significant challenges in contractor selection. This chapter will propose a clearly defined contractor selection framework, which is tailored to EPC projects, enabling clients to reach an appropriate decision on sound judgement.

5.1 DECISION MAKING IN EPC CONTRACTOR SELECTION

Clients typically consider their prioritised objectives and their attitudes towards risks when selecting the right contractor and then identify the procurement strategy that aligns with the selected delivery method. Thus, developing a contractor selection framework is like putting all the pieces of a procurement strategy jigsaw puzzle together in a structured and an interactive way. Every piece needs to be thoroughly explored considering the project's individual characteristics, risks and circumstances prior to starting to combine the pieces. Deductive reasoning helps identify appropriately fitting pieces for EPC contractor selection.

A procurement strategy which consists of policies and procedures, governs the selection of a satisfactory contractor to support a preferred project delivery method (Ruparathne & Hewage, 2015). The core focus is therefore on individual elements of the procurement process which include *contractor selection stages, tendering method, procurement policy, and tender evaluation*. Existing contractor selection practices that identify optimum choice were initially explored in Chapter 2 (Literature Review).

Depending on project nature, either a single or two-stage contractor selection process can be used to select the optimum contractor. A two-stage process

includes prequalification during the first stage and secondly, tender evaluation. Construction projects can be categorised as either new or routine. Alternatively, these projects can be categorised as simple or complex. Complex projects are typically large in value and of high risk. Routine projects can be either small or large in value, however risk can be considered as low if it is repetitive work. Therefore, routine projects are simple in nature though they are of high monetary values and large in size. Thus, risk assessment helps determine the project risk level and the project background information can be used to determine the project nature. Palaneeswaran and Kumaraswamy (2000) recommended the two-stage selection process for highly complex projects.

Then, project owners select an appropriate tendering option for inviting proposals from contractors. Tendering options can be primarily categorised as competitive and negotiation. Competitive tendering includes open tendering, prequalification and selective tendering and negotiation method involves direct negotiations. The selection of tendering method highly depends on suitability to selected project delivery method, project risk level, and market competition. Tendering time and cost involved also impact the selection of tendering method. Thus, if there are less than five potential contractors in the market, then spending extensively on prequalifying is not economical. Prequalification is typically considered where a competitive market exists. Otherwise, a contractor can be selected in a single-stage evaluation using selected tendering or negotiation (sole-source) methods. Private owners who have long-term relationships with well-established contractors often use sole-source selection.

It is essential to assess the procurement strategy objectives that best suit client needs, such as value for money, risk management, and environmental and social objectives. Three major procurement strategies are low-bid, best-value, and best-qualifications. Low-bid selection is solely based on price whereas best-qualification selection considers the best qualified tender. With the inherent disadvantages in both methods, it is less likely that the best contractor is selected for complex projects. However, the best-value procurement is defined as the selection process where price and other key factors are considered in the evaluation to minimise the impacts and enhance the long-term performance and value of construction. It consists of parameters, evaluation criteria, rating systems

and award algorithms which are the essential concepts of contractor selection. Best-value determines the optimum combination of price and non-price criteria (Molenaar & Johnson, 2003). Molenaar and Johnson (2003) further reported that the two-stage process is relatively superior in terms of cost and schedule growth to the one- stage process.

Next step is determining criteria for tender evaluation. These criteria in addition to the tender guidelines need to be included in the tender documents. It is essential to identify both price and non-price criteria, which are mutually independent and measurable in common scale.

After tendering and tender evaluation process are defined (in the pre-contracting phase), the client seeks qualification proposals (Request for Qualification) from potential contractors and assesses the information to prequalify the contractors in stage 1. The prequalification process shortlists the potential contractors who will be then invited to submit a tender proposal (Request for Tender). Competitive tendering among prequalified tenderers that provides an opportunity for more qualified contractors to bid in competitive environment is adopted in the new framework. If the proposals are requested from the prequalified contractors, the number of invitees is typically from three to six.

On receipt of proposals, a tender evaluation panel which consists of several decision makers, commences the evaluation of tenders according to the predetermined criteria in stage 2. In any tendering method, each tender is initially examined to determine a complying tender which satisfies all the requirements of the 'Conditions of Tendering' and to determine a conforming tender that conforms in all aspects to the requirements of the tender documents. As such, preliminary examination ensures the tender compliance with the tender instructions (e.g. submitted on or before nominated closing time), and confirms completeness of entries, eligibility of tenderers (e.g. legal entity), tenderer qualifications, and absence of non-conformities (Twort & Rees, 2004). The successful tenders (complying tender and confirming tenders) proceed to the comprehensive tender evaluation stage where the best tender is determined.

Tender evaluation aims to identify the tenderer offering with the best value by assessing the contractor potential on each criterion and finally arrives at the best value of each tender by combining the contractor performance scores with the

criteria weightings. After ranking the tenderers based on the best value, the top ranking tenderer is selected as the optimum choice of contractor. However, if none of the tenders represent sufficient value for money after the competitive tendering process, then negotiations can be done with the most acceptable tender based on the tender evaluation results.

5.2 PROPOSED EPC CONTRACTOR SELECTION FRAMEWORK

Depending on project nature, either a single or two-stage contractor selection process can be used to select the optimum contractor. Fig. 5.1 presents the design-build contractor selection framework proposed by Palaneeswaran and Kumaraswamy (2000). It has been adopted as the base for developing a new EPC contractor selection framework. As the EPC projects are extensively large with increased complexity and high risk levels, the new contractor selection framework thus acquires a two-stage process, consisting of prequalification and bid evaluation for successful project outcome.

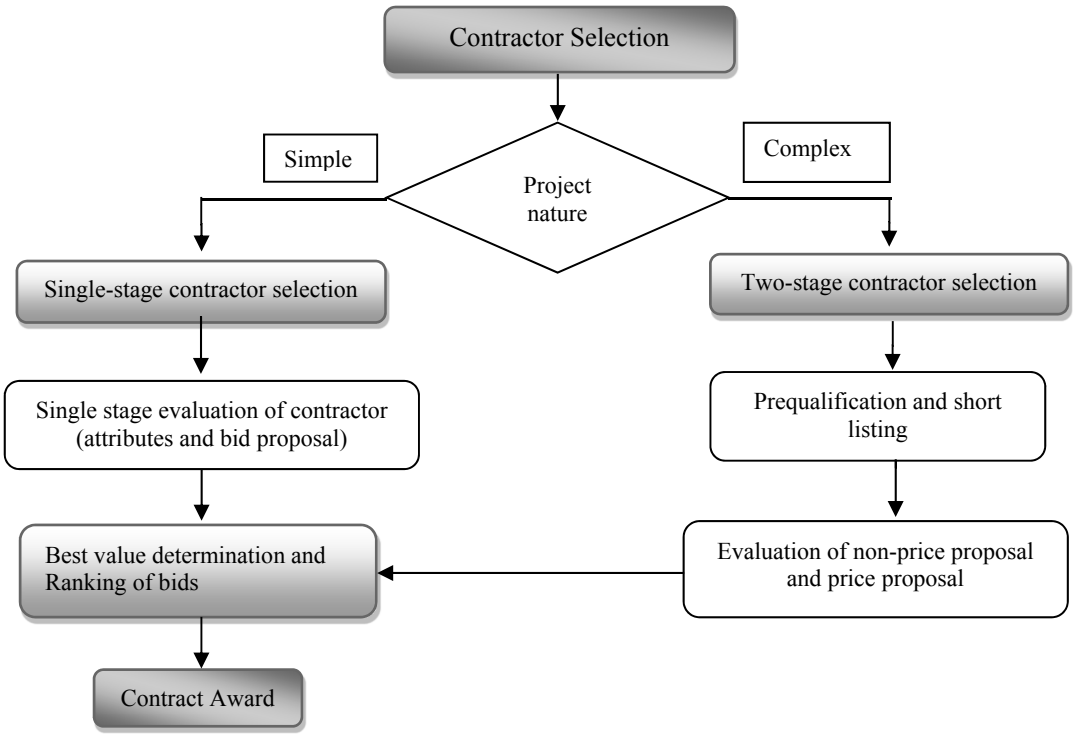


Figure 5.1 DB Contractor selection framework

Source: Palaneeswaran and Kumaraswamy (2000)

The prequalification process is used to short-list potential contractors who will be invited for submission of proposals. Among various options that project owners can select for inviting proposals, competitive bidding, which provides an opportunity for more contractors to bid in a competitive environment, is adopted in the new framework.

Though competitive bidding can be used to find the best-value or the lowest tender, traditional lowest bid selection solely based on price is regarded as the main causes of project delivery problems and it is less likely that the best performing contractor is selected. Best-value procurement emphasises quality, efficiency/effectiveness, value for money and performance standard. The best-value model, which investigates the past records of the contractor as an indicator of the contractor performance trends by evaluation of different criteria simultaneously to obtain optimum outcome, is more appropriate for EPC projects. Thus, the best value procurement considers price and other key factors in evaluation of the long-term performance of the contractor, and consists of parameters, evaluation criteria, a rating system and award algorithms which are the essential concept of contractor selection, that are adopted for the new framework.

Moreover, the EPC contractor selection framework should clearly indicate the owner's view of the importance of technical elements as well as price; as such, the criteria should align with the owner's business objectives towards cost, time, quality, safety, sustainability, local communities, etc. Therefore, appropriate criteria for EPC contractor selection and their priorities need to be further investigated.

Many researchers have developed tender evaluation models with varying degrees of success in different project environments to evaluate criteria. Palaneeswaran and Kumaraswamy (2000)'s two-stage tender evaluation model for design and build projects has used a traditional method in which a non-price proposal is firstly evaluated, and secondly, the price proposal. Then the technical evaluation score and commercial evaluation score are combined according to the pre-determined proportions of technical to commercial (e.g. 60% technical and 40% commercial) to calculate the final score (best value). Tenders that provide the best value (highest score) will be regarded as the most competitive bidder.

However, this research does not weight price score to non-price score as Palaneeswaran and Kumaraswamy (2000)'s model does. Instead, it considers price criteria and non-price criteria together with weighting based on relative importance for more objective evaluation. Palaneeswaran and Kumaraswamy (2000)'s two-stage tender evaluation model was modified to incorporate the above inputs.

Contractor selection (tender evaluation) is a complex decision-making process (Plebankiewicz, 2012; Singh & Tiong, 2005), which involves diverse criteria and multiple decision makers with a number of available options (contractors). This critical and important undertaking is performed by the project owners or their agents. The construction industry uses various contractor selection methodologies in different project environments to estimate the value of contractors using various contractor selection criteria.

However, a decision problem becomes complex and difficult with the existence of multiple criteria, multiple decision makers, uncertainty and risk associated with, incomplete information, imprecise data and vagueness in decision-making; this creates fuzziness (Singh & Tiong, 2005). Increased project complexity and higher requirements have demanded the use of multi-criteria decision-making methods for contractor selection (San Cristóbal, 2012). Multi-criteria methods based on fuzzy sets are capable of addressing the fuzziness in decision making.

To address the above problems and to deal with the complexity within the contractor selection problem, the proposed tender evaluation model is developed on the basis of two approaches (theoretical foundations) in principally.

a) Multi-criteria approach: Multi-Attribute Analysis to address the multi-criteria nature of decision making problems

b) Fuzzy approach: Fuzzy Set Theory- to address uncertainty and risk associated with, incomplete information, imprecise data and vagueness in decision making

The proposed EPC contractor selection framework is given in Fig. 5.2.

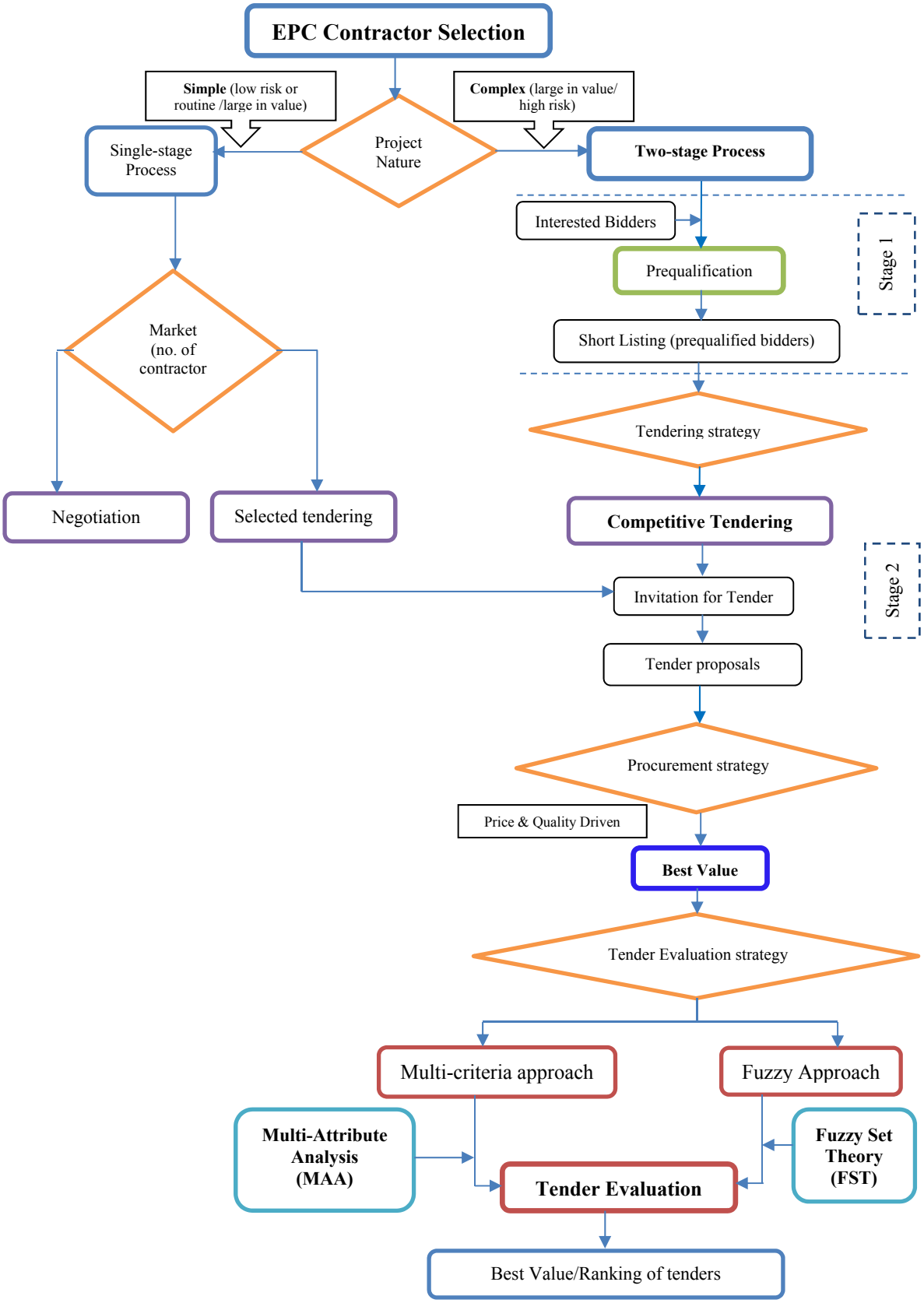

EPC Contractor Selection
Simple (low risk or routine /large in value)
Complex (large in value/ high risk)
Project Nature
Single-stage Process
Two-stage Process
Interested Bidders
Prequalification
Short Listing (prequalified bidders)
Stage 1
Market (no. of contractor
Tendering strategy
Negotiation
Selected tendering
Competitive Tendering
Invitation for Tender
Stage 2
Tender proposals
Procurement strategy
Price & Quality Driven
Best Value
Tender Evaluation strategy
Multi-criteria approach
Fuzzy Approach
Multi-Attribute Analysis (MAA)
Fuzzy Set Theory (FST)
Tender Evaluation
Best Value/Ranking of tenders

Figure 5.2 Proposed EPC contractor selection framework

5.3 SUMMARY

In this chapter, a well-structured EPC contractor selection framework, which can fulfil the demands from the EPC industry, was developed with core focus to procurement strategy, stages of achieving best value, tendering method and more importantly objective evaluation of tenders to achieve owners value for money. A new EPC contractor selection framework acquires a two-stage selection process as prequalification and bid evaluation can address the high complexity and increased risk levels on two occasions. In response to this multi-criteria decision-making problem, the framework adopts Multi-Attribute Analysis (MAA) to estimate the best value using various contractor selection criteria. Fuzzy set theory has been adopted to capture this uncertainty, aroused from the project complexity and associated risks with incomplete information, imprecise data and vagueness in decision making.

Chapter 6: EPC Tender Evaluation Criteria

This chapter investigates tender evaluation criteria for EPC projects. It starts with an introduction to selection criteria and importance weightings (section 6.1); identifying and prioritising the criteria for EPC contractor selection through three rounds of Delphi survey (section 6.2) and finally providing summary and implications of the Delphi survey findings (section 6.3).

6.1 IDENTIFICATION OF CONTRACTOR SELECTION CRITERIA AND CRITERIA WEIGHTINGS

6.1.1 Classification of Contractor Selection Criteria

Selection criteria (or evaluation criteria) are the measures used by the decision makers for selecting the most appropriate contractor in response to an invitation to offer (request for proposal). Prior to inviting suitable bidders, it is necessary to clarify and predetermine appropriate selection criteria, organise the assessment of criteria, and develop methods for evaluating in the prequalification and bid evaluation stages of procurement process (Hatush & Skitmore, 1997b).

There is no single system for setting of selection criteria. The selection criteria are generally categorised into three main categories such as (Municipal Association of Victoria, 2013):

1. Conformance requirements (or submittal requirements):

   Include requirements to submit tender documentation (e.g. schedules, statements of conformance, etc.) and certification as a part of the tender invitation

2. Mandatory requirements:

   Include confirmation of insurance policies, compliance with occupational health and safety standards, provision of financial information; quality assurance (systems/accreditation), management systems, environmental sustainability

3. Scored selection criteria (which are scored and weighted):

Included factors covering the resources of, and ability of, the tenderer to fulfil the contract, together with the tender price

This research focuses on identifying and prioritising scored selection criteria, which are referred as 'contractor selection criteria' hereafter.

6.2 IDENTIFYING CRITERIA FOR EPC CONTRACTOR SELECTION

As outlined in Chapter 3, methods employed for selection of important criteria for EPC contractor selection and derivation of criteria importance weights are given in Table 6.1.

Table 6.1 Methods achieving objectives

Objective	**Method**
Potential set of criteria (16 criteria)	Literature review
Completeness of potential criteria	Pilot study
Important criteria	Three rounds of Delphi questionnaire survey
Estimating criteria weights	Third round of Delphi Questionnaire survey

A comprehensive literature review was first conducted to identify the potential EPC contractor selection criteria in the literature. Comprehensiveness and appropriateness of these criteria for EPC contractor selection were confirmed using a pilot study. Then, three-rounds of Delphi questionnaire surveys were conducted with 64 experts in the Australian construction industry to identify and prioritise these criteria specific to the EPC projects. These 64 experts were experienced industry practitioners from public and private organisations, representing client organisations, consultancy firms and contractors working on projects in various construction sectors.

6.2.1 EPC contractor selection criteria

EPC contractor selection criteria have rarely been investigated and the existing literature on EPC contractor selection criteria is very limited. However, the research finding of industry studies that were conducted with participants from North America, Europe, and Asia from 1995 to 2005 revealed 18 contractor selection criteria from owner's perspective and criteria and their respective ranking, which are shown in Table 6.2 (Transmar Consult Inc., 2006).

Table 6.2 EPC contractor selection criteria importance - owners' perspective

		Ranking				
	Criteria	**2005**	**2003**	**2001**	**1997**	**1995**
1	Quality of key personnel	1	3	1	1	1
2	Project Management Capability	2	1	2	2	2
3	Construction capability	3	6	8	5	7
4	Detailed engineering capability	4	5	8	10	5
5	Contractor's price	5	2	3	3	6
6	Experience with similar work	6	7	5	7	8
7	Project control systems	7	4	4	8	3
8	Experience with same geographic area	8	8	7	9	10
9	Procurement capability	9	15	14	12	11
10	Quality of proposal	10	12	9	13	12
11	Responsiveness and flexibility	11	9	10	6	4
12	Size and location of office	12	11	13	11	13
13	Ability to do work in one office	13	10	12	14	14
14	Total man-hour estimates	14	14	17	17	16
15	Conceptual engineering capability	15	17	16	16	14
16	Capability of sales representatives	16	16	15	16	15
17	Start-up, training capability	17	18	18	18	16
18	Quality of senior management	18	13	11	4	9

As there are hardly any studies on EPC contractor selection criteria, it is essential to identify general contractor selection criteria initially from the literature review of scholarly publications.

6.2.2 General contractor selection criteria

Contractor selection criteria should be capable of identifying optimum choice in respect of the objectives and suitable for the multi-criteria/multi-alternative nature of contractor selection. Abdelrahman et al. (2008) highlighted that the right choice of the evaluation factors and their relevant weights is the core of a successful best value procurement.

The researchers have extensively investigated and continued to identify appropriate contractor selection criteria with the rapid changes in project procurement laws, increased complexity of projects and client needs. However, criteria has not been greatly changed over the years, and tender price, contractor

past performance and performance potential are always accounted for (Singh & Tiong, 2005).

Numerous criteria are taken into account in tender evaluation (Plebankiewicz, 2012). Hatush and Skitmore (1997a) identified the criteria for tender evaluation of prequalified contractors as bid price, and others that include quality assurance, existing workload, experience on projects, experience of working with the owner, financial stability, local knowledge, and responsible attitude towards the work. These criteria contribute in different degrees to the project success factors which include cost, time, and quality (Walraven & de Vries, 2009). The selection of criteria should be done without any bias. Therefore, owners must consider what is 'valuable', not just 'important' or 'required', in the selection process (Abdelrahman et al., 2008). Abdelrahman et al. (2008) indicated that cost, time, qualification performance, quality, and design alternatives are the primary criteria. Historically, cost is the most important factor considered by clients (Abdelrahman et al., 2008; Hatush & Skitmore, 1997b).

Watt et al. (2010) explained that no individual criteria or group of criteria are constantly being considered as more important than others and are often varied according to the purchasing situations. Table 6.3 presents the multitude of criteria considered by researchers in diverse project environments in terms of delivery method, construction sector, organisation type, and geographical location.

Table 6.3 General contractor selection criteria

	Author	**Criteria**	**Remarks**
1.	Oltean-Dumbrava, Watts, and Miah (2014)	1. Price 2. Experience 3. Technical approach 4. Management approach 5. Qualification 6. Schedule 7. Past performance 8. Financial stability 9. Responsiveness to the RFP 10. Legal status	DB contractor selection

	Author	Criteria	Remarks
2.	Alzahrani and Emsley (2013a)	1. Financial 2. Management 3. Technical 4. Past experience 5. Past performance 6. Organisation 7. Environmental 8. Health and safety 9. Quality 10. Resources	DB contractor selection
3.	Enshassi et al. (2013)	1. Financial 2. Completeness of bid document 3. Past performance in similar projects 4. Staff skills and experience 5. Contractor 's reputation/image 6. Quality of work 7. Contractor site management/execution 8. Bid understanding 9. Plant and equipment resources 10. Health and safety performance	DB contractor selection
4.	San Cristóbal (2012)	1. Cost 2. Completion time 3. Safety (as a measure of company reputation) 4. Technical capability (project specific) 5. Management capability (project specific) 6. Experience in similar jobs (level of expertise) 7. Financial status	DB contractor selection
5.	Watt et al. (2010)	1. Relevant experience 2. Track record (previous projects) 3. Quality 4. Expertise 5. Capability 6. Cost 7. Safety record 8. Capacity 9. Tender Sum	DB contractor selection
6.	Holt (2010)	1. Contractor's organisation 2. Financial considerations 3. Management resources 4. Past experience 5. Past performance	DB contractor selection

	Author	Criteria	Remarks
7.	Darvish et al. (2009)	1. Work experience 2. Technology and equipment 3. Management 4. Experience and knowledge of the operation team 5. Financial stability 6. Quality 7. Familiarity with the area (e.g. domestic) 8. Reputation 9. Creativity and innovation	DB contractor selection
8.	Watt, Kayis, and Willey (2009)	1. Workload/Capacity 2. Financial position 3. Health safety environment 4. Key personnel 5. Location 6. Project management expertise 7. Organisational experience 8. Past project performance 9. Company standing (reputation) 10. Tendered price 11. Proposal 12. Quality control 13. Client-supplier relations 14. Technical expertise 15. Method/Technical Solution	DB contractor selection
9.	Gransberg and Barton (2007)	1. Price 2. Technical 3. Qualifications 4. Schedule 5. Project management	DB contractor selection
10.	Singh and Tiong (2005)	1. Tender price 2. Past performance: financial soundness 3. Managerial capability: resources, current workloads 4. Quality: technical competence, past experience, project specific criteria 5. Health and safety aspects	DB contractor selection
11.	Mahdi et al. (2002)	1. Current capabilities (Contractor capacity, management/adaptability/co-ordination, current resources/workloads 2. Work strategy 3. Plans and method statements in terms of cash flow, manpower schedule, procurement schedule, equipment schedule, quality assurance and control plan, safety plan, organisational structure/qualifications of the staff, type of work sub-contracted	DB contractor selection

	Author	Criteria	Remarks
12.	Palaneeswaran and Kumaraswamy (2000)	1. Finance 2. Human resources 3. Organisation and management 4. Project specific requirements 5. Past experience 6. Past performance 7. Technology 8. Quality system 9. Health and safety system 10. Equipment	DB contractor selection
13.	Hatush and Skitmore (1997a)	1. Financial soundness 2. Technical ability 3. Management capability 4. Health and safety 5. Reputation 6. Past failures, length of time in business, past owner/contractor relationship, other relationships	General contractor selection
14.	Plebankiewicz (2012)	1. Experience 2. Financial situation 3. Personnel available 4. Equipment 5. Reputation	General contractor selection
15.	Abdelrahman et al. (2008)	1. Cost: Capital cost and life cycle cost 2. Time 3. Qualification and performance 4. Design alternatives 5. Technical proposal responsiveness and environmental considerations	General contractor selection
16.	Ahola et al. (2008)	1. Price 2. Experience from previous projects 3. Investments in product development 4. People involved by the supplier 5. Sub suppliers used by the supplier 6. Financial situation 7. Client's preferences 8. Risk carrying ability and willingness 7. Strategic factors	General contractor selection
17.	Fong and Choi (2000)	1. Tender price 2. Financial capability 3. Past performance 4. Past experience 5. Resources 6. Current workload 7. Past relationship 8. Safety performance	General contractor selection

	Author	Criteria	Remarks
18.	Holt et al. (1994a)	1. Organisational 2. Financial stability 3. Management 4. Past experience 5. Past performance 6. Project specific 7. Other specific 8. Current workload, prior relationship, office location	General contractor selection
19.	Holt, Olomolaiye, and Harris (1994b)	1. Contractor's organisation 2. Financial consideration 3. Management resource 4. Past experience 5. Past performance	General contractor selection
20.	Waara and Bröchner (2006)	1. Quality 2. Cost 3. Technical 4. Environmental impact 5. Operation, maintenance, technical support services 6. Project duration 7. Contractor capability & skills, past experience, past performance 8. Construction methods 9. Financial capacity 10. Health and safety 11. Conformity with bidding documents	General contractor selection

As shown in Table 6.3, various criteria have been proposed. Nevertheless, to facilitate the contractor selection process, the number of criteria should be kept as low as consistent with making a well-founded decision; the more typical arrangement is from six to twenty (Dodgson et al., 2009). Given the large number of criteria identified, it will be helpful to group them into a series of themes that relate to separate and distinguishable components of the overall objectives for the decision. The main reasons for grouping criteria are (Dodgson et al., 2009):

a) To help check whether the set of criteria selected is appropriate to the problem

b) To ease the process of calculating criteria weights in MAA applications

c) To facilitate the emergence of higher level views of the issues for example trade-offs between key objectives

The above criteria (in Table 6.3) were grouped reasonably into 18 substantial criteria as shown in Table 6. 4. The frequencies of every criterion are shown in Table 6.4.

Table 6.4 Substantial criteria for contractor selection

	Criteria	**Similar terms/Related focuses**
1	Financial Stability	Financial status/financial standing/financial soundness/ financial considerations/ financial position
2	Management Capability	Management approach/management resources/ project management (PM) expertise / management skills/ management systems/ PM organisation/management personnel/management knowledge/corporate stability
3	Experience-company	Past experience/experience in similar projects/relevant experience/work experience
4	Experience-personnel	Experience from previous projects/experience of technical personnel/key persons' years with company
5	Performance	Past performance/past project performance/track record (previous projects)/ scale and type of projects
6	Capacity	Current workload/ availability for maintenance and operation support
7	Technical	Technical approach/ /technical expertise/ technical solution/ technology and equipment /design alternatives/ technical proposal/ responsiveness/ Fitness for purpose/technical issues/ methodology/Technical ability/technical personnel /creativity and innovation /technical competence
8	Quality	Quality control/compliance with specifications and quality standards/attitude towards correcting faults or incomplete works/quality systems
9	Resources	Human resources/ management resources//plant and equipment/ personnel (key persons) and ability/knowledge of the key persons
10	Organisational	Organisation/organisational experience
11	Health, Safety & Environment	Safety record/environment/safety performance/health and safety system/health and safety policy/risk carrying ability and willingness /risk exposure
12	Relationships	Company standing/past failures/length of time in

		business/owner contractor relationships/relationships with clients/subcontractors /suppliers/company image client preferences, reputation
13	Qualifications	Professional qualifications/qualification of key persons
14	Time	Schedule/completion time/ early completion dates
15	Project understanding	Responsiveness to RFP/ Project understanding/ approach/ proposal technical/ proposal responsiveness/ project specific criteria/ Fitness for purpose/project specific requirements/ appreciation of the work
16	Legal status	Litigation tendency/attitude towards claims and counter claims
17	Location	Office location/familiarity of local environment/familiarity with the area (domestic)/ company proximity to project
18	Cost	Price/tendered price

Therefore, from the substantial list of criteria in Table 6.5, in which 20 publications were cited, the 16 potential criteria were identified out of 18 by rephrasing a few. Legal status, which gained least number of citations (<15%), was rephrased as contract and legal. Company experience was considered as 'past experience' of the contractor while experience of personnel was included under 'key personnel' as well as qualifications of contractor personnel. Capacity was included under 'organisational'. With such changes, a final list of 16 criteria were identified for likely inclusion in the round 1 questionnaire survey.

Table 6.5 Criteria usage frequency

Criteria	Oltean-Dumbrava et al. (2014)	Alzahrani and Emsley (2013a)	Enshassi et al. (2013)	San Cristóbal (2012)	Plebankiewicz (2012)	Watt et al. (2010)	G. Holt (2010)	Watt et al. (2009)	Darvish et al. (2009)	Abdelrahman et al. (2008)	Ahola et al. (2008)	Gransberg and Barton (2007)	Waara and Bröchner (2006)	Singh and Tiong (2005)	Mahdi et al. (2002)	Palaneeswaran and Kumaraswamy (2000)	Fong and Choi (2000)	Hatush and Skitmore (1997a)	Holt et al. (1994a)	Holt et al. (1994b)	% (out of 20 citations)	**Rank based on % of citation**
Management Capability	✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	85	1
Financial Stability	✓	✓	✓	✓	✓		✓	✓	✓		✓		✓	✓		✓	✓	✓	✓	✓	80	2
Experience-company	✓	✓		✓	✓	✓	✓	✓	✓		✓		✓	✓		✓	✓	✓	✓	✓	80	2
Performance	✓	✓	✓			✓	✓	✓		✓			✓	✓		✓	✓	✓	✓	✓	70	3
Technical	✓	✓		✓	✓	✓		✓	✓	✓		✓		✓	✓	✓		✓			70	3
Health Safety Environment		✓	✓	✓		✓		✓		✓	✓		✓	✓	✓	✓	✓	✓	✓		70	3
Quality		✓	✓			✓		✓	✓	✓			✓	✓	✓	✓		✓	✓		60	4
Cost	✓			✓		✓		✓		✓	✓	✓		✓	✓		✓				55	5
Resources		✓	✓				✓		✓				✓		✓	✓	✓	✓	✓		50	6
Organisational		✓	✓				✓	✓					✓		✓	✓			✓	✓	45	7
Relationships				✓	✓			✓	✓		✓			✓			✓	✓	✓		45	7
Qualifications	✓		✓		✓					✓		✓	✓						✓		40	8
Project understanding	✓		✓					✓		✓			✓	✓	✓	✓					40	8
Experience-personnel					✓				✓	✓	✓		✓					✓	✓		35	9
Capacity						✓		✓					✓	✓	✓		✓		✓		35	9
Time	✓			✓						✓		✓	✓								25	10
Location								✓	✓										✓		15	11
Legal status	✓																		✓		10	12

6.2.3 Pilot study

Before finalising the criteria for the round 1 Delphi questionnaire survey, the provisional set of criteria were assessed against a range of qualities using a pilot study to ensure:

a) Completeness: all the criteria that are necessary to compare the contractors' performance have been included and the criteria can capture all the key aspects of the objectives.

b) Redundancy: the criteria that are relatively unimportant or any duplicates have been removed and to omit criteria, it seems all contractors are likely to achieve the same level of performance economising analysis.

c) Operationality: each contractor can be judged against each criterion on a common scale of measurement reflecting objective assessment as well reflecting the subjective assessment of an expert.

d) Mutual independence of preferences: preference scores could be assigned for the contractors on one criterion without knowing what the contractors' preference scores are on any other criteria as a requirement for MAA implementation in the proposed contractor selection model. Then, to model the mutually not independent criteria by combining two criteria into a single criterion provided that new criterion's preference is independent of the remaining criteria.

e) Double counting: To check double counting because double-counted effects are likely to give more weights in the final overall decision than they deserve in MAA applications.

f) Size: the list is no longer than it needs as an excessive number of criteria can lead to extra analytical effort in assessing input data and can make the analysis more difficult.

The pilot study was conducted with four industry professionals from various disciplines with the majority from the oil and gas sector. Pilot study results are given in Table 6.6.

Table 6.6 Pilot study results of questionnaire 1

Pilot study participant	Comments
Team leader	Suggested to include 'Transport and Roads' as a separate sector in construction
Contract Engineer	No comments on the contents/corrected wordings/helped in rephrasing
Project Manager	"Survey reads well"
Project Engineer	"Looks fine"

Then the questionnaire for round 1 of the Delphi survey was developed incorporating the following 16 criteria after testing for completeness, and appropriateness using the above factors during the pilot study. These criteria formed the basis of round 1 questionnaire composition (Table 6.7).

Table 6.7 Potential criteria for EPC contractor selection for round 1questionnaire

#	Criteria
1	**Financial:** Provides details of financial capability in terms of financial statements
2	**Past Performance:** Demonstrates the performance of recently completed projects in terms of cost, time and qualities, and cooperative behaviour (conflicts/disputes)
3	**Past experience:** Provides the details of scale and type of past projects, and demonstrates the experience in similar projects, in the region, and familiarity of relevant project delivery method
4	**Technical:** Demonstrates the technical capability and capacity that include technical solution, alternative designs, expertise, specialisation, technical qualification, staffing levels, technology and equipment resources, engineering systems, creativity and innovation, and availability for operation, maintenance, repair and training needs
5	**Management:** Demonstrates the business management system that includes project management system, risk carrying ability and willingness, management personnel, and management accountabilities

6	**Organisational:** Provides the details of company size, company image, age in business, organisational structure, policies, memberships, current workload, and resources (labour, plant, equipment, human resources)
7	**Health and Safety:** Outlines the accountabilities for Occupational Health and Safety (OHS) with plans and systems and demonstrates the performance with OHS records
8	**Environment and Sustainability:** Outlines the Environmental Management Plan and commitment to sustainability
9	**Key personnel:** Provides the details of key personnel to be employed, proposed roles, their experience and skills, academic and professional qualification, years with the company, and their training
10	**Relationships:** Provides the details of subcontractors/suppliers that include the length of time with them, labour employment agreement, and maintenance of workers' compensation liabilities
11	**Time:** Provides a program indicating start and finish dates, and adherence to the dates/duration given in tender documents
12	**Cost:** Includes tendered price, and assessment of capital cost, life cycle cost, etc.
13	**Quality:** Outlines quality control and quality assurance systems, and compliance with specifications and quality standards
14	**Contractual and legal status:** Demonstrates the disputes and resolution strategy, attitude towards claims, acceptance of contract terms and conditions, and compliance with the codes
15	**Project understanding:** Responds to Request for Proposal (RFP), and demonstrates project specific criteria
16	**Geographic location:** Outlines the familiarity of local environment, and proximity to project

6.2.4 Delphi study

Round 1 of Delphi Questionnaire Survey

The first round of Delphi questionnaire was comprised of two sections. The first collected the background information of respondents, while the second defined the potential EPC tender evaluation criteria and asked the respondents to review each to identify the important criteria for EPC contractor selection. The respondents were also given the opportunity to suggest new criteria for inclusion in the next round. Round 1 Questionnaire is given in Appendix A.

Round 1 questionnaire survey was initially distributed to 272 potential participants who met at least a single requirement from the below list were selected as participants.

- Professional registration in an engineering or project management body (e.g. Engineers Australia)
- Known participation in engineering or construction project management activities particularly within EPC projects (e.g. personal contacts, networking, LinkedIn members)
- Known participation in similar research activities
- A faculty member of construction project management discipline at an institute for higher education (i.e. university)

The experts were not prequalified initially but first part of the round 1 questionnaire was intended to collect the participants' profile data by which the participants' expertise were to be verified. Then, the respondents who have over five years of EPC experience, were identified from round 1 as experts and selected to participate in round 2 of the Delphi questionnaire survey.

A total of 64 respondents replied, representing 24% response rate for the round 1 Delphi questionnaire survey. Table 6.8 shows that the response percentage has increased when the reminders were sent.

Table 6.8 Round 1 questionnaire distribution schedule and response rate

	Schedule	**Respondent (cumulative number)**	**% increase**
Invitation	20-Mar-2015	272	N/A

First Reminder	31-Mar-2015	40	48.1%
Second Reminder	10-Apr-2015	51	27.5%
Final Reminder	20-Apr-2015	64	25.5%
Close out	30-Apr-2015	64	0.0%

Quantitative data analysis of round 1 data

Graphical methods (e.g. bar charts) and statistical techniques (descriptive statistics) were used to describe and present the quantitative data using Excel and SPSS. Qualitative data from two open-ended questions and the comments were analysed using NVivo and researcher's analytical skills. The indicator that was used to describe the quantitative data is frequency.

Suggested new criteria and other important factors pertinent to contractor selection by the respondents will be presented after analysing the qualitative data (string data).

Part 1: Respondents profile

Of the 64 respondents, the majority (75.4%) of them were from private sector organisations, followed by those from public sectors (20%) and others. Table 6.9 presents the respondents' representation by sector.

These public and private sector participants were from various locations in Australia, with 52% from Queensland (this research was conducted in QLD mainly), followed by New South Wales (12%) and Western Australia (10%). Queensland, NSW and WA are the leading states for engineering constructions, which are predominantly delivered by EPC method. Fig. 6.1 shows the respondents' locale. Some of the participants were from national or even international companies, which have their business at multiple locations across Australia. This geological diversity is also important to ensure that the participants have encountered the challenges that are location specific when assessing EPC contractor performance requirements.

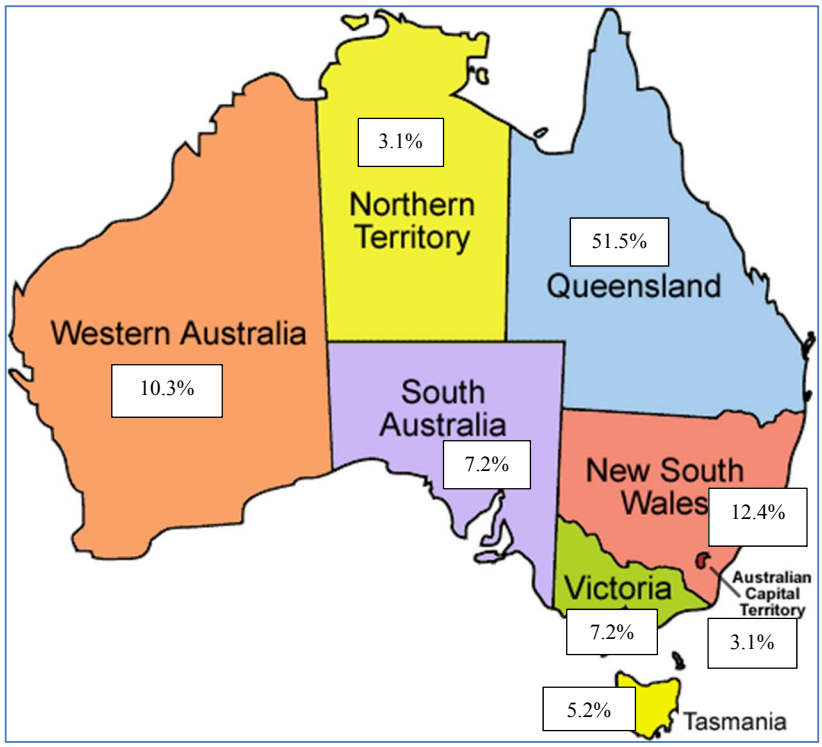


Figure 6.1 Business locale of participants' organisations

Figure 6.2 shows that majority of respondents (28) are from consultancy organisations.

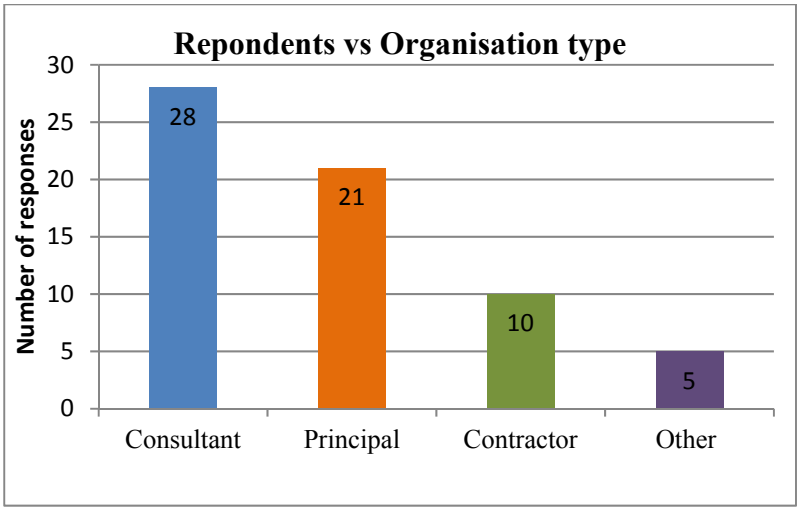


Figure 6.2 Respondents' organisation

Fig. 6.3 shows the respondents' functional roles within a project or the organisation. Twenty-three participants (36%) were project managers and 20 (31%) held other managerial positions. It implies that most of the experts hold decision-making

authority within their roles and most of them are directly involved in contractor selection activity.

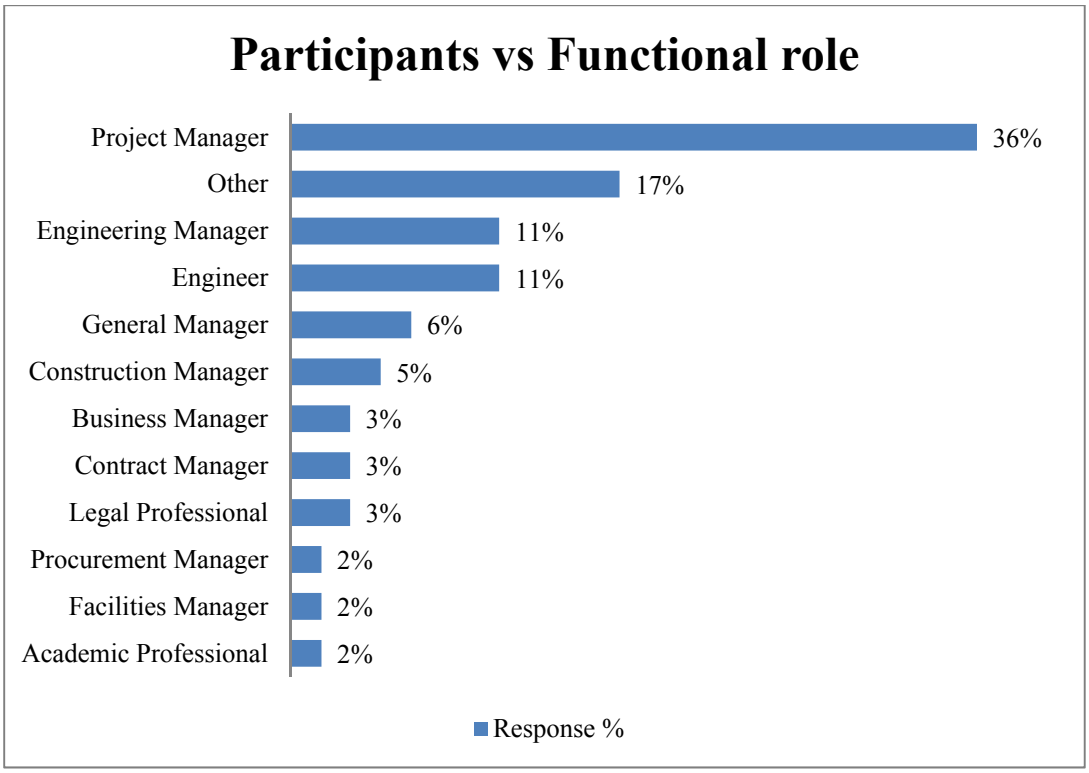


Figure 6.3 Participants' functional role

Of all the participants, most of them are well-experienced in construction industry: over 50% of them have over 20 years of construction industry experience and there were approximately 27% and 9% respondents in between 10-20 years and 5-10 years respectively. Only three respondents have less than 5 years.

The respondents were working in various construction industries such as transport, utilities, telecom, mining, and heavy industrial construction (e.g. oil & gas, chemical, and processing plants, etc.), pipelines and buildings, representing almost all the predominant EPC industries in Australia. As shown in Fig. 6.4, the majority of respondents are from the mining, heavy industrial construction and pipeline industries combined (37%), with 23% of respondents from the transport construction sector. According to EPC market review (Chapter 4), recent growth in EPC industry is mainly from infrastructure projects, in particular, transport projects.

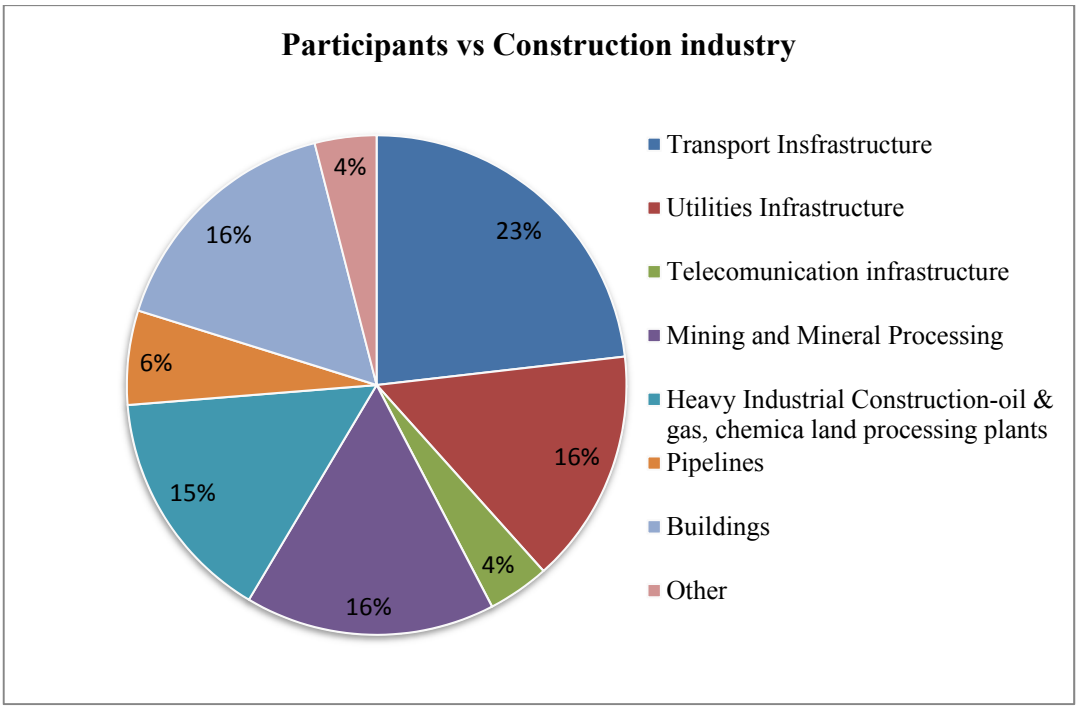


Figure 6.4 Participants' representation by industry

Participants' experience with EPC projects is vitally important. It is apparent that the majority (22) of participants have over 20 years of experience, and there are 37 participants with over 10 years of experience, suggesting that the Delphi panellists in this study are well-experienced in the subject being investigated. The least experienced (0-5 years) participants (13) have been excluded in round 2 of the Delphi questionnaire survey. Fig.6.5 presents the participants' number of years of experience with EPC or DB projects.

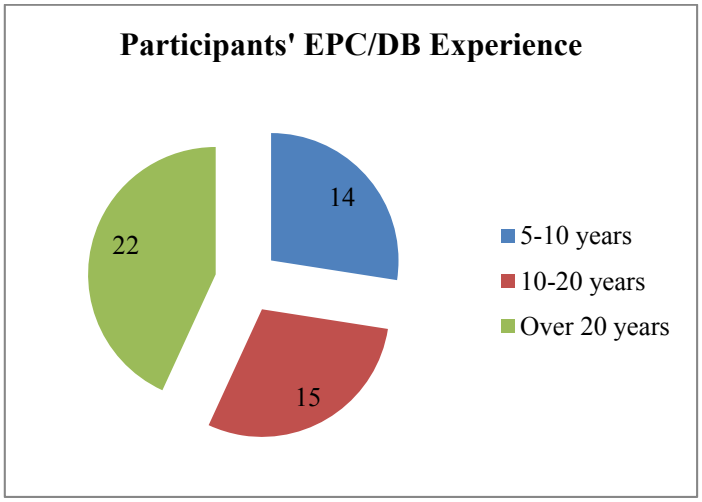


Figure 6.5 Participants' EPC/DC Work experience

A significant number of participants (34.5%) have worked or have been working on billion-dollar EPC projects and almost 14% participants have work experience with very large projects costing over A$5 billion (Fig.6.6). Only a very small proportion (<2%) of participants were not aware of the project cost. Therefore, most of them are familiar with challenges occurred in large and complex projects.

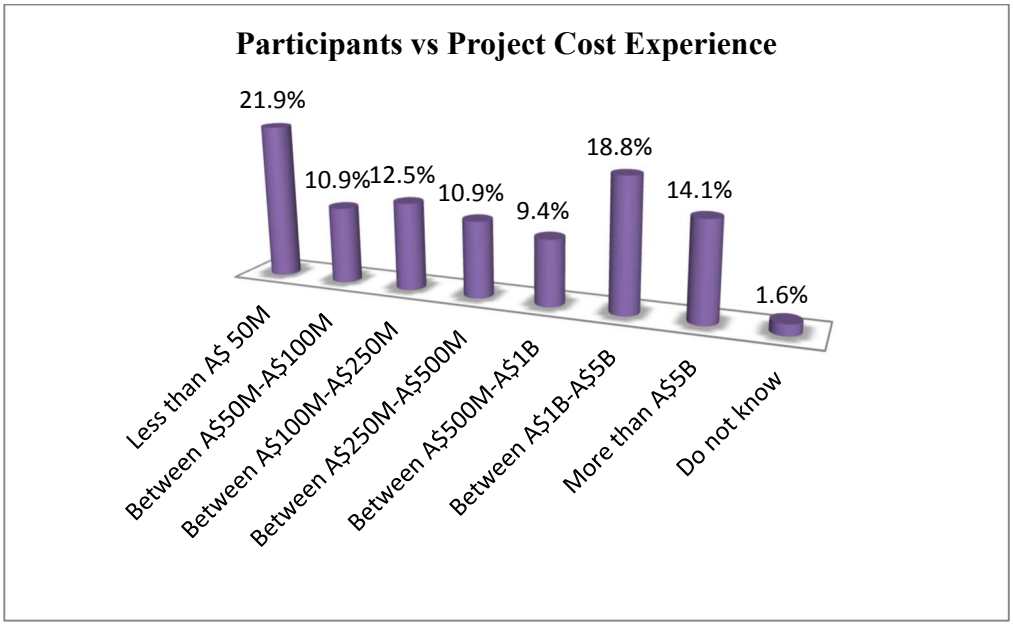


Figure 6.6 Participants' Experience in terms of project cost

Part 2: Identifying criteria for EPC tender evaluation from the experts

In part 2 of the questionnaire, the respondents were asked to select the criteria from the list that they considered to be important for EPC contractor selection. Table 6.9 shows frequency for each criterion, with more than 50% considering all 16 criteria to be important. Past performance, technical, key personnel, past experience, health and safety, financial, cost, and time are identified as the top criteria with a response rate over 90%. *Environment and sustainability* is ranked 13th even though it is broadly discussed in the local and global arena.

Table 6.9 Criteria importance results from round 1

Criteria	Response rate (as a percentage)	Rank
Past Performance	100.0%	**1**
Technical	96.88%	**2**

Key personnel	96.88%	**3**
Past experience	95.31%	**4**
Health and safety	93.75%	**5**
Financial	92.19%	**6**
Cost	92.19%	**7**
Time	90.63%	**8**
Quality	89.06%	**9**
Project understanding	87.50%	**10**
Management	82.81%	**11**
Organisational	79.69%	**12**
Environment and sustainability	75.00%	**13**
Contractual and legal	75.00%	**14**
Relationships	73.44%	**15**
Geographic location	56.25%	**16**

Additionally, public sector and private sector respondents identify some criteria that are deemed important to include in EPC tender evaluation differently. Past performance (100%) is the dominant criteria for both sectors. However, environment and sustainability has contrasting response, as 92% public sector respondents identified it as an important criterion compared with 69% of private sector respondents.

The most experienced (over 20 years of EPC/DC project experience) 32 respondents have identified the most important criteria as financial capability, past performance, health and safety, and key personnel.

Qualitative data analysis - New criteria for EPC contractor selection

In part 2, qualitative data were collected through an open-ended question to identify new criteria to be included in EPC contractor selection. The responses from all 31 respondents who suggested new criteria were taken into consideration. To analyse responses, raw string data were exported to a NVivo software program for qualitative data analysis. Then, the following analytical techniques were used to cluster these criteria to form single new criteria or to map any new criteria reasonably with the existing criteria, including

1. Coding: All the suggested criteria by 31 participants were given an individual node.
2. Cluster Analysis (CA): CA technique was used to cluster the nodes by word similarity (Appendix B). After critical review of CA results, the child nodes, which represent the criteria suggested by each individual and the parent nodes, which present the most appropriate new criteria to which the child nodes can be mapped were identified. As such, a parent node denotes a single suggested criterion derived from the suggestions.
3. In NVivo, parent nodes and child nodes were created, and the child nodes were prudently and judiciously allocated to these common nodes (parent nodes) as in Fig. 6.7.
4. Next step was to match the suggested criteria with the existing criteria manually (Appendix C). Some new criteria were included within the existing criteria because they have been phrased differently but can be fit with the content of existing criterion. Item 37 and 38 in Appendix C refer to non-confirming tenders that were excluded in response rate calculations, given that a non-confirming tender cannot be considered as a criterion and it needs be evaluated the same as a tender. Those not matched with any existing criteria were considered as new criteria.
5. Response frequencies were then calculated as a percentage of total participants (64).

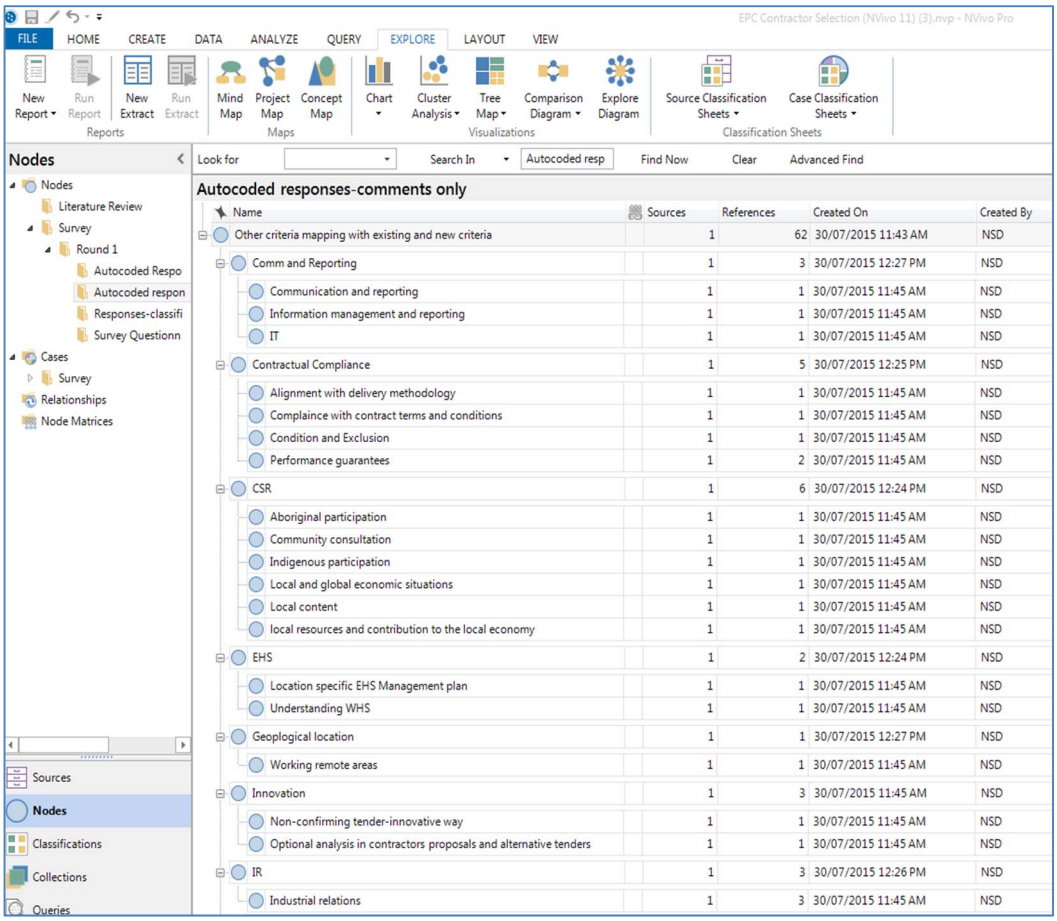


Figure 6.7 NVivo extract for suggested criteria in round

Despite the large number of criteria suggested by the participants, most of them have been already considered within the existing criteria descriptions (Appendix D). Out of these, '**Industrial relations',** which had the maximum response rate (5%) but could not be incorporated into any of existing criteria, was included in the next round of the Delphi questionnaire survey.

Additionally, the participants were asked to comment on each existing criterion. After critical review of comments, the criteria descriptions were rephrased to make them more meaningful. Finally, the respondents were given the opportunity to provide any other important information pertinent to the EPC contractor selection. Respondents highlighted other factors (not criteria) that may affect the selection of a contractor including political acceptability, corruption and fraudulent deals, which hamper the decisions. Also, some respondents' suggestions are useful to be considered when selecting EPC contractors. These include short listing from expression of interest at the earliest stage and having a sensible number of tenderers

(limited), appropriate weighting of the various criteria. Not only that, the respondents empathised that the owners'/clients' understanding and commitment are also important for successful project delivery.

Round 2 of Delphi Questionnaire Survey

A well-rounded questionnaire was developed using the results of the round 1 questionnaire. The questionnaire of round 2 contains all 16 criteria of questionnaire 1, which were refined using the round 1 feedback for each criterion in addition to the new criterion -'Industrial relations'. Each criterion was scrutinised to ensure the completeness and appropriateness, so that the participants could confidently rate each criterion in 1-7 interval scale (see Table 6.10). An additional three questions related to procurement process were also included. Round 2 questionnaire is given in Appendix D.

Table 6.10 Criteria included in round 2

	Criteria
1	**Past Performance:** Demonstrates the performance of recently completed projects with records of project cost, completion time and quality, contract claims and variation history, cooperative behaviour (conflicts/disputes), penalties, etc.
2	**Technical:** Demonstrates technical capability and capacity that includes sound engineering solutions, safety in design, creativity and innovation, constructability, engineering and technical expertise, technology and equipment resources, engineering systems, etc. Demonstrates technical support for commissioning, operation readiness, handover, maintenance, repair and training needs.
3	**Key personnel:** Provides the details of key project personnel which include proposed roles, experience and skills, academic and professional qualification, years with the company, and professional development plan. Demonstrates project team ability to work collaboratively and as a part of diverse teams, and availability for backup resources.
4	**Past experience:** Provides details of scale, complexity, and type of past projects, and demonstrates project experience of similar type(s) in a similar environment.

5	**Health and Safety:** Outlines accountabilities for Occupational Health and Safety (OHS) providing samples of site specific management plans, corporate systems, and procedures that identify and control OHS risks. Provides documentary evidence of corporate OHS performance including OHS records from recent projects.
6	**Financial:** Demonstrates contractor financial viability and financial performance over a defined period and provides financial statements, which include balance sheet, profit and loss statement, etc.
7	**Cost:** Includes tendered price, life-cycle costing, etc.
8	**Time:** Provides a project schedule with milestones, activities and deliverables with intended start and finish dates, or complies with the time constraints given in tender documents.
9	**Quality:** Outlines quality control and quality assurance systems, and complies with specifications and quality standards.
10	**Project understanding:** Demonstrates understanding of Request for Proposal (RFP), local context, project risks, unique owner standards and requirements, how the project can be executed to meet client expectations, explains exceptions from RFP, and outlines expected degree of owner involvement, approvals, etc.
11	**Project management:** Demonstrates Construction Project Management (CPM) capability (risk management strategy, procurement strategy, stakeholder management plan, logistic and supply chain management, preferred suppliers/sub-contractors, and key trade packages, etc.).
12	**Organisational:** Outlines business values and corporate commitment and provides the details of company size, company image, age of business, organisational structure, policies, memberships, current and potential future work commitments, resource optimisation (people, plant, equipment), in-house systems, etc.
13	**Environment and sustainability:** Takes the stakeholders' expectations, which include environmental requirements, social acceptances (e.g. local resources, local economy, indigenous participation,

	etc.), sustainability approach (products and processes) into account.
14	**Contract and legal:** Accepts Contract Terms and Conditions or provides clear, concise exclusions or conditional acceptances. Indicates compliance with all relevant codes and regulations.
15	**Relationships:** Demonstrates ability to develop strong and long term partnerships with clients, vendors and suppliers by providing client/subcontractor/supplier referees including information regarding the duration of the relationship, etc.
16	**Geographical location:** Outlines familiarity of local environment, and proximity to project (i.e. proposed work locale) and/or demonstrates work locations worldwide that can work together.
17	**Industrial relations:** Demonstrates employee and industrial relations plan/policy and maintenance of project agreements, multi-employer agreements, workers' compensation liabilities, etc. and provides recent industrial relations record.

A total of 47 respondents who have over five years of EPC experience, were identified from round 1 as experts and selected to participate in round 2 of the Delphi questionnaire survey.

Quantitative data analysis of round 2 survey

Round 2 respondents were asked to rate the criteria importance using 1-7 scale. To describe the central tendency of data in terms of 'mean', descriptive statistical analysis was performed using SPSS. Table 6.11 shows the results of the analysis.

Table 6.11 Round 2 result of criteria importance

Criteria	Group mean	Standard Deviation	Rank
Past Performance	6.28	1.117	**1**
Technical	6.21	1.122	**2**
Project understanding	6.17	1.167	**3**
Health and safety	6.04	1.103	**4**
Key personnel	5.85	1.398	**5**
Past experience	5.79	0.931	**6**

Contractual and legal	5.79	1.503	**7**
Financial	5.72	1.155	**8**
Cost	5.66	1.307	**9**
Management	5.64	1.223	**10**
Time	5.57	1.598	**11**
Quality	5.47	1.365	**12**
Relationships	5.21	1.587	**13**
Environment and sustainability	5.00	1.460	**14**
Industrial Relations	4.98	1.674	**15**
Organisational	4.85	1.302	**16**
Geographic location	4.57	1.298	**17**

With a mean score above 6 in a 1-7 scale, the top four criteria of past performance, technical, project understanding, and health and safety are identified as 'very important'. The least important criteria include industrial relations, organisational, and geographical location with mean values below 5.

Additionally, the data distribution of each criterion was checked using SPSS. Results show that the data was not a normally distributed data set thus non-parametric statistical techniques can be used for further investigating the data.

Reliability test

Scale reliability of round 2 was tested using Cronbach's Alpha test for the interval scale measurement on 1-7 scale. Results are given in Table 6.12. Cronbach's alpha is 0.914 for the round 2 responses measured 1-7 scale. Cronbach's Alpha is greater than 0.7 in this round and it indicates a high stability of responses in this round.

Table 6.12 Round 2 reliability statistics

Cronbach's Alpha	Cronbach's Alpha based on standardised items	N of items
0.914	0.913	17

Measure of Consensus-Non-Parametric Kendall's W test

Non-Parametric Kendall's W test (Kendall's coefficient of concordance) is used to measure group consensus. W is always between 0 and 1, where 0 indicates no disagreement and 1 indicates perfect agreement. Intermediate values show lesser or greater agreement. Non-Parametric Kendall's W test was done using SPSS and the results are given in Table 6.13.

Table 6.13 Kendall's W test result- test statistics

N	47
Kendall's W (Kendall's Coefficient of Concordance)	0.225
Assymp.Sig.	0.000

Kendall's coefficient of concordance W is 0.225 at p=0.000. In this investigation, value of W is significant with p=0.000. Cheung, Chan, and Kajewski (2010) indicated that if W is significant at a pre-determined allowable significant level (0.05), there is a reasonable degree of consensus amongst the respondents. Their study results show that values of W (0.071 and 0.325) were significant at p=0.008 and p=0.000 respectively. Accordingly, the experts in this investigation show reasonable agreement with each other determining the criteria importance with value of W =0.225 significant at p=0.000.

Round 3 of Delphi Questionnaire Survey

The mean score of each criterion from round 2 has been given as the feedback in the round 3 questionnaire. Round 3 questionnaire is given in Appendix E. Round 3 questionnaire was distributed to respondents who participated in the round 2 survey and they were asked to re-rate the criteria importance using the same scale if they wished to change their minds in light of round 2 group results. A total of 36 respondents participated in round 3 of the Delphi questionnaire survey. The result of this third round were analysed using SPSS.

Quantitative data analysis of round 3 survey

Table 6.14 shows the outcome of the descriptive statistical analysis of round 3 data. Past performance is still the most important criterion, followed by technical solution,

project understanding and key personnel, all with mean values above 6, indicating 'very important' in EPC contractor evaluation.

Table 6.14 Round 3 result (N=36)

Criteria	Group mean	Standard deviation	Rank
Past Performance	6.39	0.728	1
Project understanding	6.33	0.676	2
Technical	6.31	0.786	3
Key personnel	6.19	0.668	4
Health and safety	6.03	0.941	5
Time	5.97	0.774	6
Financial	5.92	0.841	7
Contractual and legal	5.92	0.996	8
Past experience	5.89	0.667	9
Management	5.89	0.704	10
Cost	5.75	0.874	11
Quality	5.72	0.815	12
Relationships	5.56	0.998	13
Industrial Relations	5.28	1.003	14
Environment and sustainability	5.25	0.906	15
Organisational	5.03	1.000	16
Geographic location	4.69	0.889	17

Reliability test:

Scale reliability of round 3 was tested using Cronbach's Alpha test for the interval scale measurement on 1-7 scale. As shown in Table 6.15, Cronbach's alpha is 0.819, indicating stronger stability of responses in round 3,

Table 6.15 Round 3 reliability statistics

Cronbach's Alpha	Cronbach's Alpha based on standardised items	Number of items
0.819	0.811	17

Measure of consensus- Non-Parametric Kendall's W test

Kendall's coefficient of concordance is used to measure group consensus. Non-Parametric Kendall's W test was done using SPSS and results are given in Table 6.16.

Table 6.16 Kendall's W test result- test statistics

N	36
Kendall's W (Kendall's Coefficient of Concordance)	0.285
Assymp.Sig.	0.000

Kendall's coefficient of concordance W is 0.285 and is significant at p=.000. As already mentioned in consensus measurement of round 2 data, if W is significant p=.000, there is reasonable consensus among raters. The experts in round 3 show reasonable agreement with each other determining the criteria importance with value of W =.285 significant at p=0.000. With the increased consensus from round 2 to round 3 indicates greater agreement and increased consensus.

Comparison of round 2 and 3 results

Delphi rounds and the final numbers of participants are given in Table 6.17.

Table 6.17 Delphi participants in each round

Delphi rounds	Invitation	Final participants
1	272	64
2	47	47
3	47	36

It is evident that past performance, technical and project understanding maintain their ranks at 1, 2 and 3 respectively in both round 2 and 3. Table 6.18 gives details of all criteria rankings of both rounds (round 2 and 3) for a comparison.

Table 6.18 Summary of round 2 and 3 survey results

	Round 3 (N=36)		Round 2 (N=47)	
Criteria	***Mean***	***Rank***	***Mean***	***Rank***

Past Performance	6.39	**1**	6.28	**1**
Project understanding	6.33	**2**	6.17	**3**
Technical	6.31	**3**	6.21	**2**
Key personnel	6.19	**4**	5.85	**5**
Health and safety	6.03	**5**	6.04	**4**
Time	5.97	6	5.79	11
Financial	5.92	7	5.57	8
Contractual and legal	5.92	8	5.64	7
Past experience	5.89	9	5.79	6
Management	5.89	10	5.64	10
Cost	5.75	11	5.66	9
Quality	5.72	12	5.47	12
Relationships	5.56	**13**	5.21	**13**
Industrial Relations	5.28	**14**	4.98	**15**
Environment and sustainability	5.25	**15**	5.0	**14**
Organisational	5.03	**16**	4.85	**16**
Geographic location	4.69	**17**	4.57	**17**
Kendall's W	0.285		0.225	
Cronbach's Alpha (α)	0.819		0.914	

As already mentioned, past performance, technical, project understanding, key personnel and health and safety maintain their ranks within the top 5 in both round 2 and 3 of Delphi questionnaire survey. As a measure of stability, well above 0.7 Cronbach's alpha values reflect a very good stability in criteria measurements in both rounds. The increased Kendall' W statistic in round 3 indicates that a stronger agreement is reached by the experts on the importance of the EPC contractor selection criteria.

'Geographic location'- the least important criterion - that has mean value less than 5 (4.63) in round 3 was excluded. Sixteen criteria for EPC tender evaluation were identified from this study with two main bands (or clusters) of criteria importance based on mean value (see Table 6.19).

Table 6.19 The most important to the least important criteria

Rank	Criteria	Important Bands
1	Past Performance	Very important criteria (mean vlaue>6)
2	Project understanding	
3	Technical	
4	Key personnel	
5	Health and safety	
6	Time	Important criteria (5<mean value<6)
7	Financial	
8	Contractual and legal	
9	Past experience	
10	Management	
11	Cost	
12	Quality	
13	Relationships	
14	Industrial Relations	
15	Environment and sustainability	
16	Organisational	

6.2.5 Criteria importance weightings

Criteria importance weightings were determined using the 'mean' value of each criterion from the third round of the survey. Mean value of each criterion was normalised by dividing the criterion mean by the total of mean value to obtain criteria weighting and these weighting are given in Table 6.20.

Table 6.20 Criteria weightings

Rank	Criteria	Weighting
1	Past Performance	0.0684
2	Project understanding	0.0678
3	Technical	0.0675
4	Key personnel	0.0663
5	Health and safety	0.0645
6	Time	0.0639
7	Financial	0.0634
8	Contractual and legal	0.0634

9	Past experience	0.0630
10	Management	0.0630
11	Cost	0.0615
12	Quality	0.0612
13	Relationships	0.0595
14	Industrial Relations	0.0565
15	Environment and sustainability	0.0562
16	Organisational	0.0538

6.3 SUMMARY

This chapter provides the Delphi survey findings. Three rounds of Delphi survey helped in identifying 16 selection criteria and their importance weightings for EPC tender evaluation. The findings indicate that past performance, project understanding, technical, key personnel, and health and safety are the top criteria and are 'very important' for EPC tender evaluation. The remaining 11 criteria are rated as 'important' and need to be included in EPC tender evaluations. These are past experience, time, management, financial, contractual and legal, quality, cost, relationships, environmental and sustainability, organisational, and industrial relations. Geographical location is the least important criterion and it is excluded from importance weight calculations.

Chapter 7: Fuzzy Multi-Attribute Analysis Model for EPC Contractor Selection

This chapter presents the Fuzzy Multi-Attribute Analysis model that addresses the multi-criteria nature of the EPC contractor selection problem and fuzzy nature in multi-criteria assessment. The chapter begins with implementation of MAA (section 7.1); application of the Delphi survey findings (section 7.2), and implementation of Fuzzy Set Theory (section 7.3) in the tender evaluation model of the EPC contractor selection framework developed in Chapter 5. Section 7.4 outlines the Fuzzy Multi-Attribute model; finally, section 7.5 discusses model limitations.

7.1 IMPLEMENTATION OF MULTI-ATTRIBUTE ANALYSIS (MAA)

Multi Attribute Analysis (MAA), one of the best multi-criteria analysis approaches, is capable of identifying optimum choice in terms of objective where decision alternatives are predetermined by incorporation of weights.

MAA is inherited with many advantages such as consideration of multiple attributes in respect of multiple client objectives, and a systemic approach to produce results (Holt et al., 1994a). However, results of attribute evaluation often yield incommensurable units. For example, some of the evaluation results are descriptive in nature (limited/ adequate/ excellent) or numeric. The subjectivity that prevails in MAA can be a matter of concern. In addition to the disadvantages inherited in MAA, multi-attribute evaluation is relatively difficult for decision makers to provide precise numerical values for criteria (or attributes).

Other importance aspects of MAA include that it can address (1) selection criteria, (2) importance weights, and (3) attribute evaluation in matrix form (Holt et al., 1994a), which are essential parts of the contractor selection decision making problem.

7.2 APPLICATION OF DELPHI STUDY FINDINGS (SELECTION CRITERIA AND IMPORTANCE WEIGHTS)

Criteria for EPC contractor selection and their relative importance were exclusively identified from three rounds of Delphi survey. Table 7.1 presents the EPC contractor selection and criteria importance weights which were outlined in Chapter 6.

Table 7.1 Important criteria and criteria weightings

Rank	**Criteria**	**Description**	**Importance weighting**
1	Past Performance	Demonstrates the performance of recently completed projects with records of project cost, completion time and quality, contract claims and variation history, cooperative behaviour (conflicts/disputes), penalties, etc.	0.0684
2	Project understanding	Demonstrates understanding of Request for Proposal (RFP), local context, project risks, unique owner standards and requirements, how the project can be executed to meet client expectations, explains exceptions from RFP, and outlines expected degree of owner involvement, approvals, etc.	0.0678
3	Technical	Demonstrates technical capability and capacity that includes sound engineering solutions, safety in design, creativity and innovation, constructability, engineering and technical expertise, technology and equipment resources, engineering systems, etc. Demonstrates technical support for commissioning, operation readiness, handover, maintenance, repair and training needs.	0.0675
4	Key personnel	Provides the details of key project personnel, which include proposed roles, experience and skills, academic and professional qualification, years with the company, and professional development plan. Demonstrates	0.0663

		project team ability to work collaboratively and as a part of diverse teams, and availability for backup resources.	
5	Health and Safety	Outlines accountabilities for Occupational Health and Safety (OHS) providing samples of site specific management plans, corporate systems, and procedures that identify and control OHS risks. Provides documentary evidence of corporate OHS performance including OHS records from recent projects.	0.0645
6	Time	Provides a project schedule with milestones, activities and deliverables with intended start and finish dates, or complies with the time constraints given in tender documents.	0.0639
7	Financial	Demonstrates contractor financial viability and financial performance over a defined period and provides financial statements, which include balance sheet, profit and loss statement, etc.	0.0634
8	Contractual and legal	Accepts Contract Terms and Conditions or provides clear, concise exclusions or conditional acceptances. Indicates compliance with all relevant codes and regulations.	0.0634
9	Past experience	Provides details of scale, complexity, and type of past projects, and demonstrates project experience of similar type(s) in a similar environment.	0.0630
10	Management	Demonstrates Construction Project Management (CPM) capability (risk management strategy, procurement strategy, stakeholder management plan, logistic and supply chain management, preferred suppliers/sub-contractors, and key trade packages, etc.).	0.0630
11	Cost	Includes tendered price, life-cycle costing, etc.	0.0615

12	Quality	Outlines quality control and quality assurance systems, and complies with specifications and quality standards.	0.0612
13	Relationships	Demonstrates ability to develop strong and long-term partnerships with clients, vendors and suppliers by providing client/subcontractor/supplier referees, including information regarding the duration of the relationship, etc.	0.0595
14	Industrial relations	Demonstrates employee and industrial relations plan/policy and maintenance of project agreements, multi-employer agreements, workers' compensation liabilities, etc. and provides recent industrial relations record.	0.0565
15	Environment and Sustainability	Takes the stakeholders' expectations, which include environmental requirements, social acceptances (e.g. local resources, local economy, indigenous participation, etc.), sustainability approach (products and processes), into account.	0.0562
16	Organisational	Outlines business values and corporate commitment and provides the details of company size, company image, age of business, organisational structure, policies, memberships, current and potential future work commitments, resource optimisation (people, plant, equipment), in-house systems, etc.	0.0538

7.3 APPLICATION OF FUZZY SET THEORY

Rating alternatives in a multiple-attribute decision is subjective in nature and involves linguistic terms. Fuzzy set theory is used to capture ambiguity involved in such linguistic variables. Linguistic variables can be defined quantitatively using fuzzy numbers. Since the verbal evaluations are explained by approximate values, it

is useful to implement either triangular or trapezoidal membership functions in order to reduce ambiguity of evaluation. Then fuzzy arithmetic captures these qualitative assessments using membership function belonging to the set lying between 0 and 1. Triangular fuzzy number are used in this model as they have wide use in practice and are easy to define because of their simplicity.

According to Kaufmann and Gupta (1985), a triangular fuzzy number is defined by three numbers, $x_1 < x_2 < x_3$ as in Fig.7.1.

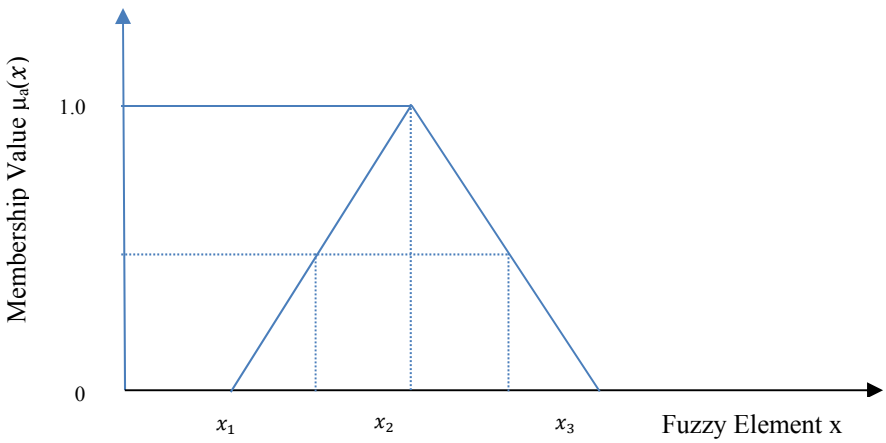


Figure 7.1 Triangular fuzzy number A

The membership function $\mu_a(x)$ is defined as

$$\mu_a(x) = \begin{cases} 0, & x<x_1 \\ (x - x_{2)/}(x_2 - x_{1),} & x_1<x<x_2 \\ 1, & x_2<x<x_3 \\ (x_2 - x)/(x_4 - x_{3),} & x_3<x<x_4 \\ 0, & x>x_4 \end{cases}$$

Fuzzy arithmetic (operations on fuzzy numbers)

Let triangular fuzzy numbers A= $(a_{1,}\ a_{2,}\ a_{3)}$ and B= $(b_{1,}\ b_{2,}\ b_{3)}$, then the operations are expressed as (Singh & Tiong, 2005),

$$A \oplus B = (a_{1,}\ a_{2,}\ a_{3)} \oplus (b_{1,}\ b_{2,}\ b_{3)}$$

$$A \oplus B = (a_1 + b_1, a_2 + b_2, a_3 + b_3) \qquad \text{Equation 7.1}$$

$$A \ominus B = (a_{1,}\, a_{2,}\, a_{3)} \ominus (b_{1,}\, b_{2,}\, b_{3)}$$

$$A \ominus B = (a_1 - b_3, a_2 - b_2, a_3 - b_1)\,) \qquad \text{Equation 7.2}$$

$$A \otimes B = (a_{1,}\, a_{2,}\, a_{3)} \otimes (b_{1,}\, b_{2,}\, b_{3)}$$

$$A \otimes B = (a_1 b_1, a_2 b_2, a_3 b_3) \qquad \text{Equation 7.3}$$

$$A \odot B = (a_{1,}\, a_{2,}\, a_{3)} \odot (b_{1,}\, b_{2,}\, b_{3)}$$

$$A \odot B = (a_1/b_3, a_2/b_2, a_{3/}b_1) \qquad \text{Equation 7.4}$$

Defuzzification

Defuzzification is defined as an operation that produces a non-fuzzy or crisp value. For a fuzzy number $(x_1\,, x_2, x_3)$ as shown in Fig. 2, its defuzzified value e is given by (Singh & Tiong, 2005):

$$e = (\,x_1 + 2x_2 + \,x_3\,)/3 \qquad \text{Equation 7.5}$$

Fuzzy MAA models which combine MAA and FST, develop a procedure for aggregating fuzzy performance levels using criteria importance weights to determine best value, thus selecting the most appropriate contractor.

7.4 PROPOSED EPC CONTRACTOR SELECTION MODEL

7.4.1 Definitions

Holt (Holt et al., 1994a) definitions for criteria, objective, and attribute are appropriate for this model.

Criteria: Measures of effectiveness that appear as either (Client) objectives or (contractor) attributes

Objective: Measures by which options may be evaluated

Attribute: Performance parameters that provide means of evaluating a decision option (contractor) in respect of an objective

Score: Score (numeric/linguistic) in preferred rating scale assigned for each alternative for each criterion assigned by the decision maker

Weighting: Numerical weights assigned for each criterion to indicate the priority of the criterion by the decision maker (or by the expertise opinion)

7.4.2 Proposed EPC Tender Evaluation Model

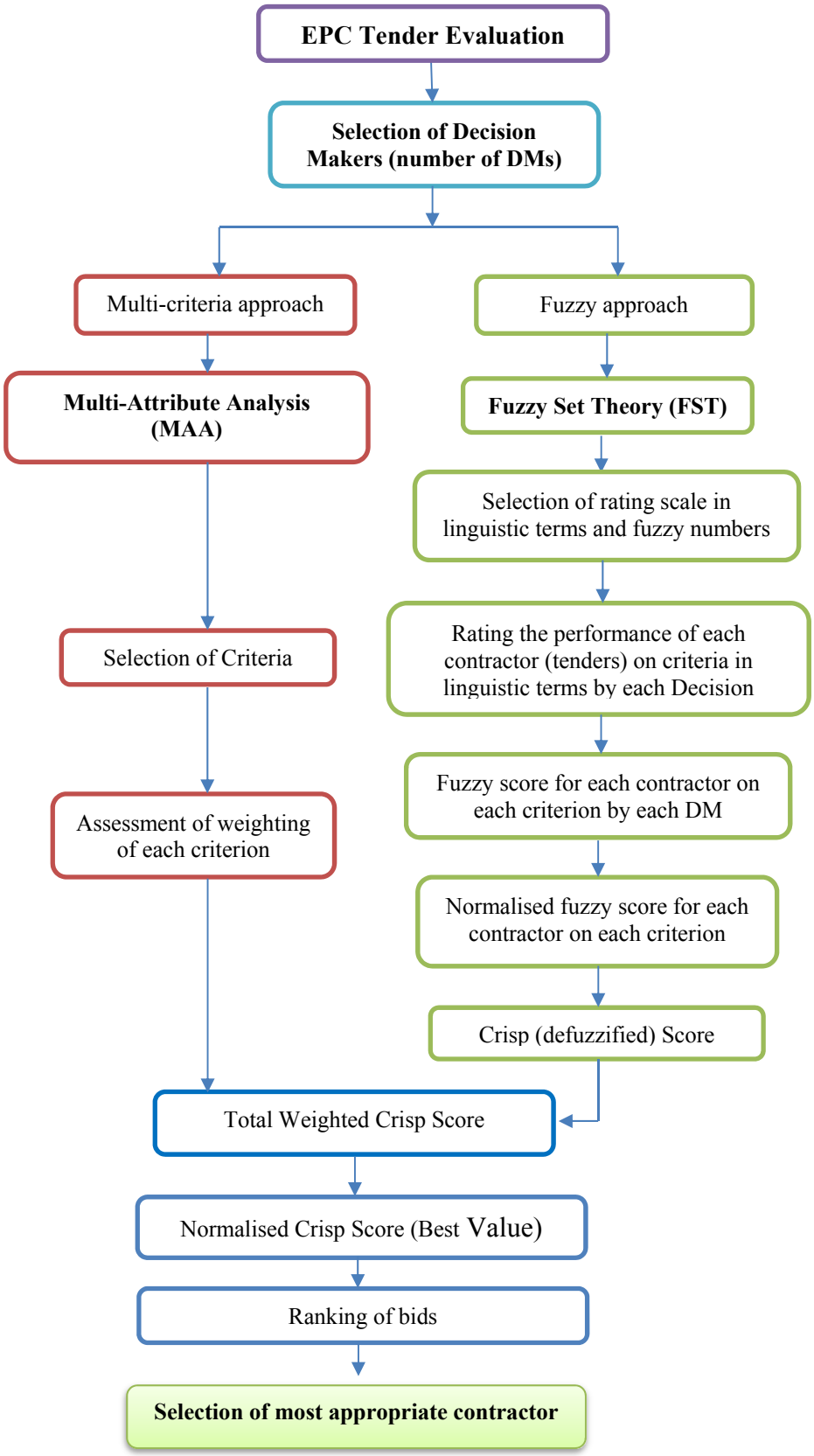


Figure 7.2 FMMA EPC Tender Evaluation Model Flow Chart

EPC Tender Evaluation model described in Fig 7.2 consists of the following steps:

Step 1: Selection of decision makers

Select the number of decision makers DM_k where k=1, ……p) that are required for evaluating tenders according to predetermined criteria.

$$DM_k = \begin{bmatrix} DM_1 & DM_2 & \cdots & DM_k & \cdots & DM_p \end{bmatrix}$$

Step 2: Identification of EPC contractor selection criteria and prioritisation of criteria by importance

Determine the attribute (or criteria)-CR_i, where (i=1, ……. n) that has impact on EPC contractor performance using the sixteen (16) criteria identified from the Delphi survey given in Table 7.1. For example, CR_1 is past performance, CR_2 is project understanding, as such the rest of the criteria.

$$CR_i = \begin{bmatrix} CR_1 & CR_2 & \cdots & CR_i & \cdots & CR_n \end{bmatrix}$$

Use criteria importance values (weightings) in (Table 7.1) which were derived using the mean values of each criterion of round 3 of the Delphi survey. Criteria weighting are given in 'weighting matrix' W_j where j=1, ……m)

$$W_j = \begin{bmatrix} W_1 \\ W_2 \\ \vdots \\ W_j \\ \vdots \\ Wm \end{bmatrix}$$

For example, Criteria weighting of CR_1 (past performance) is equal to 0.0684 and CR_2 (project understanding) is equal to 0.0678. Similarly, criteria weightings of the rest of the criteria can be obtained from the Table 7.1.

Step 3: Selection of rating scale/linguistic terms/fuzzy numbers

Select the rating scale to evaluate contractor performance against each criterion. Then, allocate linguistic variables to describe the ratings of a chosen scale. 1-7 Likert rating scale is recommended and respective linguistic terms are given in Table 7.2.

Table 7.2 Rating Scale

Crisp value	Linguistic variable
1	Very low (VL)/ Very poor (VP)
2	Low (L)/ Poor (P)
3	Medium low (ML)/ Medium poor (MP)
4	Medium (M)/ Fair (F)
5	Medium high (MH)/ Medium good (MG)
6	High (H)/ Good (G)
7	Very high (VH)/ Very good (VG)

Assign triangular fuzzy numbers (or triangular fuzzy membership) to linguistic variables (Table 7.3) - Fig. 7.3 presents fuzzy numbers graphically.

Table 7.3 Triangular fuzzy numbers

Linguistic variable (LV1, LV2, …….LV7)	**Fuzzy number (naming code)**	**Fuzzy membership (x_1, x_2, x_3)**
Very low (VL)/ Very poor (VP)	F_A	(0,0,0.1)
Low (L)/ Poor (P)	F_B	(0,0.1,0.3)
Medium low (ML)/ Medium poor (MP)	F_C	(0.1, 0.3, .05)
Medium (M)/ Fair (F)	F_D	(0.3, 0.5, 0.7)
Medium high (MH)/ Medium good (MG)	F_E	(0.5, 0.7, 0.9)
High (H)/ Good (G)	F_F	(0.7, 0.9, 1)
Very high (VH)/ Very good (VG)	F_G	(0.9, 1, 1)

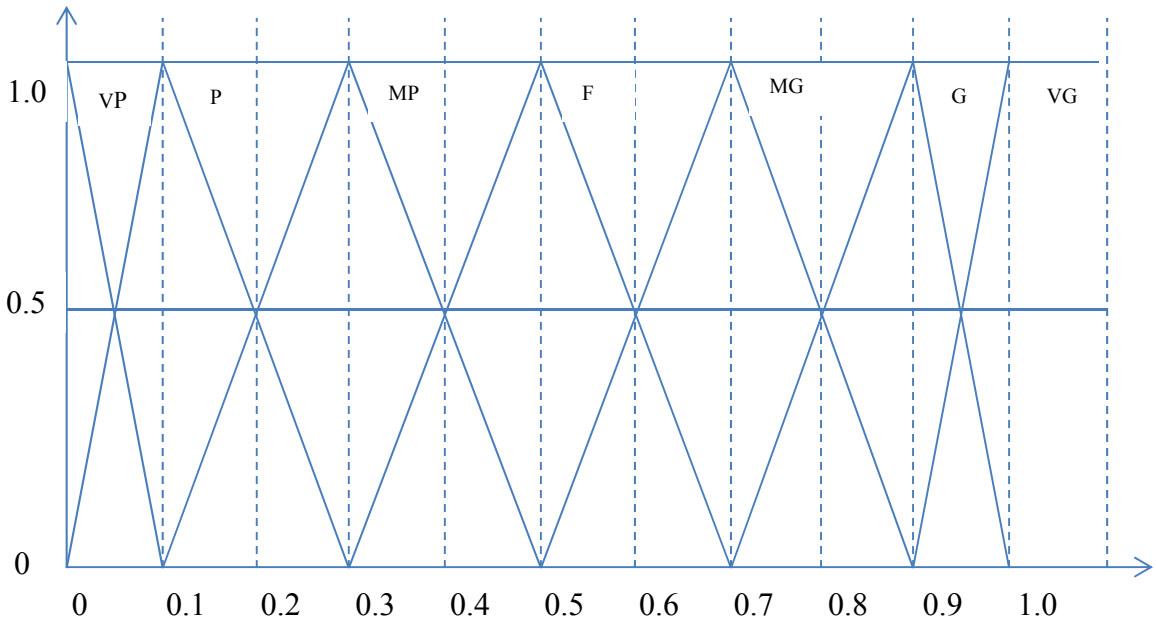


Figure 7.3 Graphical representation of fuzzy numbers:

Source: (Vahdani et al., 2013)

Step 4: Judgement of each contractor on each criterion in linguistic terms

Provide judgement of decision makers of each contractor on each criterion and develop linguistic score matrices for each criterion

Let contractors (tenders) be defined by Ar, where r=1, …. q. Each DM_k where k=1, ……. p provides judgement using linguistic terms thus developing fuzzy-linguistic-score matrix.

Criterion CR_i

$$\begin{array}{c} \\ A_1 \\ A_2 \\ \vdots \\ A_r \\ \vdots \\ A_q \end{array} \begin{array}{c} \begin{array}{cccccc} DM_1 & DM_2 & \cdots & DM_k & \cdots & DM_p \end{array} \\ \begin{bmatrix} D1A1 & D2A1 & \cdots & DkA1 & \cdots & DpA1 \\ D1A2 & D2A2 & \cdots & DkA2 & \cdots & DpA2 \\ \vdots & \vdots & \ddots & \vdots & \cdots & \vdots \\ D1Ar & D2A2 & \cdots & DkAr & \cdots & DpAr \\ \vdots & \vdots & \cdots & \vdots & \cdots & \vdots \\ D1Aq & D2Aq & \cdots & DkAq & \cdots & DpAq \end{bmatrix} \end{array}$$

DkAr=Decision maker DM_k's judgement of alternative Ar for criteria CR_i in linguistic terms.

Step 5: Fuzzy score of each contractor on each criterion

Transform linguistic variables to fuzzy numbers (as such decision makers' judgement in fuzzy numbers) and develop fuzzy score matrix for each criterion (CR_i)

Criterion CR_i

$$\begin{array}{c} \\ A_1 \\ A_2 \\ \vdots \\ \vdots \\ A_q \end{array} \begin{array}{c} \begin{array}{ccccc} DM_1 & DM_2 & \cdots & \cdots & DM_p \end{array} \\ \begin{bmatrix} F_F & F_G & \cdots & \cdots & F_F \\ F_G & F_G & \cdots & \cdots & F_F \\ \vdots & \vdots & \ddots & \cdots & \vdots \\ \vdots & \vdots & \cdots & \cdots & \vdots \\ F_H & F_G & \cdots & \cdots & F_E \end{bmatrix} \end{array}$$

$F_{A,}\ F_B, \ldots\ldots\ldots\ldots\ldots\ldots, F_H$ are respective new fuzzy numbers for the judgement of Decision makers' in linguistic terms. As such,

F_A= (a_1, a_2, a_3), F_B= (b_1, b_2, b_3), F_C= (c_1, c_2, c_3), F_D= (d_1, d_2, d_3), F_E= (e_1, e_2, e_3), F_F= (f_1, f_2, f_3), F_G= (g_1, g_2, g_3), and F_H= (h_1, h_2, h_3)

Step 6: Total Fuzzy Score for each criterion and development of Total Fuzzy Score matrix

Combine the fuzzy scores given by 'n' decision makers for each contractor on each criterion

Using equation 6.1, find Total Fuzzy Score of contractor A_r on criterion CR_i =DM_1 $_{\text{Fuzzy score}}$+DM_2 $_{\text{Fuzzy score}}$+……. DMkFuzzy $_{\text{Score}}$+……$DM_{n\text{Fuzzy score}}$)

For example: Total fuzzy score for contractor A_1 on criteria CR_1=F_F+F_G + F_F= (f_1, f_2, f_3) + (h_1, h_2, h_3) + (f_1, f_2, f_3)

Let new fuzzy score be given by new set of fuzzy numbers (calculated) F_{AA}= (A1, A2, A3), F_{BB}= (B1, B2, B3), …….... and denoted by TFS_{ArCi}, similarly scores of contractors on each criterion are given by:

$$
\begin{array}{c}
 \\ A_1 \\ A_2 \\ \vdots \\ A_q
\end{array}
\begin{array}{c}
\begin{array}{cccccc} CR_1 & CR_2 & \cdots & CR_1 & \cdots & CR_n \end{array} \\
\begin{bmatrix}
TFS_{A1C1} & TFS_{A1C2} & \cdots & TFS_{A1Ci} & \cdots & TFS_{A1Cn} \\
TFS_{A2C1} & TFS_{A2C2} & \cdots & TFS_{A2Ci} & \cdots & TFS_{A2Cn} \\
\vdots & \vdots & \ddots & \vdots & \cdots & \vdots \\
TFS_{AqC1} & TFS_{AqC2} & \cdots & TFS_{AqCi} & \cdots & TFS_{AqCn}
\end{bmatrix}
\end{array}
$$

Step 7: Normalised Fuzzy Score of each contractor on each criterion

Find the Normalised Fuzzy Score of each contractor on each criterion by dividing the Total Fuzzy Score by the number of decisions of decision makers (p)

$$
\begin{array}{c}
 \\ A_1 \\ A_2 \\ \vdots \\ A_q
\end{array}
\begin{array}{c}
\begin{array}{cccccc} CR_1 & CR_2 & \cdots & CR_i & \cdots & CR_n \end{array} \\
\begin{bmatrix}
TFS_{A1C1}/p & TFS_{A1C2}/p & \cdots & TFS_{A1Ci}/p & \cdots & TFS_{A1Cn}/p \\
TFS_{A2C1}/p & TFS_{A2C2}/p & \cdots & TFS_{A2Ci}/p & \cdots & TFS_{A2Cn}/p \\
\vdots & \vdots & \ddots & \vdots & \cdots & \vdots \\
TFS_{AqC1}/p & TFS_{AqC2}/p & \cdots & TFS_{AqCi}/p & \cdots & TFS_{AqCn}/p
\end{bmatrix}
\end{array}
$$

Let Normalised Fuzzy Score be defined by $NFS_{ArCi} = TFSS_{ArCi}/k$ for each criterion of each contractor

$$
\begin{array}{c}
 \\ A_1 \\ A_2 \\ \vdots \\ A_q
\end{array}
\begin{array}{c}
\begin{array}{cccccc} CR_1 & CR_2 & \cdots & CR_i & \cdots & CR_n \end{array} \\
\begin{bmatrix}
NFS_{A1C1} & NFS_{A1C2} & \cdots & NFS_{A1Ci} & \cdots & NFS_{A1Cn} \\
NFS_{A2C1} & & \cdots & & \cdots & \\
\vdots & \vdots & \cdots & & \cdots & \\
NFS_{AqC1} & & \cdots & NFS_{AqCi} & \cdots & NFS_{AqCn}
\end{bmatrix}
\end{array}
$$

Step 8: Crisp score (Defuzzified score)

Using equation 6.5, find the crisp score of each contractor on each criterion.

Let fuzzy number be (x_1, x_2, x_3), then

Crisp Score (CS) = $(x_1+2*x_2+x_3)/3$

$$
\begin{array}{c}
 \\ A_1 \\ A_2 \\ \vdots \\ A_q
\end{array}
\begin{array}{c}
\begin{array}{cccccc} CR_1 & CR_2 & \cdots & CR_i & \cdots & CR_n \end{array} \\
\begin{bmatrix}
CS_{A1C1} & CS_{A1C2} & \cdots & CS_{A1Ci} & \cdots & CS_{A1Cn} \\
CS_{A2C1} & & \cdots & & \cdots & \\
\vdots & \vdots & \cdots & & \cdots & \\
CS_{AqC1} & & \cdots & CS_{AqCi} & \cdots & CS_{AqCn}
\end{bmatrix}
\end{array}
$$

Step 9: Total Weighted Crisp Score of each contractor (Total Weighted Crisp Score matrix)

$$\begin{array}{c} \\ A_1 \\ A_2 \\ \vdots \\ A_q \end{array} \begin{array}{c} \begin{array}{cccccc} CR_1 & CR_2 & \cdots & CR_i & \cdots & CR_n \end{array} \\ \begin{bmatrix} CS_{A1C1} & CS_{A1C2} & \cdots & CS_{A1Ci} & \cdots & CS_{A1Cn} \\ CS_{A2C1} & CS_{A2C2} & \cdots & CS_{A2Ci} & \cdots & CS_{A2Cn} \\ \vdots & \vdots & \cdots & \vdots & \cdots & \vdots \\ CS_{AqC1} & CS_{AqC2} & \cdots & CS_{AqCi} & \cdots & CS_{AqCn} \end{bmatrix} \end{array} \times \begin{bmatrix} W_1 \\ W_2 \\ \vdots \\ W_m \end{bmatrix} = Wj$$

As such, $TWCS_{A1} = CS_{A1C1} \times W1 + CS_{A1C2} \times W_2 + \cdots\cdots\cdots\cdots\cdots\cdots + CS_{A1Cn} \times W_m$

Similarly, calculate all the TWCS of the remaining contractors.

Total Weighted Crisp Score matrix is shown below:

$TWCS_r$

$$\begin{array}{c} A_1 \\ A_2 \\ \vdots \\ A_q \end{array} \begin{bmatrix} TWCS_{A1} \\ TWCS_{A2} \\ \vdots \\ TWCS_{Aq} \end{bmatrix}$$

Ranking of contractors by Normalised Weighted Crisp Score

Calculate Normalised Weighted Score using equation 7.6 below:

Normalised Weighted Crisp Score (NWCS) for contractor is given by 'A_r'.

$$A_r = TWCS_{Ar} / \sum_{r=1}^{q} TWCS_{Ar} \qquad \text{Equation 7.6}$$

Contractors are ranked based on 'A_r' value and the contractor who tops the list is recommended for contract negotiations leading to the contract award phase.

A worked example (hypothetical) of the model has been attached (Refer to Appendix I) to explain the model steps with appropriate calculations.

7.5 SUMMARY

This chapter presents the development stages of fuzzy-multi-attribute analysis model for EPC contractor selection. Particularly, implementation of multi-attribute analysis, application of the Delphi study findings and application of fuzzy set theory were broadly discussed. This includes the tender evaluation model and its sequential steps.

Chapter 8: Validation of EPC Contractor Selection Model

8.1 INTRODUCTION

8.1.1 Validation Aim

The main aim of the model validation was to test the proposed model's practicality, usefulness, clarity, and appropriateness to the EPC industry in Australia and use the validation outcome to improve the model. Additionally, EPC contractor selection framework is also validated for appropriateness to the EPC industry.

8.1.2 Validation objective

To achieve the aims of validation, the following objectives were set.

- To identify the experts' opinion towards the elements of the EPC contractor selection framework
- To identify the appropriateness and completeness of tender evaluation criteria, and their importance weights
- To explore the opinions of experts towards the use of fuzzy set theory in this model, as such appropriateness and ease of use in practice
- To explore and discuss the ways of improving the model, if any, for industry use

8.1.3 Validation process

Validation is defined as the process of determining the degree to which a model is an accurate representation of the real world from the perspective of the intended uses of the model. Therefore, it is important to ensure that the model caters for the real world (EPC industry in Australia) demands.

There are several ways of validating a model. It can be validated using in-depth interviews or questionnaire surveys. Interviews provide a rich source of information from experts (Baroudi & Metcalfe, 2011). The semi-structured interview method was adopted for this research.

The design of the process and selection of participants plays an important role in the validation activity. The 'snowball' approach was selected to approach the potential participants. The interviewees were selected through the recommendation from the Delphi questionnaire survey respondents of this study. Priority was given to the most experienced respondents and they were asked to nominate the potential participants who have experience in selecting contractors for EPC projects.

Then, the potential participants were invited formally to take part in the interview and they were given a brief introduction of the interview expectations and about the proposed EPC tender evaluation model, and the participant information sheet, which contained the consent form. Interviews were conducted approximately for 1-2 hours at a time at a venue convenience for the interviewees. With their consent, the interviews were audio recorded.

Audio-recorded interview data were imported to NVivo qualitative data analysis software and the data were transcribed manually. Interview transcripts were analysed using coding techniques. Results of the analysis were then used to scrutinise the model for industrial use.

8.2 VALIDATION INTERVIEWS

8.2.1 Semi-structured face-to-face-interview

The face-to-face interviews enable the interviewer to directly ask questions and record the interviewee's responses. An interview package was prepared in advance and distributed a few days ahead of the interview date. The interview package consists of brief description of the proposed EPC tender evaluation model and participants' information sheet, which describes the interview ethics, timeline, etc., and semi-structured interview template (Appendix F).

8.2.2 Validation participants

Three participants who are experts in EPC contractor selection were selected for participation in the interviews. The participants were from both public and private sectors. The selected participants were from both owner and consultancy organisations in the resource sector and the infrastructure development industry where the EPC delivery method is commonly used for their major development projects.

Table 8.1 presents the profile data of the participants.

Participant	Role	Experience	Industry	Organisation	Sector
Participant 1	Project Manager	>20 years	Mining	Consultancy	Private
Participant 2	Project Director	>20 years	Transport and Roads	Owner	Public
Participant 3	Project Director	>20 years	Mining	Consultancy	Private

The three interview participants were knowledgeable and have used multi-criteria methods and criteria importance weights for selecting contractors. That was verified during the interviews.

8.2.3 Conducting the interviews

Participants who agreed to participate in validation interviews were contacted and the interviews were scheduled. Then, the interview package was sent at least three days ahead of the interview. This included a brief description of the model which explained the fuzzy set theory ahead of the interview to help them understand the rationale of the fuzzy set theory. After welcoming the interviewee at the appointment, firstly the research ethics - including confidentiality of data, purpose of interviews and how the data was applied in the research - were explained, even though they were given on the participant's information sheet. Then a request was made for participant's permission to audio-record the interview before the interview commenced. The consent to participate in the interview was obtained in written form, by completion of the consent form that was given with the participants' information sheet. All three participants gave their permission to be audio-recorded with their completed consent forms.

During the interview, the EPC tender evaluation model was briefly explained with some background information, however when the interviewee asked questions or for clarification, the details were further elaborated. As they hardly knew about fuzzy set theory, the fuzzy set theory and its application within this model was explained to all interviewees during the interview in addition to the brief description. Depending on interviewee's responses, further questions were asked. At the end, an interviewee

was asked to assess the model against five questions given in the semi-structured interview template. The questions are

Q1. To what extent is the EPC contractor selection framework practical and comprehensive? Specify the practical limitations of industry use.

Q2. Is the list of contractor selection criteria comprehensive? Are the criteria importance weights rated appropriately? Has the model addressed the new demands in the EPC industry (in terms of criteria and their importance weights)?

Q3. What would you suggest to overcome the limitations or drawbacks mentioned in answering Q1 or Q2?

Q4. Do you think the Fuzzy Multi-Attribute Analysis Model will facilitate the owner to select the most appropriate contractor, addressing the subjectivity, uncertainty, impreciseness and incompleteness in a contractor selection decision-making process?

Q5. If you were given the opportunity to use this model in your current project, would there be a significant difference in results of the contractor selection?

Q6. How would you rate the model (strongly recommended/recommended/not sure/not recommended)?

8.3 VALIDATION DATA ANALYSIS AND DISCUSSIONS

As data collected from interviews is in audio form, the first step is the preparation of data for analysis. NVivo qualitative analysis software was used for transcribing audio data to text data from the interviews. In doing so, the audio files were imported to the NVivo programme, and while the audio was playing, data was transcribed manually (hearing the recording over and again) and interview transcripts were prepared.

To simplify the data analysing process, the questions in the semi-structured interview template were labelled as in Table 8.2. These labels were later used for coding the interview responses.

Table 8.1 Question labels

#	Question description	Label

Q1	To what extent is the EPC contractor selection framework practical and comprehensive? Specify the practical limitations of industry use.	EPC contractor selection framework
Q2	Is the list of contractor selection criteria comprehensive? Are the criteria importance weights rated appropriately? Has the model addressed the new demands in the EPC industry (in terms of criteria and their importance weights)?	Selection criteria and criteria importance weights
Q3	What would you suggest to overcome the limitations or drawbacks mentioned in answering Q1 or Q2?	Limitations/drawbacks (criteria selection and weightings)
Q4	Do you think the Fuzzy Multi-Attribute Analysis Model will facilitate the owner to select the most appropriate contractor addressing the subjectivity, uncertainty, impreciseness and incompleteness in contractor selection decision making process?	Selection of contractor
Q5	If you were given the opportunity to use this model in your current project, would there be a significant difference in results of the contractor selection?	Significance of the model
Q6	How would you rate the model (strongly recommended/recommended/not sure/not recommended)?	Rating the model performance

After preparing these interview transcripts in NVivo, they were exported as Word files, and the text data was analysed using coding techniques. Each interviewee's assessments of the model against the interview questions were coded under the following three key themes:

1. Perceptions: respondent's insights
2. Ideas: suggestions only
3. Notable other insights

All the coded responses to the interview questions (in the semi-structured interview template) under the first two themes were given in Appendix G and notable other insights in Appendix H. The most important comments against each interview questions are discussed below.

EPC contractor selection framework:

Interviewee A and B emphasised that the two-stage process (prequalification + tender evaluation) is important and appropriate for EPC projects. Interviewee C said that the requirement for prequalification is highly dependent on the market - as such if there are very few (e.g. three contractors) there is no point in the short listing of contractors. However, interviewee A further indicated that the single stage selection can be adopted only for situations where the client nominates a specific contractor. Interviewee B stated that prequalification is a must for government projects. There is no selected tendering in government projects and it is always competitive bidding. However, interviewee B explained further that the open tendering exists in government procurement and it depends on the funding source. As such, open tendering is used in federally funded projects.

All three interviewees emphasised that the best value that considers long term performance is the most appropriate, not the lowest price bids. Interviewee B highlighted that the best-value is the least explained in the tender evaluations, and the EPC industry faces a dilemma as it is subjectively assessed, involving mathematical calculations, qualitative weightings, and quantitative ratings.

Moreover, Interviewee A said that this framework is practical and useful for recommending the top two contractors.

Selection criteria and criteria importance weights:

Interviewee A said that the criteria list is comprehensive and good. However, interviewee B indicated that selecting no more than 10 criteria is better otherwise a dilution effect can occur. Furthermore, he/she emphasised no point in including a criterion for which all the contractors would give a same answer. Therefore, when the industry becomes mature, some of the criteria can be eliminated. Interview C highlighted that selection of criteria cannot be generalised and it should be project specific and suggested that 'life-cycle-cost' can be a separate criterion.

All the interviewees expressed their concerns of the assessment of some of the criteria. Project variations are inevitable, therefore interview A suggested to include 'Schedule of rates' under the 'cost' criteria and evaluate 'schedule of rates'. Interviewee B indicated the company systems, processes and culture should be evaluated under 'past performance' however it is of not much importance in roads. A

single report on one project is evaluated to measure the past performance. Additionally, 'innovation' is not an important factor for road projects. Moreover, it highlighted the importance of quantifying the responses and as such, restricting the number of examples, pages of the report, etc.

Interviewee B also discussed the approaches adopted in his/her team for evaluating individual criteria, such as key personnel and experience. As highlighted, 'Key personnel' is very important to a project and the evaluating approach is conducting short interviews with the nominated project members (of the contractor). The first round would be with the individuals who compose the team and the second interview is with the team, by which the client can understand the individual players and their team culture and more importantly how they are going to work with them. Evaluating of 'past experience' is by requesting two reports: (1) client/superintendent said you did well (2) client/superintendent said you did poorly.

With regards to weights of criteria, interviewee A indicated that obtaining weightings from the industry and comparing them with the Delphi survey findings is worthy. Interviewee A discussed the possibility of using the fuzzy approach to determine weights. Interview B mentioned that they give a low rating to 'past performance' and high rating to 'key personnel' and 'project understanding'. Interview C stated that the criteria as well as weighting cannot be generalised, and criteria selection should be project specific, and also highlighted that benchmarking for criteria is essential. For example, for 'safety' criterion if the score is less than 6, that bid is eliminated from further evaluation.

Limitations/Drawbacks

Both interviewees A and C stated that understanding the criteria and weightings should be project or industry specific because weighting has a large impact on a project. Interview B highlighted that a long list of criteria can cause a dilution effect.

Selection of contractor using MAA and FST:

Interviewee A indicated that the use of a model with fuzzy numbers is not a problem in an office environment as it can be implemented using Excel spreadsheets. A's concern is, when the contractors become familiarised with MAA, that they score full marks more often in a desktop study like this. Therefore, the model can be used to recommend the best contractors (top 2 or 3) and final decisions can be made after an

interview with each of the contractors. It was suggested to use a numerical rating scale and use linguistic terms as a legend, as engineers always deal with numbers, and expressed the idea of extending the fuzzy approach to the next level (sub-criteria).

Interviewee B said that the use of FST is interesting but emphasised averaging of fuzzy numbers should not be done because averaging is all right only for close answers, when there is a low player, there is a problem. It was suggested when there are outliers, the panel members should argue and come to an agreement.

Interviewee C argued that MAA is subjective assessment and that this model captures only the uncertainty of responses of decision makers in terms of linguistic variables and does not capture how much the decision makers are sure about (certainty or uncertainty of) his or her assessment (score) because while some panel members are very much sure about their own answer (self-confidence), some are unsure. It was suggested to value the panel member uncertainly/certainty of their verbal responses and adjust fuzzy numbers accordingly.

Significance of the model:

Interviewees remarked that application of FST in tender evaluation is a new concept, so the model is significant. They inspired with the use of fuzzy set theory to address the fuzziness inherited in multi-attribute assessment with subjective, uncertain, and imprecise data when selecting contractors. They highlighted that the model enhances the tender evaluation process more systematic and realistic way to achieve value for money for EPC project owners. Another highlight was the use of linguistic terms instead numerical values allows decision makers to rate contractor performance realistically because words can help describing real situations, even some interviewers went extra length to describe this significance by match this model concept with the day to day life scenarios. This model assists decision makers by providing a list of linguistic variables and representative fuzzy numbers appropriate for contractor selection making the fuzzy set theory implementation practicable. They suggested running a few test runs in parallel with industry partners, which would be good to demonstrate the significance over the existing systems that the industry has.

Recommendation of model:

Interviewee A said, "I would recommend to anyone to use this model without doubt. I personally will use this to select the top 2 but my final selection will be through an interview". As per interviewee B, it was suggested that the model needs to change the averaging process. The Kepner Trigoe decision-making tool was suggested, which forces out the outliers and can be used instead of averaging. Interviewee C said that "it is a good model however needs running test runs".

In addition to above, the other important insights from the interviewees were grouped under 'other important insights' and are given in Appendix H. As suggested, the panel should consist of 3-4 decision makers, the Kepner Tregoe (K-T methodology) - decision making method can be used with rating scale 1-10 to eliminate outliers, and all invited contractors should have equal opportunity (i.e. level field playing concept), etc.

8.4 SUMMARY

Chapter 8 presented the validation of the EPC contractor selection framework and tender evaluation model. The aim was to test the model's practicality, usefulness, clarity, and appropriateness to the EPC industry in Australia and use the validation outcome to improve the framework/model. The validation process was implemented through face-to-face interviews. The participants represented the public sector client side and private sector consultancy services. All participants provided positive feedback on the significance of the model in selecting the best contractors, addressing the uncertainty by application of fuzzy set theory. The validation results indicate that overall framework is clear and the tender evaluation model is easy to implement.

They also highlighted the model's elements that require further research. Some of these insights were taken into consideration and the tender evaluation model was modified to reflect the feedback. Others were identified to research further in the future.

Chapter 9: **Conclusions**

9.1 OVERVIEW

Although various researchers and organisations have proposed contractor selection frameworks for different project delivery methods, few to date focus on the EPC method, despite the EPC method having gained popularity with increased use in the resource sector and infrastructure projects in recent years in Australia. Obviously, EPC projects are large and complex with high budget values, long project timelines, and involving multiple stakeholders, and this poses significant challenges for owners in selecting the most competent contractor in a highly competitive environment. Therefore, EPC projects demand a comprehensive strategy for the contractor performance evaluation.

The aim of this research is to develop a contractor selection model characterised by comprehensive evaluation strategies that align with owner objectives and contractor attributes while eliminating the weaknesses of current practices to use in Australian EPC projects. In developing the model, four objectives were identified and outlined in Chapter 1. These include (1) understanding the EPC project delivery method and the EPC market in Australia (2) identifying and prioritising the criteria for EPC contractor selection (3) developing a new EPC contractor selection model using Multi-Attribute Analysis and Fuzzy Set Theory, and (4) validating the EPC contractor selection model for industry use.

This chapter discusses the overall results of the research including directions for future research.

9.2 REVIEW OF OBJECTIVES

9.2.1 **Objective 1:** Understanding the EPC project delivery method and the EPC market in Australia

Engineering-Procurement-Construction (EPC) is a project delivery method where one or more contractors and designers combine their efforts to deliver a full and complete facility driven by engineering design (Baram, 2005; Forbes & Ahmed, 2010). EPC has become the predominant delivery method for highly complex, large-

scale and high-value private and public engineering projects. These high cost projects span for several years to deliver and involve multiple stakeholders.

This market review shows that the EPC market in Australia in recent years has been driven by engineering construction (42% of total construction). Mining, heavy industry and infrastructure construction is the dominant recipient of the EPC method. Additionally, EPC is the most popular project delivery method in the energy and natural resources sector in the global market (KPMG International, 2015).

The engineering design within EPC is a multi-disciplinary activity that creates a high level of risk and complexity. This research finds that the nature of projects has changed from one of engineering success to the delivery of more sustainable and economic outcomes. In-depth understanding, leadership, team work, and systems are required for successful project delivery. One of the most important and critical decisions for the client/owner at the early project stage is in selecting the most suitable contractor.

This review also clearly indicates that selecting the wrong contractor causes delays and cost overruns, and therefore selecting an EPC contractor through a competitive bidding process, where a client evaluates contractors based on the multi-attribute evaluation criteria that include price and other intangible factors, is important for successful project delivery. EPC projects are typically delivered by using best-value procurement and lump sum price contracts.

This comprehensive review helps contractors, project client/owners, industry professionals and other construction stakeholders to better understand the current EPC challenges, and their potential advantages and pitfalls so that proactive measures can be taken for future EPC projects. This will be of great value in planning future EPC projects. More importantly, this review highlights the requirement of developing a tender evaluation model so that the client can confidently select the best contractor through objective assessment of multi criteria.

9.2.2 **Objective 2:** Developing an EPC contractor selection framework

The EPC contractor selection framework was designed with the elements to capture the industry demands rationally using literature review findings, and core elements include procurement strategy, tendering method and tender evaluation strategy. As such, two-stage selection (prequalification first and tender request second), which is

appropriate for large, complex and high risk project, was adopted. Tender invitation through competitive bidding, where more contractors can bid in a competitive environment, was selected as the tendering method. Another essential element is selection of procurement strategy, best-value procurement which emphasises quality, efficiency/effectiveness, value for money and performance standard, and was selected for evaluating tenders. Best-value procurement adopts a multi-criteria approach. Therefore, multi-attribute analysis has been incorporated to select optimum choice through systematic assessment of multi-criteria. Fuzzy set theory has been adopted to address the fuzziness in multi-criteria assessment.

9.2.3 **Objective 3:** Investigating and prioritising the criteria for EPC contractor selection and determining criteria importance weights

Three rounds of Delphi questionnaire survey helped in identifying 16 selection criteria for EPC tender evaluation. The response rate for this Delphi questionnaire survey is high and the diverse participants represent a good cross-section of EPC experts in Australia. The Delphi findings classify three clusters of criteria importance: (1) very important (2) important and (3) least important criteria. Geographic location - the least important criterion (group mean <5) - was eliminated. Respective weightings were calculated using the group mean of the third round of Delphi questionnaire survey. The overall result of criteria and their importance weights are given below.

'**Very Important'** criteria:

1. Past Performance (0.0659)
2. Project understanding (0.0646)
3. Technical (0.0650)
4. Key personnel (0.0635)
5. Health and Safety (0.0611)

'**Important'** criteria:

6. Time (0.0609)
7. Financial (0.0596)
8. Contractual and legal (0.0596)

9. Past experience (0.0614)
10. Management (0.0603)
11. Cost (0.0582)
12. Quality (0.0582)
13. Relationships (0.0567)
14. Industrial relations (0.0513)
15. Environment and Sustainability (0.0537)
16. Organisational (0.0525)

A better understanding of criteria importance will pave the way to developing an EPC contractor selection model involving the criteria most needed to evaluate tenders objectively. The findings add significant insight to the body of knowledge of the EPC procurement system and help in the objective evaluation of tenders. The above important criteria and their ratings given here have been applied in the proposed EPC tender evaluation model which will be discussed next.

9.2.4 **Objective 4:** Developing a new EPC contractor selection model using Multi-Attribute Analysis and Fuzzy Set Theory to objectively evaluate the EPC contractor performance

The main contribution of this PhD study is developing a new EPC contractor selection model to evaluate tenders objectively using multi-attribute analysis and fuzzy set theory applications. Multi-Attribute Analysis (MAA), capable of identifying optimum choice against multiple objectives, is suitable for the multi-criteria nature of the contractor selection dilemma. MAA can address (1) selection criteria, (2) importance weights, and (3) attribute evaluation in matrix form (Holt et al., 1994a). Selection criteria and importance weights obtained from three rounds of a Delphi survey have been incorporated in this model. However, subjectivity, uncertainty and impreciseness exit in multi-attribute assessment as a matter of concern. In addition, multi-attribute evaluation is relatively difficult for decision makers to provide precise numerical values for criteria. Fuzzy Set Theory (FST) is used to address this fuzzy nature in human decision making when evaluating tenders.

In the final EPC contractor selection model, the members of a tender evaluation panel need to rate the degree of a contractor satisfying a criterion in terms of linguistic variables (such as very good, good, above average, average, very poor) corresponding to 1-7 scale. Then fuzzy arithmetic captures these qualitative assessments using membership function belong to the set lying between 0 and 1. In this model, the linguistic variables are defined quantitatively using triangular fuzzy numbers. By the application of fuzzy mathematical operations, the fuzzy score of each contractor is transformed to crisp values (defuzzified numbers). Then the best value of each contractor is obtained by combining the crisp values with criteria importance weights. Contractors are then ranked according to the best value, thus selecting the most appropriate contractor.

9.2.5 **Objective 5:** Validating the EPC contractor selection model for industry use

Fuzzy Multi-Attribute Analysis (FMAA) EPC Contractor Selection is composed of EPC contractor selection framework and tender evaluation model. The validation aim was to test the model's practicality, usefulness, clarity, and appropriateness to the EPC industry in Australia and use the validation outcome to improve the framework/model.

The validation process was implemented through three face-to-face interviews with experts from the EPC industry. All three participants were supportive of the model, even though they did not have opportunity to test the model using real tender evaluation activity.

As such, the EPC contractor selection framework has the elements to capture the industry demands, which include two-stage selection (prequalification first and tender request second), competitive bidding, and best-value procurement. Interviewees are impressed with the application of fuzzy set theory in tender evaluation to capture the subjectivity, uncertainty, and impreciseness existing in criteria assessment and the significance of the model to the industry was valued by them. Interviewees agreed that the model is practically implemented easily using excel spreadsheet which are commonly used in computing tender scores. Once the template is developed using basic excel functions for an individual organisation, that has a repeated use for evaluating tenders. The validation results also reveal the potential modifications to the model and the directions for future research. As suggested, extending the fuzzy approach to sub- criteria rating can be determined by

further research as it needs to justify whether benefits are worth more than added complexity to the contractor selection problem. However, bench marking of criteria can be determined by conducting a new set of Delphi surveys.

Another important suggestion was to replace the step in which the average of fuzzy numbers (individual score of decision maker) is considered when finding the combining score for each criterion of the tender with an appropriate decision making tools. It was indicated that there is possibility of using a method for eliminating outliers rather than averaging. This can be considered under future research.

Another notable option is to use interviews with the top two contractors at the end of this desktop study and then select the best contractor as this gives opportunity for clients to understand the team culture of the contractor and the way the contractor is going to interact with the client. This can be included as a modification for clients who wish to go beyond desktop study and the final decision can be made by an interview. This needs to be further researched to implement in this model because it requires setting the range (difference of rank 1 and 2) so that the best tenders within that range are only called for an interview. For example, it can be within 0-3% difference.

As suggested, running a few tests in parallel to a model run by an industry partner and comparing the results possibly reveals the glitches; then, modifying the model as required can be a part of future research.

9.3 RESEARCH KNOWLEDGE CONTRIBUTIONS

This research study proposed a Fuzzy Multi-Attribute Analysis Model for EPC contractor selection in the Australian construction industry where fuzzy set theory has not previously been adopted for contractor selection. The significance of this model is that it can capture the fuzziness in commonly used multi-attribute evaluation in practice. Fuzzy set theory solves the fuzziness by modelling the uncertainty using linguistic terms instead numerical values that currently used in tender evaluation exercise. That is, because fuzzy set theory can capture the idea of natural language in discussing issues which are not precise. Thus, this study enhances the EPC contractor selection approach in the construction industry.

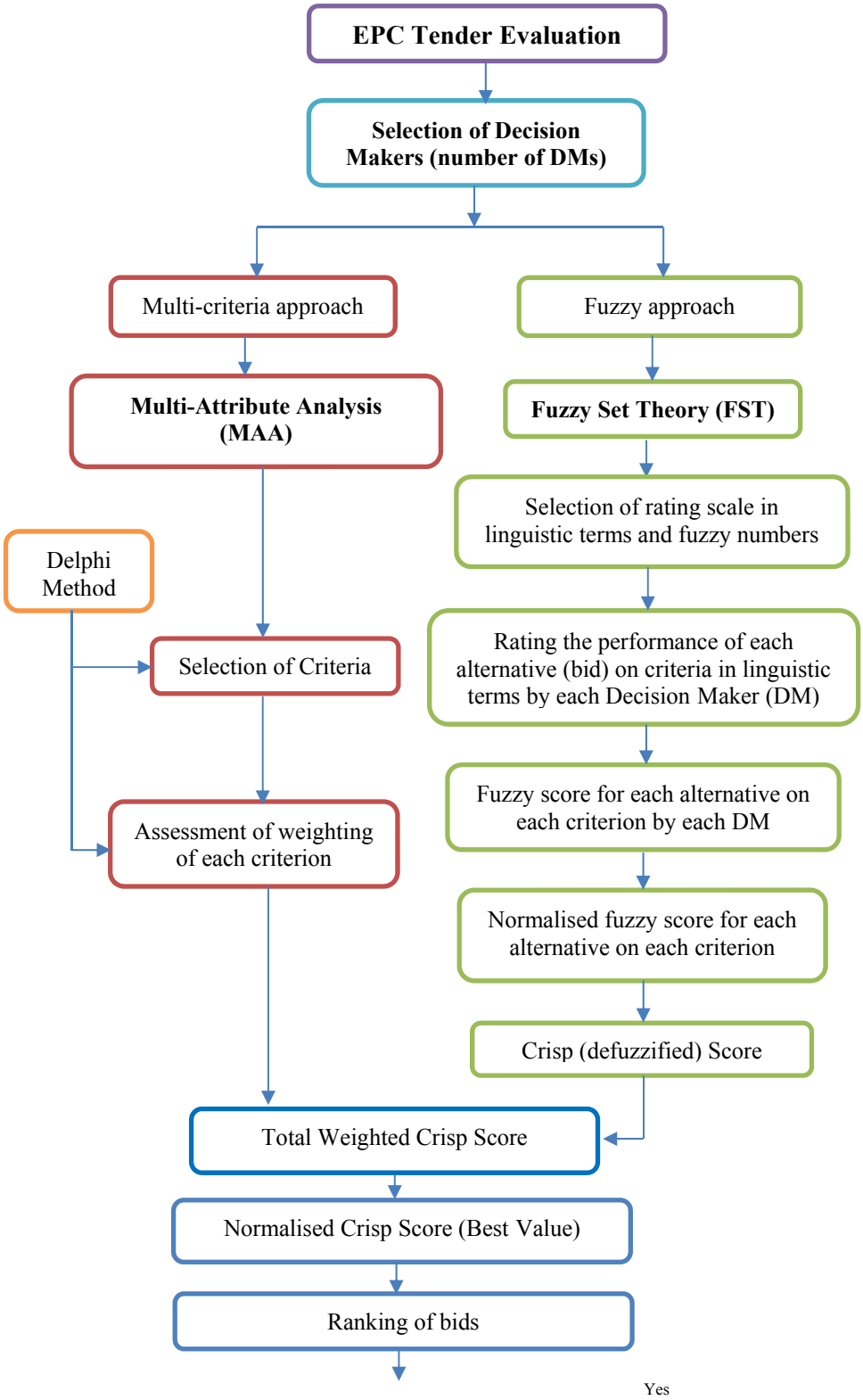
More importantly, this study contributes to the body of knowledge by filling the knowledge gap within EPC industry stakeholders with theoretical concepts and

practical evidence within the EPC industry. Nevertheless, there is hardly any criteria specific to EPC that would assist clients objectively selecting an appropriate contractor to achieve value for their investment by considering multiple attributes, this research fills the gap. The study findings present updated information to industry practitioners on EPC contractor selection criteria with relative importance weights, through a rigorous Delphi study with experts. This study has identified 16 criteria that are specific to EPC industry and the importance of each criterion when evaluating EPC tenders. Based on iindustry experts' opinion, the top five criteria are past performance, project understanding, technical, key personnel and health and safety.

Such findings are useful for both practitioners and academics. From a practitioner's point of view, existing practices for contractor selection should be further researched and modified to cater for new demands in the industry in a highly competitive environment and aligned with strong procurement objectives. For academics, this research exposes the area that needs to be further researched to cater to demands from the industry. Not having a step in this model to eliminate poorly performed tenderers until the final result (best-value) is achieved, the tender evaluation exercise can take longer than required. Time is money. An extended research to address this limitation using bench mark values for each criterion based on client preferences that facilitates early elimination of poorly performed tenders may benefit construction industry. More future research suggestions are discussed in 9.6.

9.4 FINAL FUZZY MULTI-ATTRIBUTE ANALYSIS MODEL FOR EPC TENDER EVALUATION

The final model is derived by application of validation findings. As such, to select the best contractor out of very close best-value tenders, interviews were introduced to determine the best contractor. This can be an optional step because if the gap between two tenders is widened (outside the acceptable limit), the rank 1 tender will be the best tender. However, acceptable limits need to be determined in advance by the client. or the agent. Fig. 9.1 presents the Fuzzy-Multi-Attribute-Analysis Model recommended for EPC contractor selection.


EPC Tender Evaluation
Selection of Decision Makers (number of DMs)
Multi-criteria approach
Fuzzy approach
Multi-Attribute Analysis (MAA)
Fuzzy Set Theory (FST)
Selection of rating scale in linguistic terms and fuzzy numbers
Delphi Method
Selection of Criteria
Rating the performance of each alternative (bid) on criteria in linguistic terms by each Decision Maker (DM)
Fuzzy score for each alternative on each criterion by each DM
Assessment of weighting of each criterion
Normalised fuzzy score for each alternative on each criterion
Crisp (defuzzified) Score
Total Weighted Crisp Score
Normalised Crisp Score (Best Value)
Ranking of bids
Yes

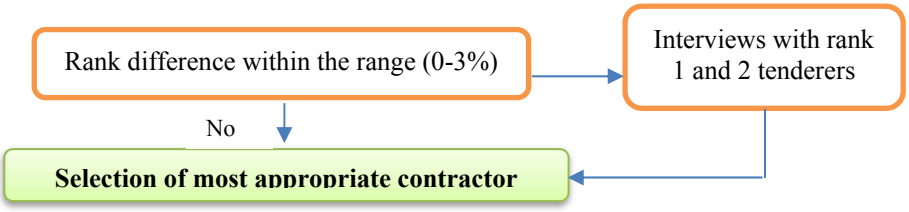


Figure 9.1 FMMA EPC Tender Evaluation Model

9.5 LIMITATION OF THE RESEARCH

Three rounds of Delphi survey with participants from various construction industries (e.g. infrastructure, oil and gas, etc.) were employed to identify and prioritise the criteria for EPC contractor selection. However, greater similarity could exist within an industry than between industries. Therefore, the criteria are not project or industry specific and some criteria may not be applicable to certain industries or specific EPC projects. This lack of generalisability is one of the limitations of this research. If the survey is possibly employed with many participants representing each industry or project types, then this industry/project specific data will potentially mitigate this limitation. Project or industry specific criteria and criteria importance weights can be determined.

Another limitation is there is no elimination until the final result (best-value) and this may result in carrying timely and costly tender evaluation of poorly performing tenderers, thus taking an unnecessarily longer time than required. Introducing an elimination step with 'benchmarking values' or allocating a 'pass mark' can help reducing the number of tenders selected for final evaluation. Only the tenders which obtain an over- minimum pass mark proceed to the next evaluation step and this will potentially help eliminate poor players in the specific criteria, which are more important to the project and where no trade-off can be endured.

Secondary data was analysed in the review of EPC market under this research. Acquiring statistical data concerning the EPC market was a challenging task and access to the full reports of the industry publication focus on EPC was also very limited. A nationwide questionnaire survey in the future should help to obtain a clearer picture of the EPC market in Australia through primary data analysis.

This model is advantageous with modelling the uncertainty, impreciseness and subjectivity in multi-attribute assessment. However, the model does not capture the confidence level of each decision maker of his/her assessment of contractor performance on each criterion, therefore some decision makers can be more

confident of their decision than others. Their working experience and the knowledge of the subject may be a factor that leads to less confident, moderately confident or overly confident decisions.

Even though, the contractors' view point is also important because they need to clearly understand the basis on which their tender is evaluated, considering the principals and consultant who represent the client and typically carry out tender evaluation exercise, only their opinions were considered during the validation process. This is another limitation of this research.

9.6 RECOMMENDATIONS FOR FUTURE WORK

The main recommendations for future research, are based on the limitations of this study identified during the validation process and are discussed below.

- This research used 16 criteria and their importance weights when developing the model. There is a recommendation for expanding the fuzzy approach to a sub-criteria level. There is a need for conducting further research to identifying the sub-criteria.
- An introduction of 'bench marking' to the main criteria, by which the low players can be eliminated under a future research can add value to the body of knowledge. It is highly recommended to carry out further research to establish bench mark values which facilitates an early elimination of poorly performed contractors.
- Another option is the use of a fuzzy approach combined with the Delphi survey to find the criteria weights despite the fact that the numerical scale (1-7) used in this model is reliable and valid statistically. Re-testing with a fuzzy approach in future research can be used to compare the findings of this model.
- Projects in the dynamic construction industry are unique. Therefore, the model needs to be further evaluated by running a few tests in a specific industry in parallel with the particular industry model, as this would help in identifying the effectiveness of the model to a specific industry. The evaluation should include several public infrastructure projects or private resource sector projects.

- Conducting similar research in other countries would improve the generalisability of the findings as well as supporting comparison of data in the global context.

9.7 SUMMARY

This chapter presented the outcome of the research achieving the aim and objectives. The aim of this study was to develop an EPC contractor selection model to evaluate tenders objectively. This aim was achieved through collection and analysis of data and literature review findings. The research objectives outlined in Chapter 1 were re-visited. Finally, the research was summarised with main findings, practical implications, and future research in this chapter. The well-argued research thesis supported with data provides significant insights in the EPC knowledge area, and provides EPC industry stake holders, owners and their agents with a structured approach to selecting the best contractor, as well as a way to future research on EPC.

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Appendices

Appendix A

Round 1 Delphi Questionnaire Survey template

SURVEY ON CRITERIA FOR ENGINEERING-PROCUREMENT- CONSTRUCTION (EPC) CONTRACTOR SELECTION- QUESTIONNAIRE 1

QUT Ethics Approval No: 1400000100

Research Overview

This research concerns the development of contractor selection framework to use when procuring an EPC project. The Delphi survey is to identify the criteria that need to be included in bid evaluation of EPC contractors within the competitive bidding in Australia through group consensus of expertise. The purpose of this first round of Delphi survey is to determine the criteria to be considered in EPC contractor selection. The purpose of the two subsequent round of Delphi will be to agree upon and validate the result of the preceding round. Please read the Participant Information sheet given here.

Please click "**NEXT**" to Continue...

PARTICIPANT INFORMATION SHEET:

PARTICIPANT INFORMATION FOR QUT RESEARCH PROJECT

Fuzzy Multi-Attribute Analysis Model for Engineering-Procurement-Construction (EPC) Contractor Selection in Australia

QUT Ethics Approval Number: 1400000100

RESEARCH TEAM

Principal Researcher:	Nayana Dissanayake	PhD student
Associate Researchers:	Dr Paul (Bo) Xia	Principal Supervisor
	Prof Martin	Associate Supervisor

Skitmore	
	Queensland University of Technology (QUT)
A/Prof Bambang Trigunarsyah	Associate Supervisor
	King Fahd University of Petroleum and Minerals (KFUPM)

DESCRIPTION

This project is being undertaken as part of Nayana Dissanayake's PhD study. The purpose of this project is to develop an EPC contractor selection model. The purpose of the survey is to identify criteria for bid evaluation of EPC contractor selection and determine the importance of the criteria. The research uses the Delphi method which will comprise of three rounds as below.

# Round	Objective
Round 1	To identify criteria for bid evaluation
Round 2	To determine importance for the criteria considering the overall results of Round 1
Round 3	To re-rate the selection criteria considering the overall results of Round 2

You are invited to participate in this project as you are recognised as being able to provide an invaluable contribution based on your extensive experiences in the construction industry.

PARTICIPATION

Participation will involve completing a survey in three rounds. The approximately time to complete each survey is given below.

# Round	Time to complete
Round 1	15 minutes
Round 2	15 minuets
Round 3	15 minutes

It would be greatly appreciated if you would participate in all three rounds of surveys. The surveys are scheduled to be completed within six months from the date of commencement. Responding to round 1 survey can be done online/via email. The responses for round 2 and 3 surveys are required to be sent via emails/online. Your participation in this project is entirely voluntary. If you agree to participate you do not have to complete any question(s) you are uncomfortable answering. Your decision to participate or not participate will in no way impact upon your current or future relationship with QUT or with your workplace. If you do agree to participate you can withdraw from the project without comment or penalty at any stage. Any identifiable information already obtained

from you will be destroyed.

EXPECTED BENEFITS

It is expected that your participation in this research project will directly benefit you in gaining advanced knowledge of EPC contractor selection criteria. It may also benefit the EPC industry. To recognise your contribution should you choose to participate the research team is offering participants a copy of research results on request.

RISKS

There are no risks beyond normal day-to-day living associated with your participation in this project.

PRIVACY & CONFIDENTIALITY

All comments and responses will be treated confidentially unless required by law. The names of individual persons are not required in any of the responses (completed surveys). To preserve confidentiality, the group responses (evaluation results) will be shared with other participants in each round of assessment and no identifiable information will be included. Any data collected as part of this project will be stored securely as per QUT's Management of research data policy. Please note that non-identifiable data collected in this project may be used as comparative data in future projects or stored on an open access database for secondary analysis.

CONSENT TO PARTICIPATE

You are requested to give your consent to participate in this research by reading this information sheet, and accepting these terms and conditions. The reply email giving consent to participate OR return of the completed survey online/via email (selecting a box in the survey) is accepted as an indication of your consent to participate in this project.

QUESTIONS / FURTHER INFORMATION ABOUT THE PROJECT

If you have any questions or require further information, please contact.

Nayana Dissanayake	07 3138 1731	nayana.dissanayake@hdr.qut.edu
Dr Paul Xia	07 3138 4373	paul.xia@qut.edu.au

CONCERNS / COMPLAINTS REGARDING THE CONDUCT OF THE PROJECT

QUT is committed to research integrity and the ethical conduct of research projects. However, if you do have any concerns or complaints about the ethical conduct of the project you may contact the QUT Research Ethics Unit on 07 3138 5123 or email ethicscontact@qut.edu.au. The QUT Research Ethics Unit is not connected with the research project and can facilitate a resolution to your concern in an impartial manner.

Thank you for helping with this research project.

*I have read the Participant Information sheet and give my informed consent to participate in this research study.

☐ Yes, I have read the information...

Part 1: Respondent Profile: This section asks the individual factors of the respondent.

1. In which sector are you currently working?

☐ Public Sector

☐ Private Sector

Other

.....

2. In which state/territory are you working?

☐ ACT

☐ NSW

☐ NT

☐ QLD

☐ SA

☐ TAS

☐ VIC

☐ WA

3. What term best describes your or your organisation involvement in a construction project?

○ Principal (e.g. project owner/initiator or their agent)

○ Consultant (e.g. professional service provider; engineering/management/technical)

○ Contractor (including sub contractors and suppliers)

Other

.....

4. What term best describes your functional role within a project or the organisation

○ General Manager

○ Project Manager

○ Engineering Manager

○ Business Manager

○ Construction Manager

○ Contract Manager

○ Procurement Manager

○ Contract Administrator

- ○ Facilities Manager
- ○ Architect
- ○ Academic Professional
- ○ Engineer
- ○ Legal Professional

Other
.....

5. How long have you been working in the construction industry?

- ○ 0-5 years
- ○ 5-10 years
- ○ 10-20 years
- ○ Over 20 years

6. What term best describes the main construction industry you work in?

- ☐ Transport Infrastructure (Roads, highways, rail projects, airport runways)
- ☐ Utilities Infrastructure (Electricity generation and supply, sewerage, drainage and water storage and supply)
- ☐ Telecommunication Infrastructure
- ☐ Mining and Mineral Processing
- ☐ Heavy Industrial Construction (Oil and gas, chemical plants, steel mills, industrial processing plants etc.)
- ☐ Pipelines
- ☐ Buildings

Other
.....

7. What are the project delivery methods that have been used to deliver previous or current major projects?

- ☐ EPC (Engineering-Procurement-Construction)
- ☐ EPCI (Engineering-Procurement-Construction-Installation)
- ☐ EPCM (Engineering-Procurement-Construction Management)
- ☐ EPCIC (Engineering-Procurement-Construction-Installation-Commissioning)
- ☐ EPCC (Engineering-Procurement-Construction-Commissioning)
- ☐ DC (Design & Construct) or DB (Design & Build)
- ☐ Turnkey
- ☐ BOT (Built-Operate-Transfer)
- ☐ BOOT (Built-Own-Operate-Transfer)

Other
.....

8. How many years of experience do you have in relation to EPC or Design and Construct (DC) project delivery?
○ 0-5 years
○ 5-10 years
○ 10-20 years
○ Over 20 years

9. Approximately what is the largest estimated project cost of current or previous EPC or DC projects that you have involved in?
○ Less than A$ 50M
○ Between A$50M-A$100M
○ Between A$100M-A$250M
○ Between A$250M-A$500M
○ Between A$500M-A$1B
○ Between A$1B-A$5B
○ More than A$5B
○ Do not know

Part 2: Key Criteria for EPC Contractor Selection (Bid Evaluation)

10. This section asks you select the criteria from the list that you would consider to be important for EPC contractor selection.		
	Check all that apply	Comments
1. Financial:	☐	.....
Provides details of financial capability in terms of financial statements		
2. Past Performance:	☐	.....
Demonstrates the performance of recently completed projects in terms cost, time and qualities, and cooperative behaviour (conflicts/disputes)		
3. Past experience:	☐	.....
Provides the details of scale and type of past projects, and demonstrates the experience in similar projects, in the region, and familiarity of relevant project delivery method		

4. Technical:	☐	.....
Demonstrates the technical capability and capacity that include technical solution, alternative designs, expertise, specialisation, technical qualification, staffing levels, technology and equipment resources, engineering systems, creativity and innovation, and availability for operation, maintenance, repair and training needs		
5. Management:	☐	.....
Demonstrates the business management system that include project management system, risk carrying ability and willingness, management personnel, and management accountabilities		
6. Organisational:	☐	.....
Provides the details of company size, company image, age in business, organisational structure, policies, memberships, current workload, and resources (labour, plant, equipment, human resources)		
7. Health and Safety:	☐	.....
Outlines the accountabilities for Occupational Health and Safety (OHS) with plans and systems and demonstrates the performance with OHS records		
8. Environment and Sustainability:	☐	.....
Outlines the Environmental management plan and commitment to sustainability		
9. Key personnel:	☐	.....
Provides the details of key personnel to be employed, proposed roles, their experience and skills, academic and professional qualification, years with the company, and their training		
10. Relationships:	☐	.....
Provides the details of subcontractors/suppliers that include the length of time with them, labour employment agreement, and maintenance of workers' compensation liabilities		

11. Time:	☐	.....
Provides a program indicating start and finish dates, and adherence to the dates/duration given in tender documents		
12. Cost:	☐	.....
Includes tendered price, and assessment of capital cost, life cycle cost, etc.		
13. Quality:	☐	.....
Outlines quality control and quality assurance systems, and compliance with specifications and quality standards		
14. Contractual and legal status:	☐	.....
Demonstrates the disputes and resolution strategy, attitude towards claims, acceptance of contract terms and conditions, and compliance with the codes		
15. Project understanding:	☐	.....
Responds to Request for Proposal (RFP), and demonstrates project specific criteria		
16. Geographic location:	☐	.....
Outlines the familiarity of local environment, and proximity to project		

11. What other criteria do you think to be included in EPC bid evaluation process?
……

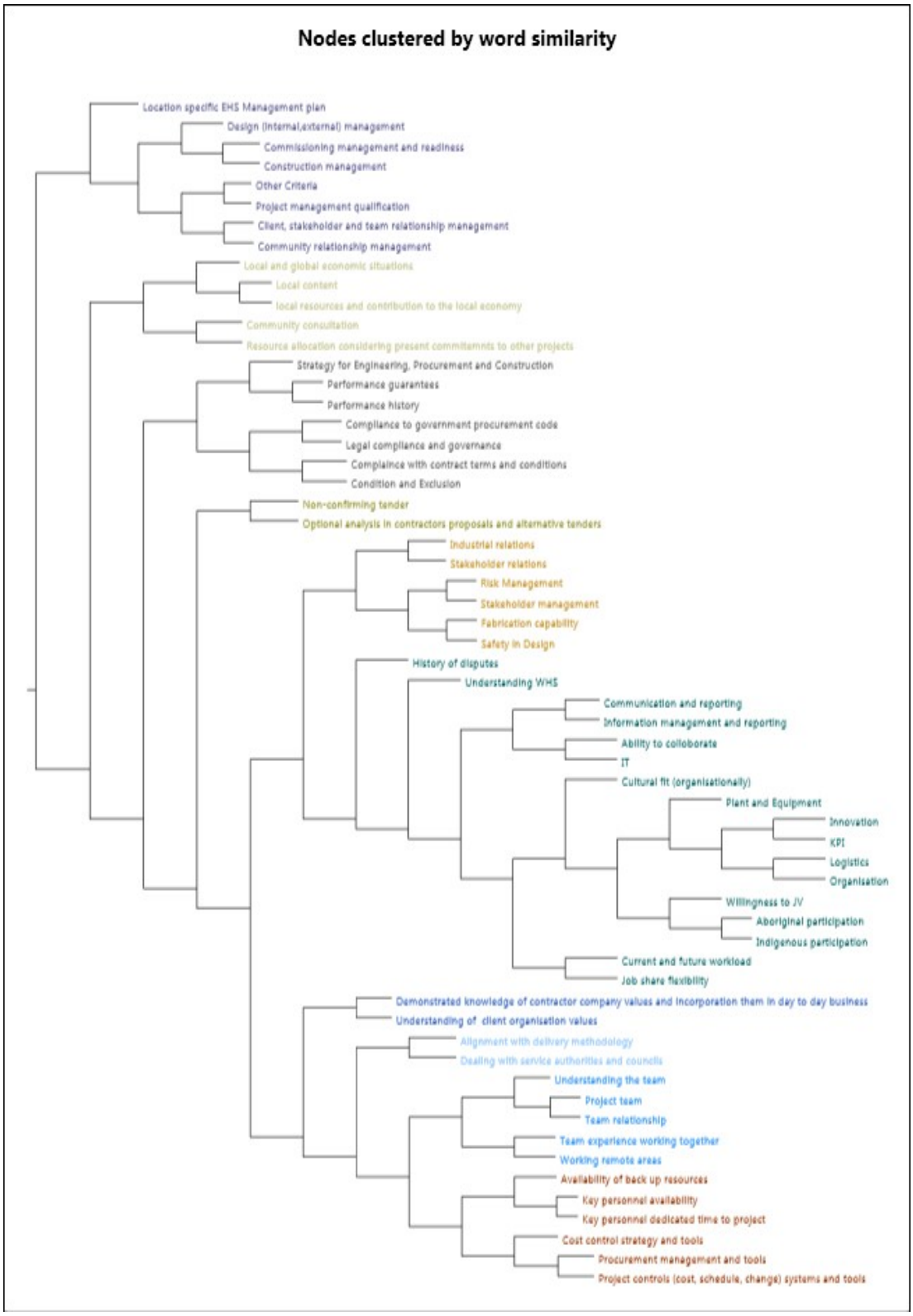
12. Do you have any other comments pertinent to contractor selection for an EPC project?
……

Please click "SUBMIT" to upload your responses.

Please do not hesitate to contact Nayana Dissanayake by email nayana.dissanayake@hdr.qut.edu.au or on 07 3138 1731 should you have any questions.

Appendix B

Round 1 Questionnaire: Cluster Analysis using NVivo on 'other criteria'



Appendix C

Mapping of respondents' suggestions with existing criteria

#	Respondents suggestions	Existing criteria										Potential new criteria							
		Management*	**EHS***	**Relationships***	**Key personnel***	**Sustainability & Environment***	**Contractual ***	**Past performance***	**Technical***	**Organisational***	**Geographic location***	**Collaboration**	**Present commitments**	**Procurement**	**Legal**	**Industrial Relations**	**Comm. & reporting**	**Innovation**	**Logistic & Supply chain**
1	Ability to collaborate				✓					✓		✓							
2	Aboriginal participation					✓													
3	Alignment with delivery methodology						✓												
4	Availability of back up resources				✓					✓									
5	Client, stakeholder and team relationship management			✓															
6	Commissioning management and readiness	✓										✓							
7	Communication and reporting																✓		
8	Community consultation			✓															
9	Community relationship management			✓															
10	Compliance to government procurement code													✓					
11	Compliance with contract terms and conditions						✓												
12	Condition and Exclusion						✓												
13	Construction management	✓																	
14	Cost control strategy and tools	✓																	
15	Cultural fit (organisationally)				✓					✓									

16	Current and future workload																		
17	Dealing with service authorities and councils	✓							✓										
18	Demonstrated knowledge of contractor company values and incorporation them in day to day business									✓									
19	Design (internal/external) management	✓							✓										
20	Fabrication capability								✓										
21	History of disputes							✓											
22	Indigenous participation					✓													
23	Industrial relations															✓			
24	Information management and reporting																✓		
25	Innovation																	✓	
26	IT																✓		
27	Job share flexibility				✓							✓							
28	Key personnel availability				✓					✓			✓						
29	Key personnel dedicated time to project				✓					✓			✓						
30	KPI	✓																	
31	Legal compliance and governance														✓				
32	Local and global economic situations					✓													
33	Local content					✓													
34	local resources and contribution to the local economy					✓													
35	Location specific EHS Management plan		✓																
36	Logistics																		✓
37	Non-confirming tender																		
38	Optional analysis in contractor's proposals and alternative tenders																		

39	Organisation									✓									
40	Performance guarantees						✓												
41	Performance history							✓											
42	Plant and Equipment									✓									
43	Procurement management and tools													✓					
44	Project controls (cost, schedule, change) systems and tools	✓																	
45	Project management qualification				✓														
46	Project team				✓														
47	Resource allocation considering present commitments to other projects									✓			✓						
48	Risk Management	✓																	
49	Safety in Design								✓										
50	Stakeholder management	✓																	
51	Stakeholder relations			✓															
52	Strategy for Engineering, Procurement and Construction	✓							✓										
53	Team experience working together			✓	✓							✓							
54	Team relationship			✓															
55	Understanding of client organisation values									✓									
56	Understanding the team			✓															
57	Understanding WHS		✓																
58	Willingness to JV			✓			✓												
59	Working remote areas				✓					✓	✓								
	Response %											5%	6%	3%	2%	5%	5%	3%	2%

Appendix D

Round 2 Delphi Questionnaire Survey template

SURVEY ON CRITERIA FOR ENGINEERING-PROCUREMENT- CONSTRUCTION (EPC) CONTRACTOR SELECTION- QUESTIONNAIRE 2

QUT Ethics Approval No: 1400000100

Research Overview

This research concerns the development of contractor selection framework to use when procuring an EPC project. The Delphi survey is to identify the criteria that need to be included in bid evaluation of EPC contractors within the competitive bidding in Australia through group consensus of expertise. The purpose of this second round of Delphi survey is to determine the importance of the criteria (which identified by the first round) in EPC contractor bid evaluation. The purpose of the third round of Delphi will be to validate the results of the preceding round. Please read the Participant Information sheet given here.

Please click "NEXT" to Continue...

PARTICIPANT INFORMATION SHEET:

PARTICIPANT INFORMATION FOR QUT RESEARCH PROJECT

Fuzzy Multi-Attribute Analysis Model for Engineering-Procurement-Construction (EPC) Contractor Selection in Australia

QUT Ethics Approval Number: 1400000100

RESEARCH TEAM

Principal Researcher:	Nayana Dissanayake	PhD student
Associate Researchers:	Dr Paul (Bo) Xia	Principal Supervisor
	Prof Martin Skitmore	Associate Supervisor
		Queensland University of Technology (QUT)
	A/Prof Bambang	Associate Supervisor

Trigunarsyah

King Fahd University of Petroleum and Minerals (KFUPM)

DESCRIPTION

This project is being undertaken as part of Nayana Dissanayake's PhD study. The purpose of this project is to develop an EPC contractor selection model. The purpose of the survey is to identify criteria for bid evaluation of EPC contractor selection and determine the importance of the criteria. The research uses the Delphi method which will comprise of three rounds as below.

# Round	Objective
Round 1	To identify criteria for bid evaluation
Round 2	To determine importance for the criteria considering the overall results of Round 1
Round 3	To re-rate the selection criteria considering the overall results of Round 2

You are invited to participate in this project as you are recognised as being able to provide an invaluable contribution based on your extensive experiences in the construction industry.

PARTICIPATION

Participation will involve completing a survey in three rounds. The approximately time to complete each survey is given below.

# Round	Time to complete
Round 1	15 minutes
Round 2	15 minuets
Round 3	15 minutes

It would be greatly appreciated if you would participate in all three rounds of surveys. The surveys are scheduled to be completed within six months from the date of commencement. Responding to round 1 survey can be done online/via email. The responses for round 2 and 3 surveys are required to be sent via emails/online. Your participation in this project is entirely voluntary. If you agree to participate you do not have to complete any question(s) you are uncomfortable answering. Your decision to participate or not participate will in no way impact upon your current or future relationship with QUT or with your workplace. If you do agree to participate you can withdraw from the project without comment or penalty at any stage. Any identifiable information already obtained from you will be destroyed.

EXPECTED BENEFITS

It is expected that your participation in this research project will directly benefit you in gaining advanced knowledge of EPC contractor selection criteria. It may also benefit the EPC industry. To recognise your contribution should you choose to participate the research team is offering participants a copy of research results on request.

RISKS

There are no risks beyond normal day-to-day living associated with your participation in this project.

PRIVACY & CONFIDENTIALITY

All comments and responses will be treated confidentially unless required by law. The names of individual persons are not required in any of the responses (completed surveys). To preserve confidentiality, the group responses (evaluation results) will be shared with other participants in each round of assessment and no identifiable information will be included. Any data collected as part of this project will be stored securely as per QUT's Management of research data policy. Please note that non-identifiable data collected in this project may be used as comparative data in future projects or stored on an open access database for secondary analysis.

CONSENT TO PARTICIPATE

You are requested to give your consent to participate in this research by reading this information sheet, and accepting these terms and conditions. The reply email giving consent to participate OR return of the completed survey online/via email (selecting a box in the survey) is accepted as an indication of your consent to participate in this project.

QUESTIONS / FURTHER INFORMATION ABOUT THE PROJECT

If you have any questions or require further information, please contact.

Nayana Dissanayake	07 3138 1731	nayana.dissanayake@hdr.qut.edu.au
Dr Paul Xia	07 3138 4373	paul.xia@qut.edu.au

CONCERNS / COMPLAINTS REGARDING THE CONDUCT OF THE PROJECT

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Thank you for helping with this research project.

*I have read the Participant Information sheet and give my informed consent to participate in this research study.

☐ Yes, I have read the information...

Part 1: Key Criteria for EPC Contractor Selection (Bid Evaluation)

1. In this section you will be asked to rate each criteria importance for EPC contractor selection. Please rate how important the following criteria are to you in practice on a scale of 1 to 7, where 1 is 'not at all important' and 7 is 'extremely important'. Please note that round 1 result of frequency for each criterion is given for your consideration. A new criterion (Industrial Relations) has been included based on round 1 feedback.

	1	2	3	4	5	6	7	Do not know
1. Past Performance: Round 1 frequency =100%	○	○	○	○	○	○	○	○
Demonstrates the performance of recently completed projects with records of project cost, completion time and quality, contract claims and variation history, cooperative behaviour (conflicts/disputes), and penalties, etc.								
2. Technical: Round 1 frequency = 97%	○	○	○	○	○	○	○	○
Demonstrates technical capability and capacity that includes sound engineering solutions, safety in design, creativity and innovation, constructability, engineering and technical expertise, technology and equipment resources, engineering systems, etc. Demonstrates technical support for commissioning, operation readiness, handover, maintenance, repair and training needs.								
3. Key personnel: Round 1 frequency = 97%	○	○	○	○	○	○	○	○
Provides the details of key project personnel which include proposed roles, experience and skills, academic and professional qualification, years with the company, and professional development plan. Demonstrates project team ability to work collaboratively and as a part of diverse teams, and availability for backup resources.								
4. Past Experience: Round 1 frequency = 95%	○	○	○	○	○	○	○	○

Provides details of scale, complexity, and type of past projects, and demonstrates project experience of similar type(s) in a similar environment.								
5. Health and Safety: Round 1 frequency = 94%	○	○	○	○	○	○	○	○
Outlines accountabilities for Occupational Health and Safety (OHS) providing samples of site specific management plans, corporate systems, and procedures that identify and control OHS risks. Provides documentary evidence of corporate OHS performance including OHS records of recent projects.								
6. Financial: Round 1 frequency = 92%	○	○	○	○	○	○	○	○
Demonstrates contractor financial viability and financial performance over a defined period and provided financial statements, which include balance sheet, profit and loss statement, etc.								
7. Cost: Round 1 frequency = 92%	○	○	○	○	○	○	○	○
Includes tendered price, life-cycle costing, etc.								
8. Time: Round 1 frequency = 91%	○	○	○	○	○	○	○	○
Provides a project schedule with milestones, activities and deliverables with intended start and finish dates, or complies with the time constraints given in tender documents.								
9. Quality: Round 1 frequency = 89%	○	○	○	○	○	○	○	○
Outlines quality control and quality assurance systems, and complies with specifications and quality standards.								
10. Project understanding: Round 1 frequency = 88%	○	○	○	○	○	○	○	○
Demonstrates understanding of Request for Proposal (RFP), local context, project risks, unique owner standards and requirements, how the project can be executed to meet client expectations,								

and explains exceptions from RFP, outlines expected degree of owner involvement, approvals, etc.								
11. Management: Round 1 frequency = 83%	○	○	○	○	○	○	○	○
Demonstrates Construction Project Management (CPM) capability (risk management strategy, procurement strategy, stakeholder management plan, logistic and supply chain management, preferred suppliers/subcontractors, and key trade packages, etc.								
12. Organisational: Round 1 frequency = 80%	○	○	○	○	○	○	○	○
Outlines business values and corporate commitment and provides the details of company size, company image, age in business, organisational structure, policies, memberships, current and potential future work commitments, resource optimisation (people, plant, equipment), in-house systems, etc.								
13. Environment and Sustainability: Round 1 frequency = 75%	○	○	○	○	○	○	○	○
Takes the stakeholders' expectations, which include environmental requirements, social acceptances (e.g. local resources, local economy, indigenous participation, etc.), sustainability approach (products and processes) into account.								
14. Contractual and Legal: Round 1 frequency = 75%	○	○	○	○	○	○	○	○
Accepts Contract terms and Conditions or provides clear, concise exclusions or conditional acceptances. Indicates compliance with all relevant codes and regulations.								
15. Relationships: Round 1 frequency = 73%	○	○	○	○	○	○	○	○
Demonstrates ability to develop strong and long term partnerships with clients, vendors and suppliers by providing client/subcontractor/supplier referees								

including information regarding the duration of the relationship, etc.								
16. Geographic location: Round 1 frequency = 56%	○	○	○	○	○	○	○	○
Outlines familiarity of local environment, and proximity to project (i.e. proposed work locale) and /or demonstrates work locations worldwide that can work together.								
17. Industrial Relations: New (Round 1 frequency = 5%)	○	○	○	○	○	○	○	○
Demonstrates employee and industrial relations plan/policy and maintenance of project agreements, multiemployer agreements, workers' compensation liabilities, etc. and provides recent industrial relations record.								

Please click "**SUBMIT**" to upload the responses

Please do not hesitate to contact Nayana Dissanayake by email nayana.dissanayake@hdr.qut.edu.au or on 07 3138 1731 should you have any questions.

Appendix E

Round 3 Delphi Questionnaire Survey template

SURVEY ON CRITERIA FOR ENGINEERING-PROCUREMENT- CONSTRUCTION (EPC) CONTRACTOR SELECTION- QUESTIONNAIRE 3

QUT Ethics Approval No: 1400000100

Research Overview

This research concerns the development of contractor selection framework to use when procuring an EPC project. The Delphi survey is to identify the criteria that need to be included in bid evaluation of EPC contractors within the competitive bidding in Australia through group consensus of expertise. The purpose of this (third) round of Delphi survey is to re-rate the criteria considering the overall results of round 2.

Please click "**NEXT**" to Continue...

Part 1: Key Criteria for EPC Contractor Selection (Bid Evaluation)

Please review the round 2 ratings (your rating with the group rating) for each criterion. If you consider to alter your previous rating in light of group response to each criterion, please use same rating scale from 1 to 7, where 1 is not at all important and 7 is extremely important.

	1	2	3	4	5	6	7	Do not know
1. Past Performance: Your Round 2 Rating =_____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Demonstrates the performance of recently completed projects with records of project cost, completion time and quality, contract claims and variation								

history, cooperative behaviour (conflicts/disputes), and penalties, etc.								
2. Technical: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Demonstrates technical capability and capacity that includes sound engineering solutions, safety in design, creativity and innovation, constructability, engineering and technical expertise, technology and equipment resources, engineering systems, etc. Demonstrates technical support for commissioning, operation readiness, handover, maintenance, repair and training needs.								
3. Key personnel: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Provides the details of key project personnel which include proposed roles, experience and skills, academic and professional qualification, years with the company, and professional development plan. Demonstrates project team ability to work collaboratively and as a part of diverse teams, and availability for backup resources.								
4. Past Experience: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Provides details of scale, complexity, and type of past projects, and demonstrates project experience of similar type(s) in a similar environment.								
5. Health and Safety: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Outlines accountabilities for Occupational Health and Safety (OHS) providing samples of site specific management plans, corporate systems, and procedures that identify and control OHS risks. Provides documentary evidence of corporate OHS performance including								

OHS records of recent projects.								
6. Financial: Your Round 2 Rating = _____; Group Rating (mean)= _____	○	○	○	○	○	○	○	○
Demonstrates contractor financial viability and financial performance over a defined period and provided financial statements, which include balance sheet, profit and loss statement, etc.								
7. Cost: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Includes tendered price, life-cycle costing, etc.								
8. Time: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Provides a project schedule with milestones, activities and deliverables with intended start and finish dates, or complies with the time constraints given in tender documents.								
9. Quality: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Outlines quality control and quality assurance systems, and complies with specifications and quality standards.								
10. Project understanding: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Demonstrates understanding of Request for Proposal (RFP), local context, project risks, unique owner standards and requirements, how the project can be executed to meet client expectations, and explains exceptions from RFP, outlines expected degree of owner involvement, approvals, etc.								
11. Management: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Demonstrates Construction Project Management (CPM) capability (risk management strategy, procurement								

strategy, stakeholder management plan, logistic and supply chain management, preferred suppliers/subcontractors, and key trade packages, etc.								
12. Organisational: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Outlines business values and corporate commitment and provides the details of company size, company image, age in business, organisational structure, policies, memberships, current and potential future work commitments, resource optimisation (people, plant, equipment), in-house systems, etc.								
13. Environment and Sustainability: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Takes the stakeholders' expectations, which include environmental requirements, social acceptances (e.g. local resources, local economy, indigenous participation, etc.), sustainability approach (products and processes) into account.								
14. Contractual and Legal: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Accepts Contract terms and Conditions or provides clear, concise exclusions or conditional acceptances. Indicates compliance with all relevant codes and regulations.								
15. Relationships: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Demonstrates ability to develop strong and long term partnerships with clients, vendors and suppliers by providing client/subcontractor/supplier referees including information regarding the duration of the relationship, etc.								
16. Geographic location: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○

Outlines familiarity of local environment, and proximity to project (i.e. proposed work locale) and /or demonstrates work locations worldwide that can work together.								
17. Industrial Relations: Your Round 2 Rating = _____; Group Rating (mean) = _____	○	○	○	○	○	○	○	○
Demonstrates employee and industrial relations plan/policy and maintenance of project agreements, multiemployer agreements, workers' compensation liabilities, etc. and provides recent industrial relations record.								

Please click "**SUBMIT**" to upload your responses (even if you have not altered your previous responses).

Please do not hesitate to contact Nayana Dissanayake by email nayana.dissanayake@hdr.qut.edu.au or on 07 3138 1731 should you have any questions.

Appendix F

Semi-structured interview template

Queensland University of Technology Brisbane Australia	Interview Template

Fuzzy Multi Attribute Analysis Model for Engineering-Procurement-Construction (EPC) Contractor Selection in Australia

QUT Ethics Approval Number 1400000100

Interviewee Name:		Date of Interview
Interview start and end times:	Start:	End:
Interviewer:		Location of interview:

Interviewee Profile		
Organisation type	Principal/consultant/other	
Functional Role	Project Manager/Procurement Manager/etc.	
Experience in EPC project experience	No of years	

Interview questions

Q1	To what extent is the EPC contractor selection framework practical and comprehensive? Specify the practical limitations of industry use.

Q2	Is the list of contractor selection criteria comprehensive? Are the criteria importance weights rated appropriately? Has the model addressed the new demands in the EPC industry (in terms of criteria and their importance weights)?

Q3	What would you suggest to overcome the limitations or drawbacks mentioned in answering Q1 or Q2?

Q4	Do you think the Fuzzy Multi-Attribute Analysis Model will facilitate the owner to select the most appropriate contractor addressing the subjectivity, uncertainty, impreciseness and incompleteness in contractor selection decision making process?

Q5	If you were given the opportunity to use this model in your current project, would there be a significant difference in final results of the contractor selection?

Q6	How would you rate the model (strongly recommended/recommended/not sure/not recommended)?

Appendix G

Responses to interview questions

#	Question label/	Perceptions (on model)	Ideas (suggestions to improve model)
Q1	EPC contractor selection framework		
	Interviewee A	• Practical, • Useful • Two stage process is recommended, never get lowest price and	• Single stage can be used if the client nominated contractor for certain type of contracts • Model can be used to recommend the best contractors (top 2) and final decision after an interview with each of the contractor
	Interviewee B	• Prequalification/best value	• Prequalification only for government projects (no selected tendering) • Always competitive bidding • Open tendering depends on funding source-for federal funded projects -open bidding • Normally best-value is least explained-it is subjectively assessed/mathematical calculation/qualitative weighting/quantitative weighting-that's the dilemma we faced
	Interviewee C	• Requirement for prequalification depends on the market-if there are only three no need to short list • Not lowest bid- long term performance is important	
Q2	Selection criteria and criteria importance weights		
	Interviewee A	• Criteria-good	• Get weightings from a few people in the industry and compare with the weighting from the Delphi survey findings • Include schedule of rates under cost because contracts always have

			variations and this needs to be evaluated • Fuzzy numbers may be used to find weightings
	Interviewee B	• Past performance-low rating • Key personnel -very important-high rating • Project understanding-high rating • Past performance and innovation are not of much importance particularly in roads	• Past performance is assessment of company systems, processes and culture • Key personnel-interview key personnel for approx. 8 min (team members individually and as team with the boss) as a part of evaluation-this test how the team is going to work together and with me • Past experience-we use reports on past experience-two reports; 1. Client/superintendent said you did well 2. Client/superintendent said you did poorly because everyone has failure and the report on a similar project and three learnings from these projects and how the learnings be applied in this project • Assessed whether company has interest about the project • No more than 10 criteria-as it causes dilution effect and not point of inclusion of a criteria if all the contractors give same result • Past performance-newest approach in B's team-one project example • Should give model answers-panel members scare to give high marks • 'Pass mark' is important -who gets below the pass mark is not proceeded to the next level-that's a way to cut the people-pass mark is the risk • Always quantify the responses-restrict the number of examples, limit number of pages and guide them
	Interviewee C	• Some of the criteria may not be suitable -you can't generic like this-depends on individual project • Weighting highly depend on what you asked	• Benchmarking of criteria is required (e.g. for safety if score is less than 6, it is not considered) • Life-cycle-cost of the asset should be very much a separate criterion-to assess life-cycle cost and performance

		(criteria)-no generalisation	
Q3	Limitations/drawbacks		
	Interviewee A	• Weighting is not generic as in this model	• Obtain weighting by giving rating 1-5 scale by the decision makers as each organisation has different view and it is different from person to person-this can be done during a company workshop • Fuzzy number approach to weighting because weighting has large impact on a project
	Interviewee B	• Averaging not permitted-robust discussion should be	
	Interviewee C	• Criteria is in general-needs understanding criteria properly to the project and owner-selecting criteria in general is dangerous	
Q4	Selection of contractor using MAA and FST		
	Interviewee A	• Model can be implemented by using of Excel spreadsheet for calculation, in that point of view this is easy to use in office • Contractors become familiarised with MAA	• When contractor becomes familiar with the MAA, they score full marks-so use the model to recommend contractors for face-to-face interview by which final decision is made • Engineers typically work with numbers so use numerical rating scale to rate the contractor performance and use linguistic terms as a legend • Also, introduce fuzzy to next level (sub-criteria)
	Interviewee B	• Use of Fuzzy set theory is interesting • No averaging (fuzzy numbers)-it is correct only when everyone gives close answers if there is a low player, it is a problem	• Panel members should argue and come to an agreement for example 3 or 7 • Think about how you going to assess when two say 7 and other two say 3-is it 5? If contractor is very good, is it fair?
	Interviewee C	• MAA is subjective assessment based on judgement • Model shows uncertainty in only linguistic terms, it does not address the how sure are they	• It is worth valuing how the panel members are sure about their verbal responses and accordingly adjusting the fuzzy numbers to reflect uncertainty.

		doing assessment (some panel members may sure but some are not)	
Q5	Significance of the model		
	Interviewee A	• Very good at selecting the two top contractors but not selecting the best contractor • Need to run a few tests runs to see the significant difference from existing systems • Weighting system is not new but application of fuzzy set theory is new	• Final selection should be through interviews • It would be interesting to get companies like John Holland, or Brisbane City Council who have a lot of infrastructure work on board as there is certainly a function to play in the model. • Run this model in parallel to theirs (company's)
	Interviewee B	• Apart from averaging process, other parts are good and significant	
	Interviewee C	• Needs to do comparison with industry use weighted model and this fuzzy introduced model to see the significance	
Q6	Rating the model		
	Interviewee A	• I would recommend to any one to use model like this without any doubt-I personally use this to select for top 2 but this wouldn't be my final decision, final decision will be through an interview specially when the tenders are mathematically very close to • Doing this desktop study for a final decision is incorrect	• Final decision by interview-you need to meet the people • Most important is people who are doing the project is important- it is subjective criteria-however, very, very little you can do as an individual but much more as a team
	Interviewee B	• Need changing averaging process	
	Interviewee C	• Good but need to demonstrate the significance by testing with MAA only and with MAA+Fuzzy	

Appendix H

Other important insights from the interviews

Interviewee	Insight from the interviewee
Interviewee A	
Interviewee B	• Contractors are prequalified nationally in every 3 years by Financial (F level) in B's organisation. Re-assessed at locally (e.g. Brisbane office) • EPC road industry in Australia is unique with pre-qualification • In some organisations, contractor should guarantee the people whose names given in the tender document are the project team otherwise there will be penalties-considered as trust-worthiness factor • Project understanding-all invited contractors should have equal opportunity to apply-i.e. level field playing concept (if they don't know something you can't asked that question) • Should know the market first, then you know the responses -if your questions get the same answer, it is not good • B's team use ...number of criteria • Panel members should have good understanding of project, experience and knowledge • 3-4 decision makers in a panel-unfortunately no one want to be there • Interview B uses Kepner Tregoe method (K-T methodology) - decision making method using rating scale 1-10
Interviewee C	• Well-argued research supported by data is the most important-without data you do not have opinions-so good doing this type of research

Appendix I

Worked Example

Following numerical example illustrates the FMAA model use in EPC contractor selection. Three contractors (A1, A2 and A3) have submitted for an EPC project (hypothetical) in Australia.

Step 1: Selection of Decision makers

Let number of decision makers be 3 which means k=3 in this case.

$$DM_k = [DM_1 \qquad DM_2 \qquad DM_3]$$

Step 2: Identification of EPC contractor selection criteria and prioritisation of criteria by importance

Criteria (CRi) for contractor selection where i=1,2, ……., 16

$$CR_i = [CR_1 \qquad CR_2 \qquad CR_3 \qquad \cdots \qquad CR_{16}]$$

Criteria weighting are given in 'weighting matrix' W_j where j=1,……m)

$$W_j = \begin{bmatrix} W_1 = 0.0684 \\ W_2 = 0.0678 \\ \vdots \\ \vdots \\ W16 = 0.0538 \end{bmatrix}$$

Step 3: Selection of rating scale/linguistic terms/fuzzy numbers

Linguistic terms given in Table 7.2 were selected for rating the contractor performance against each criterion in this numerical example. Using the triangular fuzzy numbers given in Table 7.3, these linguistic terms were transformed to fuzzy numbers.

Step 4: Judgement of each contractor on each criterion in linguistic terms

Decision makers' judgments on contractor performance on each criterion in linguistic terms were given in Table 1.

Table AI.1 Decision makers rating in linguistic terms

		Tender	DM_1	DM_2	DM_3
CR_1	Past Performance	A1	MG	G	MG
		A2	G	G	MG
		A3	VG	G	F
CR_2	Project understanding	A1	F	G	G
		A2	VG	VG	G
		A3	G	MG	VG
CR_3	Technical	A1	G	MG	F
		A2	VG	VG	VG
		A3	MG	G	VG
CR_4	Key personnel	A1	VG	G	VG
		A2	VG	VG	VG
		A3	G	VG	MG
CR_5	Health and Safety	A1	MG	G	MG
		A2	G	G	MG
		A3	VG	G	F
CR_6	Time	A1	G	MG	F
		A2	VG	VG	VG
		A3	MG	G	VG
CR_7	Financial	A1	VG	G	VG
		A2	VG	VG	VG
		A3	G	VG	MG
CR_8	Contractual and legal	A1	F	F	F
		A2	VG	MG	G
		A3	G	G	MG
CR_9	Past experience	A1	F	F	F
		A2	VG	MG	G
		A3	G	G	MG
CR_{10}	Management	A1	F	G	G
		A2	VG	VG	G
		A3	G	MG	VG

CR_{11}	Cost	A1	G	MG	F
		A2	VG	VG	VG
		A3	MG	G	VG
CR_{12}	Quality	A1	MG	G	MG
		A2	G	G	MG
		A3	VG	G	F
CR_{13}	Relationships	A1	F	G	G
		A2	VG	VG	G
		A3	G	MG	VG
CR_{14}	Industrial relations	A1	MG	G	MG
		A2	G	G	MG
		A3	VG	G	F
CR_{15}	Environment and Sustainability	A1	VG	G	VG
		A2	VG	VG	VG
		A3	G	VG	MG
CR_{16}	Organisational	A1	F	F	F
		A2	VG	MG	G
		A3	G	G	MG

Step 5: **Fuzzy score of each contractor on each criterion**

Triangular fuzzy numbers in Table 7.3 were used to transform the linguistic terms to fuzzy numbers and fuzzy score matrix for criteria 1 (CR_1) is shown below.

Fuzzy score matrix for criterion (CR_1=Past Performance)

$$\begin{array}{c} \\ A_1 \\ A_2 \\ A_q \end{array} \begin{array}{c} \begin{array}{ccc} DM_1 & DM_2 & DM_3 \end{array} \\ \begin{bmatrix} F_E & F_F & F_E \\ F_F & F_F & F_E \\ F_G & F_F & F_D \end{bmatrix} \end{array}$$

Where F_F = (0.7, 0.9, 1), F_G = (0.9, 1, 1), F_E = (0.5, 0.7, 0.9) and F_D = (0.3, 0.5, 0.7) are the respective fuzzy numbers.

Similarly, all the fuzzy numbers for each contractor (A1, A2 and A3) on each criterion (CR1, CR2 ……CR16) by the three decision makers (DM1, DM2 and DM3) were obtained and tabulated in columns A-F of Table AI.2.3 using Excel spreadsheet.

Step 6: Total Fuzzy Score for each criterion and development of Total Fuzzy Score matrix

Total fuzzy Score (TFS) of each contractor on each criterion was calculated using fuzzy addition equation (Equation 7.1).

Total fuzzy score (TFS) of contractor A1 on criterion C1 is given by:

$TFS_{A1C1} = (F_E + F_F + F_E) = (0.5, 0.7, 0.9) + (0.7, 0.9, 1) + (0.5, 0.7, 0.9) = (1.7, 2.3, 2.8)$

Refer Column 'G' of Table AI.2 for all the TFSs.

Step 7: Normalised Fuzzy Score of each contractor on each criterion

Normalised fuzzy score (NFS) of each contractor on each criterion was found by:

Normalised fuzzy combined score(NFS) = Total fuzzy score (TFS) of each criterion/p (number of DMs)

NFS of contractor A1 on criterion C1 is equal to = (1.7, 2.3, 2.8)/3 = (0.5667, 0.7667, 0.9333). Similar all NFS were calculated and tabulated in column H of Table AI.2.

Step 8: Crisp score (Defuzzified score)

Using equation 7.5, the crisp scores of each contractor on each criterion were calculated.

Crisp Score (CS) of Contractor A1 on criterion C1 is given by:

$CS_{A1C1} = (0.5667 + 2*.7667 + .9333)/3 = 1.0111$

Similar, all CSs were calculated. Refer Table AI.3 for calculations.

Next, develop the Crisp score matrix using all CSs:

$$CR_1 \qquad CR_2 \qquad CR_3 \qquad \cdots \qquad CR_{16}$$

$$\begin{array}{c} A_1 \\ A_2 \\ A_3 \end{array} \begin{bmatrix} 1.0111 & 1.000 & 0.9222 & \cdots & 1.0111 \\ 1.0889 & 1.2556 & 1.3000 & \cdots & 1.0889 \\ 1.0444 & 1.1333 & 1.1333 & \cdots & 0.9667 \end{bmatrix}$$

Table AI.1 Fuzzy combine score results

A	B	C	D	E	F	G	H
			DM_1	*DM_2*	*DM_3*	*Total Fuzzy Score TFS=(DM_1+DM_2+DM_3)*	*Normalised Fuzzy Score NFS =TFS/p (no. of DMs)*
CR_1	*Past Performance*	*A1*	*(0.5, 0.7, 0.9)*	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(1.7, 2.3, 2.8)*	*(0.5667, 0.7667, 0.9333)*
		A2	*(0.7, 0.9, 1)*	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(1.9, 2.5, 2.9)*	*(0.6333, 0.8333, 0.9667)*
		A3	*(0.9, 1, 1)*	*(0.7, 0.9, 1)*	*(0.3, 0.5, 0.7)*	*(1.9, 2.4, 2.7)*	*(0.6333, 0.8000, 0.9000)*
CR_2	*Project understanding*	A1	*(0.3, 0.5, 0.7)*	*(0.7, 0.9, 1)*	*(0.7, 0.9, 1)*	*(1.7, 2.3, 2.7)*	*(0.5667, 0.7667, 0.9000)*
		A2	*(0.9, 1, 1)*	*(0.9,1, 1)*	*(0.7, 0.9, 1)*	*(2.5, 2.9, 3)*	*(0.8333, 0.9667, 1.0000)*
		A3	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(0.9, 1, 1)*	*(2.5, 2.9, 3)*	*(0.7000, 0.8667, 0.9667)*
CR_3	*Technical*	A1	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(0.3, 0.5, 0.7)*	*(1.5, 2.1, 2.6)*	*(0.5000, 0.7000, 0.8667)*
		A2	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(2.7, 3, 3)*	*(0.9000, 1.0000, 1.0000)*
		A3	*(0.5, 0.7, 0.9)*	*(0.7, 0.9, 1)*	*(0.9, 1, 1)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
CR_4	*Key personnel*	A1	*(0.9, 1, 1)*	*(0.7, 0.9, 1)*	*(0.9, 1, 1)*	*(2.1, 2.6, 2.9)*	*(0.8333, 0.9667, 1.0000)*
		A2	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(2.7, 3, 3)*	*(0.9000, 1.0000, 1.0000)*
		A3	*(0.7, 0.9, 1)*	*(0.9, 1, 1)*	*(0.5, 0.7, 0.9)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
CR_5	*Health and Safety*	A1	*(0.5, 0.7, 0.9)*	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(1.7, 2.3, 2.8)*	*(0.5667, 0.7667, 0.9333)*
		A2	*(0.7, 0.9, 1)*	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(1.9, 2.5, 2.9)*	*(0.6333, 0.8333, 0.9667)*
		A3	*(0.9, 1, 1)*	*(0.7, 0.9, 1)*	*(0.3, 0.5, 0.7)*	*(1.9, 2.4, 2.7)*	*(0.6333, 0.8000, 0.9000)*
CR_6	*Time*	A1	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(0.3, 0.5, 0.7)*	*(1.5, 2.1, 2.6)*	*(0.5000, 0.7000, 0.8667)*
		A2	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(2.7, 3, 3)*	*(0.9000, 1.0000, 1.0000)*
		A3	*(0.5, 0.7, 0.9)*	*(0.7, 0.9, 1)*	*(0.9, 1, 1)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*

CR_7	*Financial*	A1	*(0.9, 1, 1)*	*(0.7, 0.9, 1)*	*(0.9, 1, 1)*	*(2.1, 2.6, 2.9)*	*(0.8333, 0.9667, 1.0000)*
		A2	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(2.7, 3, 3)*	*(0.9000, 1.0000, 1.0000)*
		A3	*(0.7, 0.9, 1)*	*(0.9, 1, 1)*	*(0.5, 0.7, 0.9)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
CR_8	*Contractual and legal*	A1	*(0.3, 0.5, 0.7)*	*(0.3, 0.5, 0.7)*	*(0.3, 0.5, 0.7)*	*(0.9, 1.5, 2.1)*	*(0.3000, 0.5000, 0.7000)*
		A2	*(0.9, 1, 1)*	*(0.5, 0.7, 0.9)*	*(0.7, 0.9, 1)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
		A3	*(0.7, 0.9, 1)*	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(1.9, 2.5, 2.9)*	*(0.6333, 0.8333, 0.9667)*
CR_9	*Past experience*	A1	*(0.3, 0.5, 0.7)*	*(0.3, 0.5, 0.7)*	*(0.3, 0.5, 0.7)*	*(0.9, 1.5, 2.1)*	*(0.3000, 0.5000, 0.7000)*
		A2	*(0.9, 1, 1)*	*(0.5, 0.7, 0.9)*	*(0.7, 0.9, 1)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
		A3	*(0.7, 0.9, 1)*	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(1.9, 2.5, 2.9)*	*(0.6333, 0.8333, 0.9667)*
CR_{10}	*Management*	A1	*(0.3, 0.5, 0.7)*	*(0.7, 0.9, 1)*	*(0.7, 0.9, 1)*	*(1.7, 2.3, 2.7)*	*(0.5667, 0.7667, 0.9000)*
		A2	*(0.9, 1, 1)*	*(0.9,1, 1)*	*(0.7, 0.9, 1)*	*(2.5, 2.9, 3)*	*(0.8333, 0.9667, 1.0000)*
		A3	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(0.9, 1, 1)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
CR_{11}	*Cost*	A1	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(2.7, 3, 3)*	*(0.9000, 1.0000, 1.0000)*
		A2	*(0.7, 0.9, 1)*	*(0.9, 1, 1)*	*(0.5, 0.7, 0.9)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
		A3	*(0.3, 0.5, 0.7)*	*(0.3, 0.5, 0.7)*	*(0.3, 0.5, 0.7)*	*(0.9, 1.5, 2.1)*	*(0.3000, 0.5000, 0.7000)*
CR_{12}	*Quality*	A1	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(0.7, 0.9, 1)*	*(1.0, 2.5, 2.9)*	*(0.8333, 0.9667, 1.0000)*
		A2	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(0.9, 1, 1)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
		A3	*(0.9, 1, 1)*	*(0.5, 0.7, 0.9)*	*(0.7, 0.9, 1)*	*(2.5, 2.9, 3)*	*(0.8333, 0.9667, 1.0000)*
CR_{13}	*Relationships*	A1	*(0.9, 1, 1)*	*(0.7, 0.9, 1)*	*(0.7, 0.9, 1)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
		A2	*(0.7, 0.9, 1)*	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(2.5, 2.9, 3)*	*(0.8333, 0.9667, 1.0000)*
		A3	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(0.9, 1, 1)*	*(2.5, 2.9, 3)*	*(0.8333, 0.9667, 1.0000)*
CR_{14}	*Industrial relations*	A1	*(0.7, 0.9, 1)*	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(1.9, 2.5, 2.9)*	*(0.6333, 0.8333, 0.9667)*
		A2	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(0.7, 0.9, 1)*	*(2.5, 2.9, 3)*	*(0.8333, 0.9667, 1.0000)*

		A3	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(0.9, 1, 1)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
CR_{15}	*Environment and Sustainability*	A1	*(0.7, 0.9, 1)*	*(0.5, 0.7, 0.9)*	*(0.9, 1, 1)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
		A2	*(0.9, 1, 1)*	*(0.7, 0.9, 1)*	*(0.9, 1, 1)*	*(2.5, 2.9, 3)*	*(0.8333, 0.9667, 1.0000)*
		A3	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(0.9, 1, 1)*	*(2.7, 3.0, 3)*	*(0.9000, 1.0000, 1.0000)*
CR_{16}	*Organisational*	A1	*(0.7, 0.9, 1)*	*(0.9, 1, 1)*	*(0.5, 0.7, 0.9)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*
		A2	*(0.3, 0.5, 0.7)*	*(0.3, 0.5, 0.7)*	*(0.3, 0.5, 0.7)*	*(0.9, 1.5, 2.1)*	*(0.3000, 0.5000, 0.7000)*
		A3	*(0.9, 1, 1)*	*(0.5, 0.7, 0.9)*	*0.7, 0.9, 1)*	*(2.1, 2.6, 2.9)*	*(0.7000, 0.8667, 0.9667)*

Table AI.2 Crisp score of each alternative

A	*Fuzzy score*			*Crisp score (Eq. 7.5)*		
	A_1	A_2	A_3	A_1	A_2	A_3
CR_1	(0.5667, 0.7667, 0.9333)	(0.6333, 0.8333, 0.9667)	(0.6333, 0.8000, 0.9000)	1.0111	1.0889	1.0444
CR_2	(0.5667, 0.7667,0.9000)	(0.8333, 0.9667, 1.0000)	(0.7000, 0.8667, 0.9667)	1.0000	1.2556	1.1333
CR_3	(0.5000, 0.7000, 0.8667)	(0.9000, 1.0000, 1.0000)	(0.7000, 0.8667, 0.9667)	0.9222	1.3000	1.1333
CR_4	(0.8333, 0.9667, 1.0000)	(0.9000, 1.0000, 1.0000)	(0.7000, 0.8667, 0.9667)	1.2556	1.3000	1.1333
CR_5	(0.5667, 0.7667, 0.9333)	(0.6333, 0.8333, 0.9667)	(0.6333, 0.8000, 0.9000)	1.0111	1.0889	1.0444
CR_6	(0.5000, 0.7000, 0.8667)	(09000, 1.0000, 1.1000)	(0.7000, 0.8667, 0.9667)	0.9222	1.3000	1.1333
CR_7	(0.8333, 0.9667, 1.0000)	(0.9000, 1.0000, 1.0000)	(0.7000, 0.8667, 0.9667)	1.2556	1.3000	1.1333
CR_8	(0.3000, 0.5000, 0.7000)	(0.7000, 0.8667, 0.9667)	(0.6333, 0.8333, 0.9667)	0.6667	1.1333	1.0889
CR_9	(0.3000, 0.5000, 0.7000)	(0.7000, 0.8667, 0.9667)	(0.6333, 0.8333, 0.9667)	0.6667	1.1333	1.0889
CR_{10}	(0.5667, 0.7667,0.9000)	(0.8333, 0.9667, 1.0000)	(0.7000, 0.8667, 0.9667)	1.0000	1.2556	1.1333
CR_{11}	(0.9000, 1.0000, 1.0000)	(0.7000, 0.8667, 0.9667)	(0.3000, 0.5000, 0.6667)	1.3000	1.1333	0.6667
CR_{12}	(0.8333, 0.9667, 1.000)	(0.7000, 0.8667, 0.9667)	(0.8333, 0.9667, 1.0000)	1.2556	1.1333	1.2556
CR_{13}	(0.7000, 0.8667,0.9667)	(0.6333, 0.8333, 0.9667)	(0.8333, 0.9667, 1.0000)	1.1333	1.0889	1.2556
CR_{14}	(0.6333, 0.8333, 0.9667)	(0.8333, 0.9667, 1.0000)	(0.7000, 0.8667, 0.9667)	1.0889	1.2556	1.1333
CR_{15}	(0.7000, 0.8667, 0.9667)	(0.7667, 0.9333, 1.0000)	(0.9000, 1.0000, 1.0000)	1.1333	1.2111	1.3000
CR_{16}	(0.7000, 0.8667, 0.9667)	(0.3000, 0.5000, 0.7000)	(0.7000, 0.8667, 0.9667)	1.1333	0.6667	1.1333

Step 9: Total Weighted Crisp Score of each contractor (Total Weighted Crisp Score matrix)

Importance weights matrix W_j was derived using the importance weighting of each criterion given in Table 7.1 as in step2:

$$W_j = \begin{bmatrix} 0.0684 \\ 0.0678 \\ \vdots \\ 0.0538 \end{bmatrix} \begin{matrix} W_1 \\ W_2 \\ \vdots \\ W_{16} \end{matrix}$$

Multiplying two matrices gives the weighted crisp score of each alternative.

$$\begin{matrix} & CR_1 & CR_2 & CR_3 & \cdots & CR_{16} \\ A_1 & & & & & \\ A_2 & & & & & \\ A_3 & & & & & \end{matrix} \begin{bmatrix} 1.0111 & 1.0000 & 0.9222 & \cdots & 1.1333 \\ 1.0889 & 1.2556 & 1.3000 & \cdots & 0.6667 \\ 1.0444 & 1.1333 & 1.1333 & \cdots & 1.1333 \end{bmatrix} \times W_j = \begin{bmatrix} 0.0684 \\ 0.0678 \\ \vdots \\ 0.0538 \end{bmatrix} \begin{matrix} W_1 \\ W_2 \\ \vdots \\ W_{16} \end{matrix}$$

Total weighted crisp score (TWCS) of Contractor A1 is given by =1.0111*.0684+1.0000*0.0678+…….. 1.1333*.0538 = 1.044

$TWCS_r$

$$\begin{matrix} A_1 \\ A_2 \\ A_3 \end{matrix} \begin{bmatrix} 1.044 \\ 2.716 \\ 2.509 \end{bmatrix}$$

Ranking of contractors by Normalised Weighted Crisp Score (NWCS)

Weighted average method is implemented to find Normalised weighted crisp score (NWCS) using equation 7.6.

Normalised weighted crisp score for contractor A1:

$A_1 = TWCS_{A1} / \sum_{r=1}^{3} TWCS_{Ar}$

Where ΣTWCS=6.270

Table AI.3 Tender Evaluation summary result

	A_1	A_2	A_3
NWCS	0.166	0.433	0.400
Ranking	3	1	2
Recommendation		Selected for contract negotiations	